Swarm Search and Rescue with Differing Scales of Autonomy

Abstract

[To write. Draft facts for it: 33 operators, two 8-minute sessions each (Strategic Heavy and Tactical Heavy scenarios, counterbalanced), a heterogeneous 33-drone fleet, two optional assistants at two decision tiers. Headline results: reliance is a stable personal style; hand-over rises with mission size on both tiers; momentary load does not push operators onto the assistants; the Tactical Assistant is trusted more; a visible flaw on the strategic tier and an invisible one on the tactical tier.]

1 Introduction

The deployment of large numbers of robots has advanced from speculation from practice in a short time frame. Present applications of drones and similar uncrewed vehicles are increasingly requiring either large numbers of well-organised operators [23], or else provide early demonstrations of semi-autonomous command-and-control software [3]. There is significant appetite for the incorporation of autonomy for reasons including but not limited to the impracticalities of supporting one pilot for each asset, and a desire for a single operator to maintain a complete picture of the entire operation [29]. As swarms scale further, the one-to-one or many-to-one models of human control must give way to a one-to-many paradigm, and some degree of autonomy becomes not merely desirable but unavoidable.

How that autonomy is best integrated, however, remains an open question. Autonomy is far from new; defence operations have long been organised around autonomy at different levels of a hierarchy. Auftragstaktik (mission command) requires strategic command to communicate high-level goals to subordinates and grants them latitude over low-level execution, and indeed further delegation. A similar logic can be applied to robot swarms: tactical decisions may be delegated to individual assets or sub-groups, while strategic decisions remain with a human commander. Yet it is not obvious that the best configuration places the tactical under full autonomy while leaving the strategic entirely manual. Lower-level interventions may benefit heavily from human oversight; robots still struggle with physical execution, while autonomous planners are near-perfect at many allocation problems. Of course, higher-level decisions carry greater consequence and arguably warrant closer human control; it is natural to prefer an autonomy-at-tactical approach, however the best possible combination of autonomy levels may not be so simple. Each of these judgements also depends on operator workload and situational context. The need for interfaces and control paradigms which allow autonomy is therefore high [29], but the question of *where* in a command hierarchy that autonomy should sit, and whether this should be fixed or operator-configurable, has received little attention. Previous works have explored many forms of autonomy [18], but these have typically been treated at the level of a single decision layer.

This work represents a first step toward a longer-term vision: We aim to understand how autonomy effects both user and mission when deployed at each level of a multi-layered command hierarchy, and move towards a system where each level of concern from strategic down to tactical details, can be automated to varying degrees. Finally, we propose a "co-pilot" that, instead of attempting to fully automate the mission, identifies where the user is most needed, and suggest to them an autonomy profile which adapts on-the-fly.

We develop a simulated search-and-rescue scenario using a heterogeneous drone swarm, in which a two-tier assistive system (a strategic tier and a tactical tier) is available to the operator throughout.

[Step 4 note (kept from the old draft): this is the natural place to make the *ORCHID lineage* explicit — name `jennings2014human` (human-agent collectives / flexible autonomy) and `ramchurn2021trustworthy` alongside the already-cited `ramchurn2016disaster`. See cluster C1 in the Related Work notes box. Not drafted.] Neither assistant has errors injected: the manipulation is not the assistants' reliability but *where the operational demand sits*. In a within-subjects design each participant plays two 8-minute sessions — **Strategic Heavy** (frequent, small

missions, placing demand on cross-mission allocation) and **Tactical Heavy** (fewer, larger missions, placing demand on within-mission planning) — with the fleet and the assistants identical in both. Using either assistant is always optional, so the study observes when operators choose to hand a decision over, at which tier, and how that relates to the mission, the moment, the person and their trust.

[RQ block — REVISED to match what the analysis found (the old block assumed a perfect-agent "value of intervention" RQ4 that the data cannot answer). Candidate framings, each mapped to a results section; merge or cut as you like:

- **RQ1 (who).** How much of an operator's use of the assistants is a stable personal style, and is it predicted by attitudes measured beforehand? → Section 7.2.
- **RQ2 (when).** What makes an operator hand a decision over — the mission (its size), the moment (load, the queue), or practice — and does the answer differ between the strategic and tactical tiers? → Sections 7.3–7.7.
- **RQ3 (flaws).** How do operators respond to an assistant's flaw that is visible (the illogical strategy card) versus one that is not (the tactical assistant's wrong refusal)? → Section 7.8.
- **RQ4 (trust by tier).** Is one tier's assistant trusted more, where does the difference come from, and does stated trust match use? → Section 7.9.

Note: "failure" in this study means hardware drone failure during execution, plus the two side effects of the assistants' rules (Section 4.3) — not injected agent error. Operator-built deadlocks were auto-repaired and never surfaced.]

2 Related Work

Essential to any autonomous system is the notion of trust. Trust in this context is defined by Lee and See as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" [11]. Such qualities are abundant in swarm contexts, where the positions, motions, and goals of the robots are rife with uncertainty, and there is a significant risk of failure stemming from the attritable nature of many such assets and the high levels of risk in their environment [27]. It is an important consideration, as a system which is not trusted will likely not be used by the operator, and a system which is trusted too much will be relied on without appropriate checking and oversight. A system which balances under- and over-trust is said to be well calibrated [13].

It has been observed that once a user falls into the trap of under- or over-trust, they have a tendency to remain there unless actively coaxed back into a well-calibrated relationship [20]. The standard tool for this is explanations, however these are not always given due attention [17, 31], and often-times users are biased in favour of action over inaction so accept suggestions readily [15]. In the domain of HSI the notion of neglect benevolence describes the observed advantage from leaving the swarm to resolve its own behaviour instead of intervening overly often [16], additionally supporting the notion of autonomy which manages itself under human supervision instead of constant interaction with the recommender. Equally important to explanations, then, may be a design pattern where users work in tandem with autonomous tools and can elect to deploy them selectively, with varying levels of autonomy on varying scales of a swarm or sub-swarm.

Traditionally swarm robotic systems are characterised by autonomy, however with the importance of human oversight a question arises as to what level of autonomy should be given, and how the human's actions interface with some level of autonomy. Traditional scales such as Sheridan and Verplank's LOAs are a common metric [26], as well as more swarm-focussed scales of a similar form [7]. An important note in more developed robotic systems is that the degree of autonomy can vary at different parts of the robot's role (for example information acquisition may be more autonomous than decision-making) [21].

An important note is that the terms "automation" and "autonomy" are often used interchangeably but are in fact different things, with the former describing well-defined processes of tasks deferred to an *automated system*, and the latter describing *autonomous systems* which can make "decisions independent of human control" [12]. As we are referring to swarm robots in this work, the terms are somewhat more interchangeable, as to automate a swarm on a particular task requires the granting of some degree of autonomy (for example performing decentralised allocation, where an automated algorithm runs across robots who then autonomously resolve their tasks [4]). We tend to prefer

the term autonomy as it is more relevant to classical swarm paradigms, but remind the reader that there can be some distinction dependent on use case.

Similar to calibrating trust, it is important that any assistive autonomous system is sufficiently performant to merit use. Human performance in the presence of such an agent is highly sensitive to the reliability and performance of that agent [25]. Some degree of imperfection is an expectation (otherwise there would be no need for a human), and humans are typically willing to tolerate occasional failure if it represents an adequate time- and effort-save. In fact, the most trusted systems are often those which sit in between 'too poor to trust' and 'too reliable to bother checking' [10].

Another important factor is the level of workload. Prior works have found that under high task load, more reliable automation increases operator performance, however under low task load, more reliable automation actually *decreases* operator performance. This is likely due to the over-trust issue discussed above, and presents a difficult challenge to designers, as users may defer to the agent even when not overloaded [22].

The level of chosen autonomy will necessarily represent a trade-off between different desirable qualities as alluded to above, and no single solution is likely to best suit every scenario [14]. This motivates a flexible approach where the level of autonomy is not fixed, rather it can be chosen by the operator. Such flexible approaches have been explored to the extent where agents can be granted autonomy where desirable [5], or else autonomous agents can be overridden at any level while they attempt to perform autonomously [30]. So-called mixed-initiative approaches to swarm control allow a human to switch the entire swarm between manual and autonomous modes [18]. To our knowledge, however, no prior work has examined the behaviour of an operator when autonomous assistance is structured into a two-tier hierarchy and the operator is free to use it selectively. Such a design allows us to ask not only whether the assistive system improves performance, but *when* operators choose to engage it and what advantage that engagement confers.

We term the two structural dimensions of selective autonomy the **Depth** and **Breadth** of autonomy. **Depth** refers to how autonomous some agent is, pertaining directly to Sheridan and Verplank's LOAs [26]. **Breadth** refers to the scale of autonomy, whether that be an individual asset, a mission's worth of assets, or the entire reserve. The present work extends this framework by stacking two agents vertically in a hierarchy—one operating at the within-mission (tactical) level and one at the cross-mission (strategic) level—and asking how the operator engages with, and elects to intervene in, autonomy at each level.

Metrics exist to assess the difficulty of tasks in the context of swarms, such as efficiency and effectiveness, and it has been suggested that autonomy should vary depending on these [8]. As well as the complexity of the task, a more direct assessment of cognitive complexity via operator state metrics such as EEG has been employed [1], and such signals could in principle inform a meta-level recommendation system.

[Drafted gap paragraph — my pass, for your check; rewrite freely. Intended as the closing paragraph of Related Work. Requires the supplement bib for the four keys.]

Most recently, the one-operator, many-robot paradigm has moved from concept to field demonstration. Adams et al. showed that a single trained commander can supervise a swarm of one hundred heterogeneous ground and aerial robots across real urban field exercises, albeit with frequent workload peaks [2], while the MOSAIC framework demonstrated a single operator coordinating a heterogeneous five-robot team in a lunar-analogue mission by shifting individual platforms between driver-, task-, and mission-level autonomy [19]. In parallel, variable autonomy has consolidated into a recognised subfield in its own right, with dedicated venues [28] and emerging design guidelines and research agendas [24]. Across this line of work, however, autonomy is treated as a single dial — a *level* to which a robot or team is set — and the evidence base consists of demonstrations and design agendas rather than controlled operator studies. What remains unexamined is the setting in which assistance is available simultaneously at *distinct decision tiers* — cross-mission resource allocation and within-mission planning — and the operator chooses, moment to moment, how far to engage each. The present study addresses exactly this gap: a controlled, within-subjects examination of how the locus of operational demand shifts operator engagement, workload, and trust across a two-tier assistive hierarchy.

Step 4 — Reference integration notes (proposals, not drafted text). New keys live in `refs_supplement_new.bib` (add it via `\bibliography{refs,refs_supplement_new}`). Each cluster gives *where* to place it and *what* it buys; the wording is yours. Key nuance for the whole trust literature below: our agent is now **perfect** — cite the trust-in-*imperfect*-autonomy work as calibration/per-tier-trust *backdrop and motivation*, not as error-detection, since we inject no agent error.

- **C1 — ORCHID lineage** (Intro near `ramchurn2016disaster`, and/or opening Related Work). Keys: `jennings2014human` (HAC vision; "flexible autonomy"/"agile teaming"), `ramchurn2021trustworthy` (successor framing — its UAV disaster-response pipeline mirrors our two-tier setup and its "errors enter at every pipeline stage" point motivates the per-tier question), `soorati2022intelligent` (group agenda). *Angle:* frame the study as building on the ORCHID human-agent-collectives programme; `ramchurn2016disaster` (already cited) is its flagship.
- **C2 — task-allocation substrate** (short new paragraph in Related Work, by the automation/autonomy discussion). Keys: `ramchurn2010cfstp` (CFSTP: heterogeneous agents, deadline/workload tasks — the formal root of our strategic/tactical allocation), `capezzuto2020anytime` & `capezzuto2021largescale/capezzuto2021ar` (modern scalable extensions), `macarthur2011distributed` (already in bib, uncited), `ramchurn2010robocup`, `wu2015agile`, `baker2016planning` (UAV SAR planning). *Angle:* one or two sentences establishing the agent's allocation task as a well-studied MAS problem; cite the lineage, don't dwell.
- **C3 — single-operator / heterogeneous-swarm HSI** (near the LOAs / one-to-many discussion). Keys: `soorati2021designi` (in-group HSI interface; HARIS precursor), `abioye2024adaptive` (your own group; workload-adaptive HSI → RQ2/RQ3), `adams2025supervise` (single operator, 100 heterogeneous robots; now also in the gap paragraph — best used as an Intro feasibility anchor, an RQ2 workload-at-scale *contrast* (their physiological multi-dimensional estimation vs. our NASA-TLX), and qualitative motivation for the tactical tier: operators were "red-lined" precisely when tactical allocation autonomy was unavailable. Not a design comparator — no tiers, no trust, n=2 field observation), `chandarana2018determining` (multi-job-type swarm sizing). *Angle:* situate the one-operator + heterogeneous-asset setup and the workload stakes.
- **C4 — variable/adjustable autonomy** (attach to the Depth/Breadth paragraph and the mixed-initiative cites). Keys: `methnani2024charge` (survey — anchor), `theodorou2024editorial`, `ramesh2023experimental`, `verhagen2024meaningful` (firefighting; SAR-flavoured), `reinmund2024variable`, `oberacker2026mosaic` (MOSAIC; now also in the gap paragraph — the depth axis operating in the field: one operator slides a heterogeneous five-robot team across driver/task/mission autonomy levels, but allocation is folded into mission-level autonomy with no separately-operable strategic tier; arXiv preprint, verify status at submission); `tecuci_seven_2007` (already in bib, mixed-initiative). *Angle:* the RQ3 grounding ("where should autonomy sit across levels"); links Depth/Breadth to the variable-autonomy subfield.
- **C5 — trust across tiers** (extend the trust-calibration paragraph; motivates RQ3 per-tier trust and RQ4). Keys: `wischnewski2023measuring` (subsystem-failure → whole-system trust spillover — the key precedent for measuring trust *per tier*), `wildman2024trust` (multilevel trust in 3+-member HATs), `rittenberg2024trust` (between-subjects reliability manipulation), `naiseh2023explanation`, `centeiojorge2023exploring`; `parasuraman2010complacency` (already in bib). *Angle:* frame as system-wide vs. component-specific trust; justify measuring trust separately for each tier. NB the closest analogue — a multilevel "agent reliability pairing" study (Am. J. Psychology) — I could not fully verify; see the supplement TODO before citing.
- **C6 — selective use / recommendation acceptance** (the "elect to deploy them selectively" sentence). Key: `fink_let_2024` (already in bib) — giving users control over whether to accept a recommendation raises acceptance; direct support for the selective-deployment design and RQ2.
- **Optional (thesis-level, not this paper):** the AtomicOrchid platform papers and Fischer's "in-the-loop vs on-the-loop" work sit better with your HARIS/platform chapter (RQ1) than here; omitted from the supplement to keep scope tight.

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3 Scenario and Task Design

3.1 Operational Scenario

We built a simulated search-and-rescue operation in which a single operator directs a heterogeneous fleet of 33 drones from one hub. Missions keep arriving at positions on a map; each consists of two to six tasks, and each task needs a

fixed team of drone types. For every mission the operator makes two decisions in sequence:

- a **strategic** decision — *how many* drones of each type to commit to the mission, taken from the shared reserve that every other mission also draws on; and
- a **tactical** decision — *which* committed drone does *which* task, and in what order.

An assistant is available for each decision (Section 4); using either is always optional. Drones can fail mid-task, in which case the operator repairs the plan or abandons the mission. The score rewards completed tasks and penalises the time missions spend unfinished, so both getting it right and getting it done quickly matter.

3.2 Asset Types

The fleet comprises 11 drones of each of three types (Table 4). The types are named to the operator by role rather than colour — **Fast** (blue; 11 u/s), **Lifter** (red; 10 u/s) and **Camera** (green; 9 u/s) — and drones are labelled accordingly (e.g. `Fast-7`). Speeds are not shown to participants; the tutorial describes the types only as fastest, standard and slowest. Drones committed to a mission leave the reserve at once and return to it when their task chain is complete. The mission generator's archetype weights (below) were tuned in simulation so that aggregate demand, in drone-seconds, is approximately equal across the three types in both scenarios, so no single type is the systematic bottleneck.

3.3 Task Types

There are five task types of increasing complexity and value (Table 1). Each has a primary composition; three have a substitute composition with one fewer drone of the paired type, which takes about 2.5 times as long. The planner shows the substitute as an "OR" line on the task badge.

Table 1: Task types. Execution times are for the primary composition / the substitute.

Type	Name	Primary	Substitute	Time (s)	Points
T1	Recce	1 Fast	—	10 / —	20
T2	Recon	2 Fast	1 Fast	15 / 38	40
T3	Supply Drop	2 Lifter + 1 Camera	1 Lifter + 1 Camera	25 / 62	60
T4	Precision Supply Drop	1 Lifter + 2 Camera	1 Lifter + 1 Camera	30 / 75	80
T5	Search & Service	1 Fast + 1 Lifter + 1 Camera	—	45 / —	100

3.4 Missions

Missions arrive by a Poisson process. Each has a **category** that sets its number of tasks and how quickly its penalty grows — A Routine (2 tasks), B Moderate (3), C Significant (4), D Critical (5), E Mass Casualty (6) — and an **archetype** drawn at generation (Fast-, Lifter-, Camera-leaning or Mixed, weights 38/26/21/15) that sets which task types make up that count (Table 2). A mission therefore needs one drone type (e.g. an all-Recce Fast mission), or all three; only 12 of 606 strategic decisions concerned a two-type mission. Mission zones are circles of radius 80 units at least 150 units from the hub and 200 units from any other active zone. Where the findings speak of **mission size** they mean the number of drones the mission needs: the sum of its tasks' primary requirements, which the task badges display and which is the same for every operator who met that mission.

3.5 Score

The score S is completion reward minus an accrued time penalty, floored at zero:

$$S = \max\left(0,\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} w_k\,\mathbf{1}[k \text{ completed}]\ -\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} r_m\,\frac{w_k}{\sum_{k'\in\mathcal{T}_m} w_{k'}}\left(t_k^{\text{end}} - t_m^{a}\right)\right) \tag{1}$$

where w_k is the points value of task k (Table 1), r_m the penalty rate of the mission's category (0.05–0.40 points/s; Table 4), t_m^a the mission's arrival time and t_k^{end} the task's completion time, or the current time while it is unfinished. The penalty is thus charged per task, weighted by the task's share of its mission's points, from the mission's arrival

Table 2: Task list for each mission (category × archetype), as generated. Task counts grow with category; the archetype fixes the mix.

Cat.	#	Fast-lean	Lifter-lean	Camera-lean	Mixed
A	2	T1, T2	T3, T5	T4, T5	T1, T3
B	3	T1, T2, T2	T3, T3, T5	T1, T4, T4	T1, T3, T5
C	4	T1, T1, T2, T2	T3, T3, T5, T5	T1, T4, T4, T4	T1, T2, T3, T5
D	5	T1, T1, T2, T2, T5	T3, T3, T3, T5, T5	T1, T4, T4, T4, T5	T1, T2, T3, T3, T5
E	6	T1, T1, T2, T2, T2, T5	T3, T3, T3, T5, T5, T5	T4, T4, T4, T1, T5, T5	T1, T2, T3, T3, T4, T5

until that task completes; it is displayed in 15 s steps. If a mission is abandoned, its parent's penalty clock stops and its unfinished tasks are re-queued as a new *residual* mission (id suffix `-R`) that accrues from that moment, so abandoning transfers work rather than losing it. Because the four mission streams (Section 6.1) differ in how many points they offer, raw scores are compared only within a game.

3.6 Drone Failures

Any deployed drone can fail. In the main build (`study-v1.4` onwards; Appendix B) failures are a live hazard of $1/900$ per drone per second, rolled on a fixed one-second simulated grid, identical for every drone type and mission. A drone that fails after its part of a task is done (e.g. the Lifters of a Supply Drop after the drop) exits gracefully. Otherwise the task reverts to pending and the mission enters **recovery**: its card turns red and the tactical planner opens in recovery mode (Figure 5). The operator may reassign the drones still on that mission (never the hub reserve, which is what makes abandoning a real option), press Suggest for a repair, or abandon the mission. A mission with an open recovery takes no further failures until 30 s after it is resolved (`v1.5` onwards). On average a session had about four drone failures (Table 3).

4 The Two Assistants

Both assistants are deliberately simple, transparent rules rather than learned models, and in every session analysed neither had errors injected (Section 4.3). In the interface they are called the *Strategic Assistant* and the *Tactical Assistant*; the words "AI" and "agent" are not used on screen.

4.1 Strategic Assistant: Strategy Cards

When the operator clicks *Allocate* on a queued mission, a panel opens beneath its card and two strategy cards load, each after an independent delay of about 4–5 s (Figure 3b). Each card shows the drones it would commit per type, the projected mission completion time (ETA), the reserve that would remain, and three bars: Speed (the best ETA of the two divided by this card's ETA), Reserve (share of the current reserve left behind) and Resilience (spare drones above the minimum needed).

- **Aggressive** — enough drones of each type for every task to run at once (the parallel sum of the primary compositions), with at least one spare of each type the mission uses; capped by the reserve.
- **Conservative** — the smallest team that completes every task in series (primary compositions, or substitutes where the reserve cannot cover a primary), plus one spare of each type the mission uses, plus 15% of the remaining reserve of *every* type. Tasks run mostly in series.

The 15% top-up is applied to every drone type, including types the mission does not use. On a one-type mission the Conservative card therefore often includes, say, a Camera drone on an all-Fast mission. We call this the **illogical card**: it was on screen for 132 of the 606 strategic decisions, almost all on one-type missions, and every participant met it. It was discovered during data collection and deliberately left unchanged so as not to split the cohort; Section 7.8 studies it.

The operator can deploy a card as offered (*Deploy Selected*), choose *Edit this allocation* (a counter per type, pre-filled

from the selected card), or choose *Set manually instead* (the same counters, empty), which is possible before the cards have loaded (Figure 3c). On deployment the committed drones leave the reserve at once and fly to the mission zone, where they wait at its edge for the tactical plan.

4.2 Tactical Assistant: Suggest

Once a mission has its drones, it appears as pending in the tactical planner on the second screen (Figure 4). The planner opens **empty**: no plan is shown until the operator asks for one. The operator can build the plan by hand — drag a drone onto a task; shift-drag chains a drone on to a further task after its first, and the sequence numbers update live — or press **Suggest**. Suggest builds a plan with a virtual-timeline scheduler: tasks are taken most-constrained first (T5 before T1) and each is given the earliest-free drones of the required types from the committed team, chaining drones across tasks where that finishes sooner. The plan appears one drone→task entry per second, and *Deploy* unlocks when it is complete. The operator can edit any part of it before deploying. *Deploy* is enabled only when every task has at least its substitute composition. The planner works in "plan-all" mode: the whole sequence is confirmed once.

Because Suggest routes every drone in one global task order, its plans cannot deadlock. An operator-built plan can (a cycle of chained drones each waiting on the other's task); in every session analysed, such deadlocks were detected and silently re-routed onto one order by the system, so a participant could never encounter one.

In recovery mode Suggest proposes a repair from the drones on the mission. It considers only drones that are not currently executing a task. When the only fix needs a busy drone — chaining it on after its current task, which the operator could do by hand — Suggest answers "No workable plan". We call this a **wrong refusal**. To the operator it is indistinguishable from a correct refusal. It was found after data collection and also left unchanged.

4.3 Assistant Reliability

The platform can inject errors into either assistant (a per-session error rate per tier). This was switched off ($\varepsilon_S = \varepsilon_T = 0$) for every session analysed; the one participant who ran with errors switched on (P09) is excluded. Neither assistant was therefore ever deliberately wrong. The assistants' only flaws are the two side effects of their rules described above: the illogical card on the strategic tier, which is visible on the card face, and the wrong refusal on the tactical tier, which is not. [The old draft's RQ4 ("value of intervention into a near-optimal agent") and the per-tier error manipulation do not apply to these data; see the RQ note in the Introduction.]

5 Platform and Interface

The platform is a browser application (React/TypeScript; no server) designed for two monitors side by side (Figure 1). All randomness is seeded, so a seed and a configuration reproduce a session exactly, and every operator action and assistant output is logged with a millisecond timestamp.

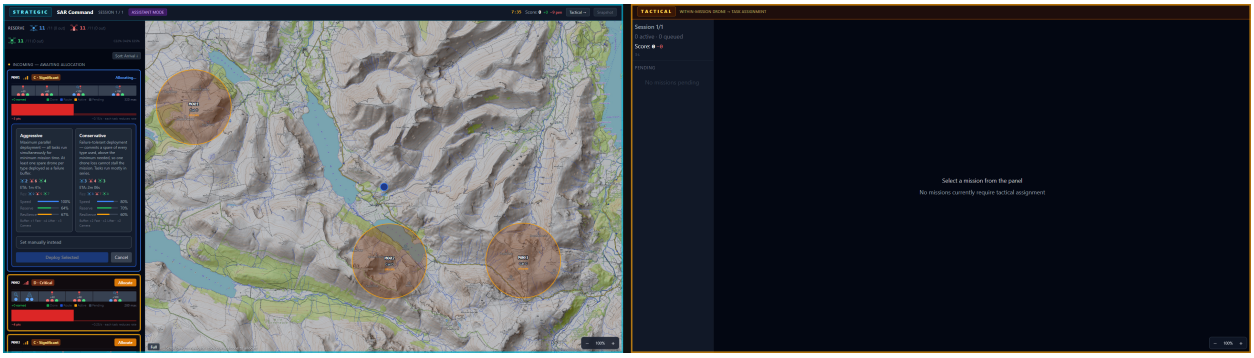


Figure 1: The two screens side by side. **Left, strategic screen:** reserve (top left), the queue of missions awaiting allocation (left column; the first is being allocated and shows its two strategy cards), and the strategic map with each mission's zone and the hub. **Right, tactical screen:** the tactical planner, here with no mission pending.

Strategic screen. The strategic screen (Figure 2) holds the reserve count per drone type, the mission queue and the strategic map. Each mission card (Figure 3a) shows the mission's id and category, a badge per task with its required drones and points, a progress legend (done, en route, active, pending) and a red penalty bar that grows while the mission is unfinished. The queue can be sorted by arrival or by criticality. About three and a half cards are visible at once; further waiting missions appear only on the map. The map can show zones only or every drone's path.

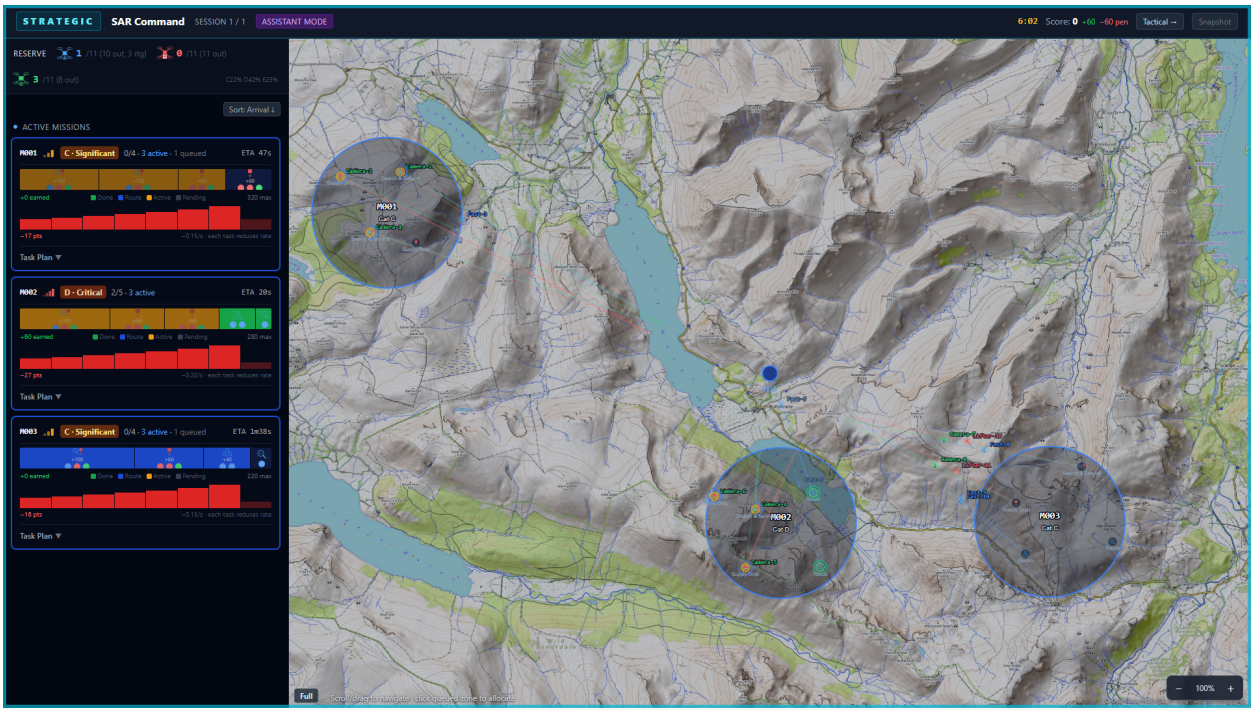


Figure 2: Strategic screen mid-session: missions in progress, drones in flight between the hub and the mission zones, and the penalty bars on the queued and active mission cards.

Tactical screen. The tactical screen lists missions awaiting a plan; selecting one opens the planner (Figure 4): the mission zone with each task as a waypoint labelled with its type and staffing (e.g. `1/2L 0/1C`, with its substitute below), the committed drones waiting at the zone edge, and a right-hand panel listing unassigned drones and the task schedule with each task's assigned drones and projected start and finish. Hovering a drone shows its full route.

Failures. When a drone fails, the mission's card turns red with a banner naming the lost drone, and the mission is listed as *reassign pending* on the tactical screen. The recovery planner (Figure 5) marks executing and travelling tasks, shows the failed drone crossed out, and offers *Suggest*, *Reassign* (enabled once every pending task is covered) and *Abandon Mission*.

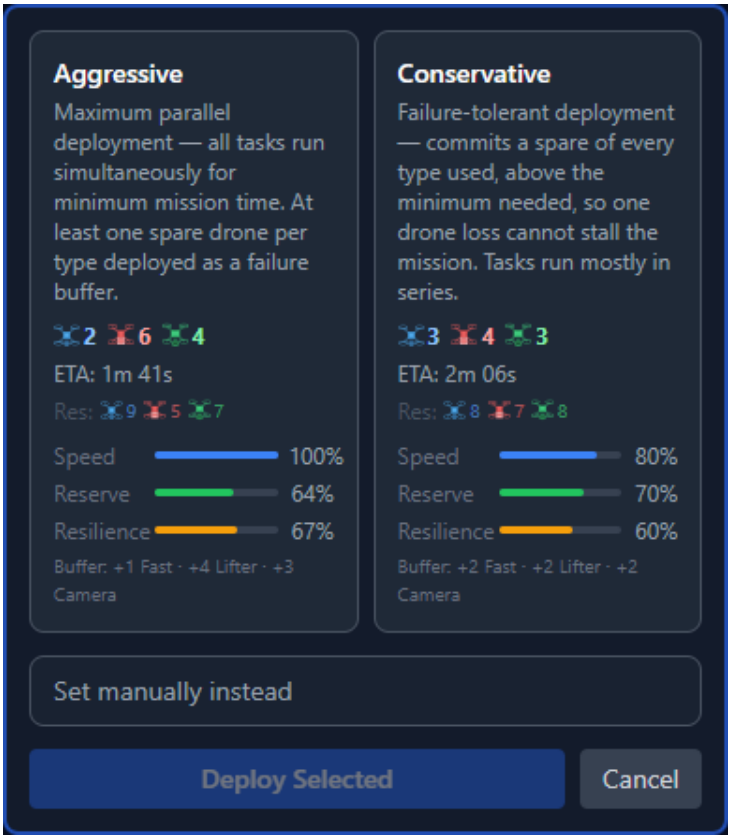
Tutorial. Before session 1 every participant completed a guided tutorial of 48 steps on a smaller fleet (6/6/6), with its own clock. It teaches the manual workflow first — allocate, plan, chain, deploy, recover from a failure by hand, and abandon a mission that cannot be recovered (two scripted failures) — and only then introduces the two assistants; the Suggest button is hidden until that lesson. The tutorial runs with error-free assistants.

6 Study Design

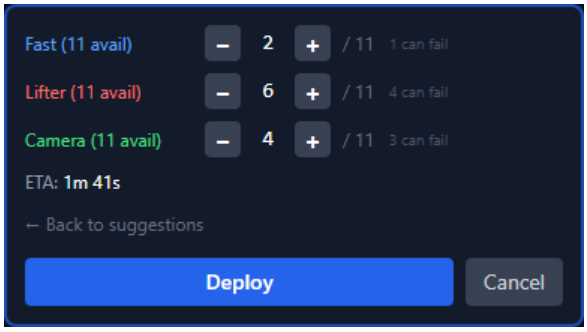
6.1 Design

The study used a within-subjects design with one factor, the **scenario**, i.e. where the operational demand sits. Each participant played one 8-minute (480 s) session of each scenario, in counterbalanced order (17 Strategic Heavy first,

(a) Mission queue: three missions awaiting allocation. Each badge shows a task's required drones and its points; the red bar is the accruing penalty.

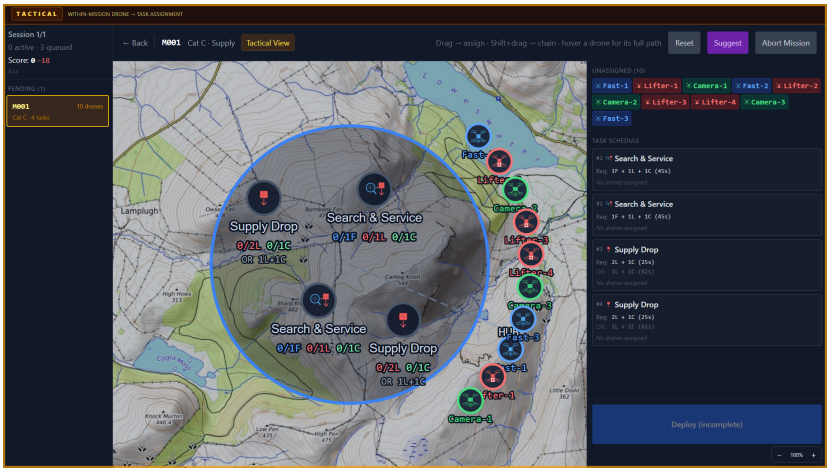


(b) The Strategic Assistant's two strategy cards for a Category C mission: drones per type, ETA, reserve left ("Res"), and the Speed, Reserve and Resilience bars.

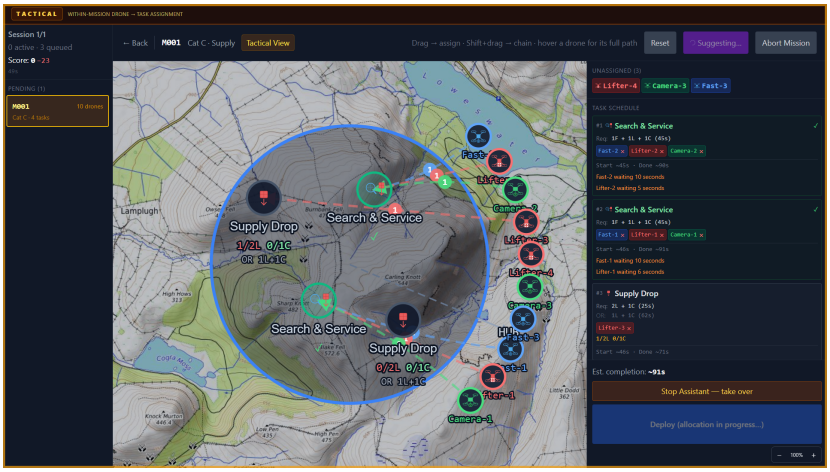


(c) Manual allocation (*Edit this allocation* or *Set manually instead*): a counter per type, how many of each could fail without stalling the mission, and the live ETA.

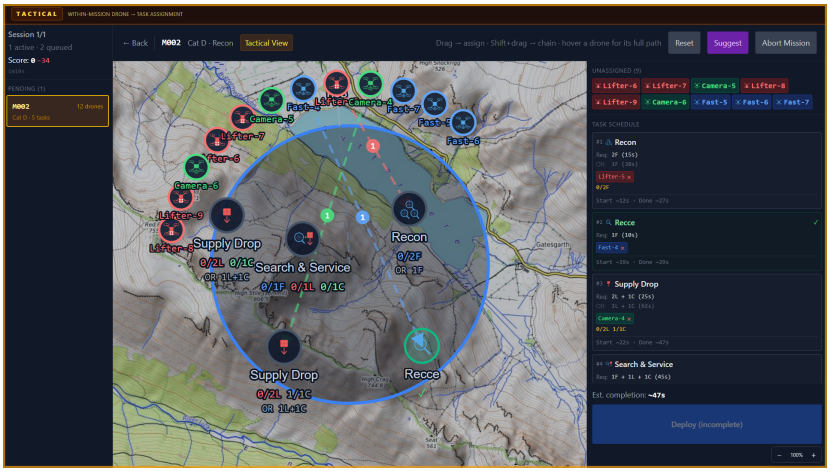
Figure 3: The strategic decision.



(a) The planner as it opens: every committed drone unassigned.



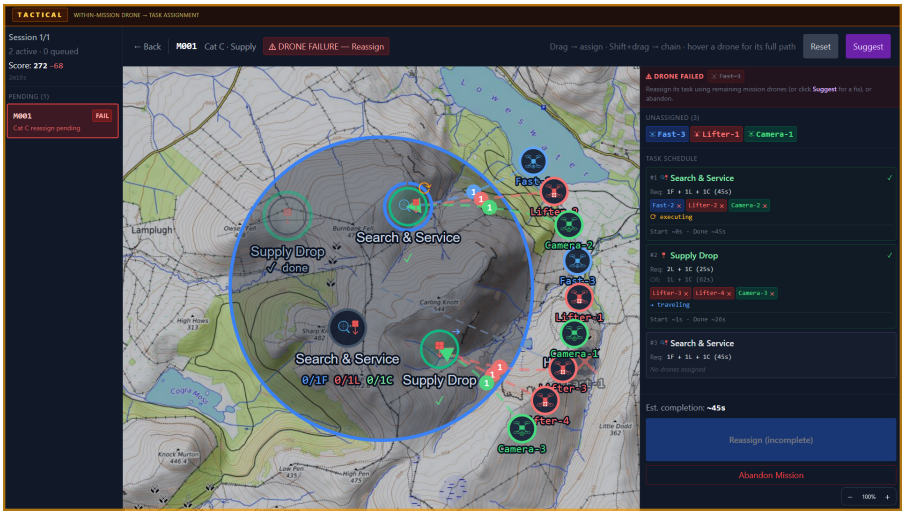
(b) Suggest filling in the plan, one drone→task entry per second; dashed lines and numbers show each drone's assigned task and order. *Deploy* stays locked until the fill completes.



(c) A plan being built by hand by dragging drones onto tasks.

Figure 4: The tactical decision in the tactical planner.

(a) Failure flagged in the queue.



(b) Recovery planner after Fast-1 failed: one Search & Service task is unstaffed; the remaining mission drones are listed as unassigned.

Figure 5: Drone failure and recovery.

16 Tactical Heavy first). The fleet, the assistants and every other parameter were the same in both.

- **Strategic Heavy** — missions arrive often (mean inter-arrival 37 s) and are mostly small (category weights A–E 40/38/16/5/1): about 13.5 missions per session, loading the cross-mission allocation decision.
- **Tactical Heavy** — missions arrive less often (75 s) and are mostly large (5/13/28/38/16): about 7.5 missions per session, loading the within-mission planning decision.

The two scenarios were tuned in simulation to comparable difficulty: a scripted near-optimal operator completes about 83% of tasks in both at the final failure rate.

Four games, not two. Every session used the same seed (42). A session's mission stream is seeded from the seed and a plan index, so there are exactly four fixed mission streams ("games"): Strategic Heavy 1 and 2 and Tactical Heavy 1 and 2. Up to P30 the plan index was the session number, so a participant's session-1 game was always the index-1 game of its scenario; from P31 (`study-v1.13`) the order was swapped, so later participants played the same games in the other position. The games differ in workload (tasks completed per session: SH1 25.1, SH2 28.9, TH1 21.5, TH2 32.8; Table 6), so every between-person comparison in the results holds the game fixed (z-scores within game, game means removed, or game as a fixed effect).

Figure 6: Four games, not two scenarios. Workload per session in each of the four fixed mission streams (Strategic Heavy 1 and 2, Tactical Heavy 1 and 2).

Build versions. The platform was updated during data collection (builds `study-v1.0` to `v1.13`); 24 of 33 participants played `v1.12` or `v1.13`. The changes that bear on the data are listed in Appendix B; the main one is that P01 and P02 played with half the later task points, which we rescale wherever a raw score is used.

6.2 Participants

Thirty-four people took part; 33 are analysed. P09 was run with assistant errors switched on (a pilot of the error manipulation) and did not follow the task, and is excluded throughout. The 33 were aged 21–44 (mean 28.7, SD 6.0); 25 men and 8 women. 28 held a postgraduate degree (23 master's, 5 doctorate) and 5 an undergraduate degree; 32 of 33 were from computing, AI, robotics or electronic engineering, and one from the armed forces. Self-reported experience was mostly moderate with simulations (20 moderate or extensive) and autonomous systems (24), and low with drones (21 none) and command or emergency management (24 none). [Recruitment route, ethics approval number, payment, and dates (sessions ran 2026-09).]

6.3 Procedure

1. **Consent and pre-study questionnaire.** Background and experience; a short comprehension check; and, from `study-v1.3`, three attitude scales: the AI Attitude Scale (AIAS-4, 10-point), a bespoke verification-propensity scale and a bespoke delegation-boundary scale (both 7-point).
2. **Tutorial** (Section 5).
3. **Session 1** (8 minutes; one scenario).
4. **Post-session questionnaires**: NASA-TLX and trust in each assistant (Section 6.4).
5. **A 30-second break screen.**
6. **Session 2** (8 minutes; the other scenario), then the same questionnaires.
7. **Debrief interview**, audio-recorded, with the researcher's written notes.

Participants were asked to think aloud while playing, and the researcher occasionally asked a short question during play; quotes that answer such a question are marked as following a leading question. Both screens, the cursor and the participant's voice were recorded throughout. [Total duration per participant; setting (lab, monitors); the debrief question list.]

6.4 Measures

Event log. Every session's first event dumps the full parameter set, and every operator action and assistant output is logged as it happens (606 strategic decisions and 630 tactical plans in all). The main behavioural measures are defined as follows.

- **Chose a strategy card** (used the Strategic Assistant): a strategic decision deployed from a strategy card as offered. A card that was edited before deploying counts as an allocation built by hand.
- **Pre-emptive**: an allocation built by hand and committed before the strategy cards had loaded.
- **Requested a tactical suggestion** (used the Tactical Assistant): the operator pressed the planner's Suggest button at least once on that tactical plan (the planner starts empty, so this is always a deliberate request). **Kept unmodified**: a suggested plan deployed with no change.
- **Handed over**: either of the two.
- **Decision time**: from opening the strategic panel (or the tactical planner) to the choice, minus the forced wait (the card reveal; one second per Suggest entry), unless stated otherwise.
- **Queue depth**: other missions that had arrived and had not yet received a strategic decision when the panel opened, rebuilt from the log.
- **Mission size**: drones needed (Section 3.4); also tasks, drone types, drones committed.

Questionnaires. After each session: **NASA-TLX** [6], six sliders from 0 to 20, reported as the unweighted mean (performance runs from 0 = perfect to 20 = failure; from `v1.12` the slider carried a note that it runs the other way, after two participants were confused by it). **Trust** in each assistant separately: six statements rated 1–7 ("The [Strategic/Tactical] Assistant is reliable", "I trust the … Assistant's recommendations", "… performs well", "I feel confident using …", "… provides useful guidance", "I would follow … suggestions without hesitation"), averaged.

[These six items are bespoke (adapted in spirit from [9], but not that scale) — describe and justify, or cite a source.] Technology-acceptance scales exist in the platform but are only given after a third session, so none were collected.

Qualitative material. Think-aloud and debrief audio was transcribed automatically (faster-whisper, word timestamps) and aligned to the event log by the on-screen session clock. Debriefs and the written interview notes were coded against a codebook of 28 codes written before coding (C01–C28; e.g. C03 "small myself, big to the assistant", C11 "hands over when overloaded", C12 "noticed over-allocation or unneeded types"). Each code is present or absent per participant, with every supporting excerpt checked to occur verbatim in the source. Quotes in the results were selected from these sources and checked verbatim. [Coding was done by a single coder, an AI model (Claude), with a second AI pass ($\kappa = .96$); the only independent check (stated preference, round 1) gave $\kappa = .27$. A human second coder is needed before the "X of 33 said" counts are reported as more than descriptive.]

6.5 Analysis

The analysis is exploratory, with a pre-specified core: ten hypotheses (H1–H10) were written and signed off before any test was run, each with one primary test, Holm-corrected together; everything else is reported as exploratory. The findings were then refined over five review passes in which every result was checked and re-specified. [Describe the analysis process honestly here: the analyses were scripted with the help of AI coding agents and every result re-run and reviewed by the researcher (register: `docs/analysis/register`). Decide how much of this belongs in the paper.] Common elements:

- **Decisions within people.** Decision-level outcomes are modelled with generalised estimating equations (GEE; logistic for yes/no, Gaussian for times on the log scale) clustered on participant, with small-sample bias-reduced standard errors. Predictors of interest are split into a *within-person* part (the decision's value minus the person's own mean) and a *between-person* part (the person's mean), so "within person" estimates compare each operator only with themselves.
- **Game held fixed**: game (one of the four) and session as fixed effects, or z-scores within game.
- **Expected and adjusted rates.** Where a raw rate could reflect who reaches a situation rather than the situation itself, a logistic model predicts each decision's chance of handing over from the mission (drones, tasks, types), the minute of the session, the game and the person (their rate on all their *other* decisions, so a decision never predicts itself). *Adjusted* = the overall rate plus the mean of (observed − expected) in a bin.
- **Intervals**: 95% intervals by resampling people (whole participants, with replacement; 400–2,000 draws), so one heavy user cannot make an interval look tight.
- **Nulls** are reported with the effect size the data could have detected.

7 Results

[v4. The results below, Section 8 and Section 9 are draft prose (`scripts/paper/prose.yaml`, built by `scripts/paper/build` into `results.tex`, `qual.tex` and `practitioner.tex`), modelled on your edit of the first finding, which is kept as you wrote it apart from typos and terms. Each finding is one block ending in a page break: a paragraph, then its figure with a statement as the caption, then the next paragraph. Quotes are woven in, one or two at a time, as you trimmed them. The blue *options* block at the end of each finding holds everything the prose leaves out: the other graph views, the highlighted quotes not used, the key numbers, the register's text for sections not used, and your pass-5 roles and notes. Hide every options block at once by changing `\includecomment{options}` to `\excludecomment{options}` in the preamble. Terms are made consistent throughout, including the graph labels: a strategy card is *chosen*, a tactical suggestion is *requested*, an allocation is *built by hand*. A † after a quote means it has not been ticked as checked against the recording. Sidelined analyses are in Appendix C.]

7.1 Overview

Table 3 summarises each scenario, and Table 6 each game. Strategic Heavy sessions had nearly twice as many missions and decisions as Tactical Heavy ones, but about the same number of tasks completed; Tactical Heavy offered more points and accrued more penalty. Operators chose a strategy card on 60% of decisions in Strategic Heavy and 73% in

Tactical Heavy, and requested a tactical suggestion on 72% and 79% of plans. Workload (NASA-TLX) did not differ between the scenarios.

Table 3: Descriptive overview by scenario: mean (SD) over the 33 sessions of each scenario. Every participant played one session of each, so p is a paired Wilcoxon signed-rank test (33 pairs; fewer where a measure is undefined, e.g. no manual decisions). Behavioural measures use the definitions of Section 6.4. [a]P01 and P02 ran at half the later task weights; their points are doubled here.

Measure (per session)	Strategic Heavy	Tactical Heavy	p
Missions arrived (non-residual)	13.5 (0.5)	7.5 (1.5)	<.001
Strategic decisions made	11.4 (2.3)	6.9 (1.5)	<.001
Tactical plans confirmed	11.2 (2.5)	6.8 (1.5)	<.001
Tasks completed	26.5 (6.0)	27.2 (9.0)	.902
Completion points[a]	1276 (316)	1627 (608)	.003
Time penalty (points)	276 (59)	395 (182)	.001
Score (points − penalty)[a]	1000 (364)	1233 (564)	.042
Strategy card chosen (% of decisions)	60 (27)	73 (28)	.004
Built by hand before cards loaded (%)	19 (23)	11 (20)	.039
Strategy card edited, then deployed (%)	6 (16)	6 (14)	.508
Tactical suggestion requested (% of plans)	72 (26)	79 (29)	.029
Suggested plan kept unmodified (% of plans)	69 (28)	75 (32)	.084
Drone failures	4.0 (2.6)	3.8 (2.6)	.636
Missions abandoned	1.2 (1.6)	0.8 (0.9)	.165
NASA-TLX, raw mean (0–20)	8.8 (3.1)	8.6 (3.5)	.607
Mental demand	11.7 (4.8)	11.0 (6.1)	.456
Temporal demand	13.6 (3.9)	12.4 (5.0)	.226
Performance (0 = perfect)	7.5 (4.9)	8.8 (4.9)	.152
Effort	9.6 (4.7)	8.9 (4.7)	.352
Frustration	6.0 (5.0)	6.2 (5.8)	.467
Trust, Strategic Assistant (1–7)	4.38 (1.52)	4.84 (1.33)	.098
Trust, Tactical Assistant (1–7)	5.52 (0.88)	5.36 (0.93)	.537

7.2 Each operator has a personal style

Operators differ a great deal in how they use each assistant, and the differences appear personal. We describe each session by ten habits (how often a strategy card is chosen, how often a tactical suggestion is requested, how long a decision takes, etc). From those habits alone, a participant's second session resembles their own first (median profile $r = .48$) and essentially nobody else's ($r = -.01$): their own session is in the closest three out of the 33 people far more often than chance (41% vs 9%), even though the second session is a different scenario. Who the operator is explains approximately six times more of strategy card choice than the mission does. The styles do not fall into types, and nothing measured beforehand (attitude to AI, views on human authority, experience) predicts them.

To compare styles we describe every 8-minute session by ten habits. Figure 7 shows each habit's spread over all 66 sessions. Most habits vary enormously between sessions (strategy card choice runs from 0% to 100% of a session's decisions); two are nearly universal (most sessions keep every tactical suggestion unchanged, and most never edit a strategy card). The right-hand column shows the correlation (Spearman's rank) between each habit's session 1 and session 2; i.e. how much a person's habit in one run carries over to the next.

Figure 7: The habits describing each user and how they vary. These ten habits accurately describe a user and many carry over to that user's following scenario.

When asked to describe their approach, 15 of 33 named a consistent strategy of their own. In Figure 8, each person's ten-habit profile from session 1 is compared with all 33 session-2 profiles, and the 33 are ranked from most to least alike (and the same the other way round, session 2 among the session 1s) by Euclidean distance over the 10 dimensions. If habits were not personal, a person's own other session would be equally likely to land anywhere in that ranking (dashed line). Instead it is the single closest match 12 times out of 66 (2 expected by chance) and in the closest three 27 times (6 expected), even though the second session is a different scenario ($p < .001$).

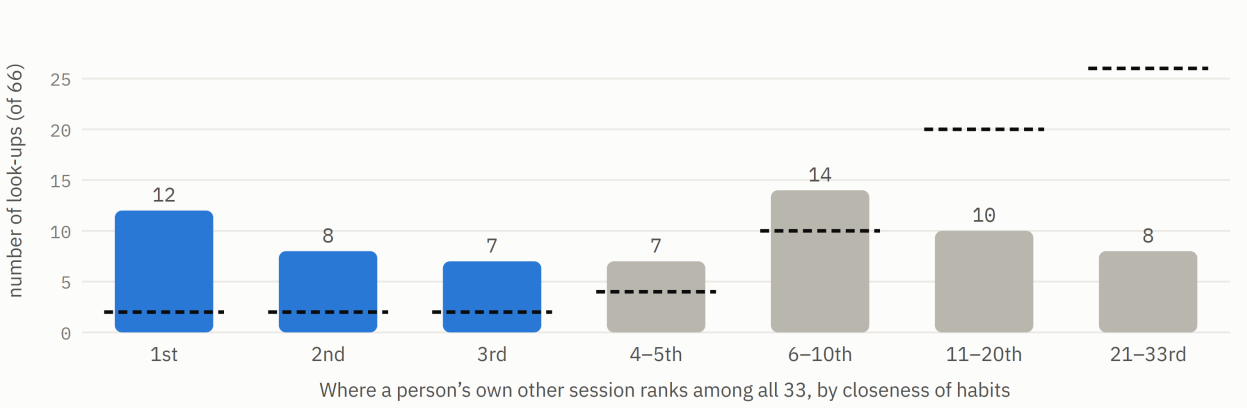


Figure 8: A person's own other session is usually among their closest matches. Where each person's own session ranks among all 33 by ten-habit profile; dashed line: where it would fall if habits were not personal.

When inspecting every instance of a user choosing or not choosing a strategy card, knowing *who* is deciding predicts whether a card is chosen about six times better than knowing everything about the mission and when it arrived. We built three logistic models with access to different priors about the decision. Knowing only *who* made the choice, the model classified the right one 75% of the time (a coin: 50%; knowing only the mission: 63%). A person's session-1 rate on its own still does it 71% of the time for their session-2 decisions. Figure 9 shows this as a set of R^2 values.

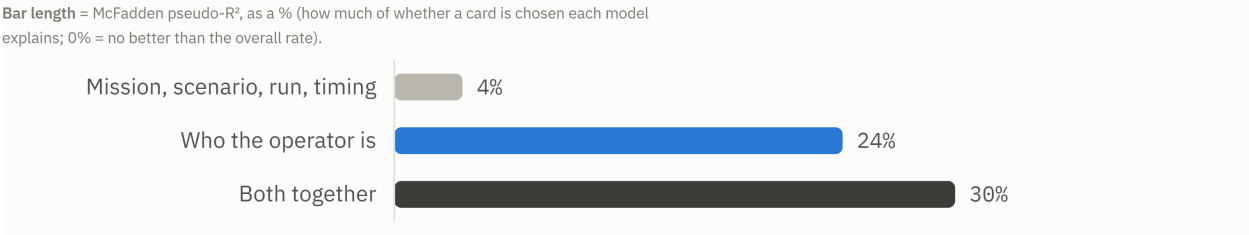


Figure 9: Who decides explains far more of strategy card choice than what is being decided. McFadden pseudo-R^2 of models predicting whether a strategy card is chosen from the mission and its timing, from the person, and from both.

People often described their use of the assistants as part of who they are, and in a highly varied sense: 'naturally conservative' (P32), 'I'm more conservative' (P26), 'I always let the agent do it first' (P13). At the extremes, P14 used the assistants 'for almost every allocation' and P16 'manually overrode most of the time'. P32 calls themself 'naturally conservative' and also says their use was 'very much about workload', so a person's style and the situation are both in play.

Participants' strategic and tactical reliance correlated only slightly. Figure 10 shows each participant's proportional use of each assistant. The correlation between use of the two assistants is only $\rho = 0.37$.

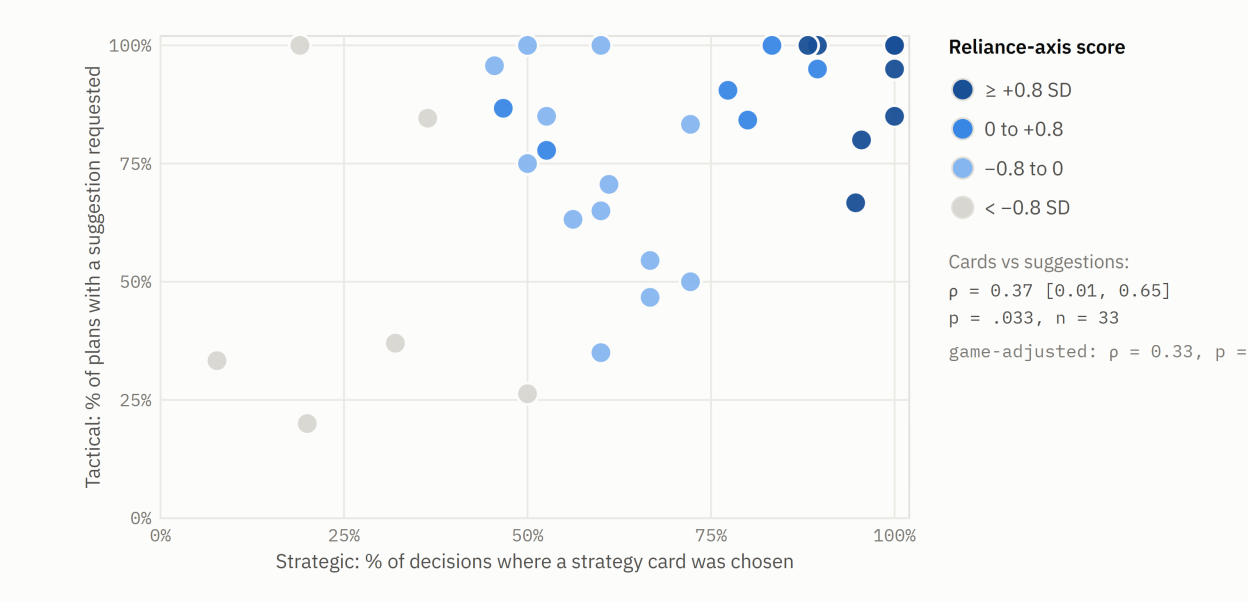


Figure 10: Use of one assistant says little about use of the other. Each dot is a person: the share of their strategic decisions where a strategy card was chosen, against the share of their tactical plans where a suggestion was requested.

The styles are individual rather than types. Placed on their two largest habit dimensions, the 33 operators form one continuous cloud that looks like a set of simulated operators with no types by construction (Figure 11). However many groups we ask for (two, three or four), the real people separate no better than the simulations do (silhouette .17–.20 against .19–.20), and groups found in one session barely come back in the other (adjusted Rand index .08–.09, against .13 for a smooth cloud in which people keep their place).

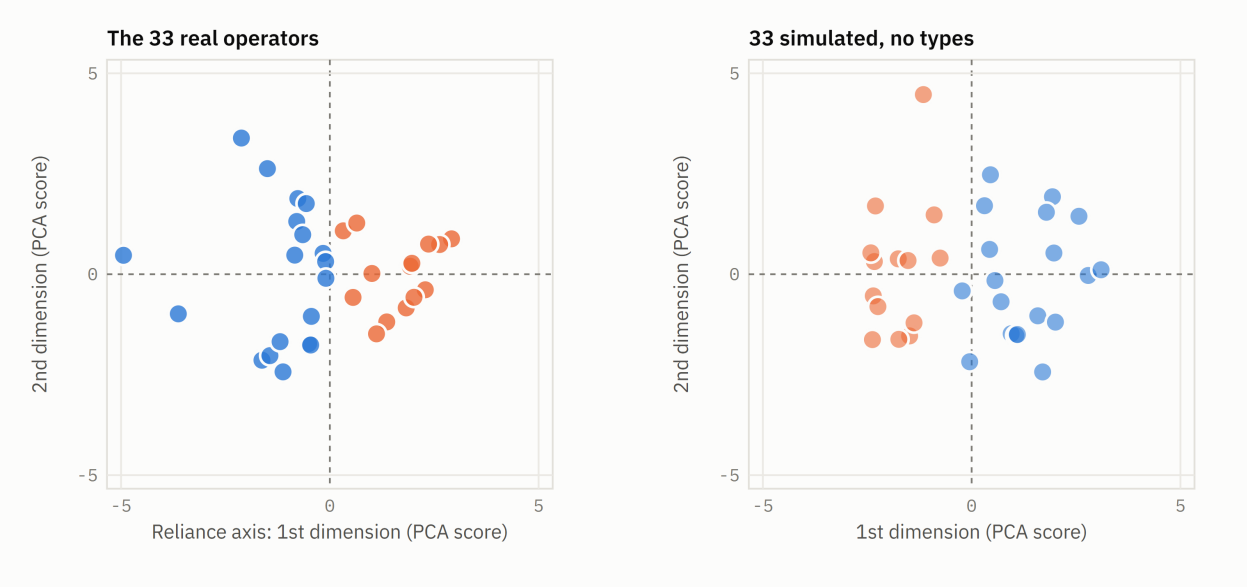


Figure 11: No types: the operators form one continuous cloud. Each person placed on the first two principal components of the ten habits.

Nothing measured before the study predicts a person's style. Of 40 tests of attitude to AI (AIAS-4), verification propensity, views on human authority and prior experience against five behaviours, three reach $p < .05$ and none survives Holm correction; chance alone gives two such hits on average, and three or more in a third of shuffles (Figure 12). This is an absence of evidence rather than evidence of no link: with 33 people only correlations of $|\rho| \geq .48$ could be detected, and the sample is uniformly positive about AI (AIAS-4 mean 8.0 of 10). In their own words, people's general views on AI did not line up with what they did, and most explained their use by time pressure and job size rather than by a view about AI.

Each cell: Spearman ρ across people, questionnaire score (row) × behaviour (column, mean of both sessions)
Colour: blue = positive ρ, amber = negative; stronger colour = larger |ρ|

	Chose a strategy card	Requested a tactical plan	Kept the tactical plan unmodified	Strategic decision time	Session score
AIAS-4 attitude to AI	-.15	.01	.05	.16	.07
Verification propensity	.02	-.21	-.33	.06	**-.41**
Delegation boundary	-.21	**-.38**	**-.40**	.04	-.23
"Trust AI more for suggesting than decidi…	-.16	-.11	-.18	.05	-.09
"A human should make the final decision"	-.08	.06	.06	.13	.19
"Hard to tell when AI has erred"	-.10	.12	.17	-.07	.21
Prior experience (mean of five items)	.19	-.09	-.11	-.31	-.12
Experience with automation / AI	.21	.07	.02	-.19	-.08

Outlined = p < .05 before correction (3 of 40); after Holm: 0. Detectable at 80% power: |ρ| ≥ 0.48.

Figure 12: Nothing measured beforehand predicts style. Correlations between the pre-study scales and five behaviours; none survives correction for the 40 tests.

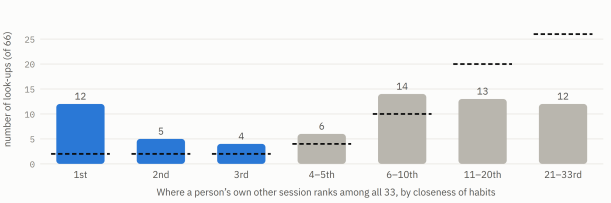
Options for this finding

The habits we grade them on [pass 5: numbers only]

Key numbers:

- Strategy card chosen as offered: median 67% of a session's strategic decisions (middle half 47–91%, range 0–100%)
- suggestion requested: median 86% of tactical plans (55–100%); suggested plan kept unchanged: median 100% (middle half 100–100%)
- Decision time beyond the card wait: median 6.5 s (5.4–10.3 s, range 1.6–43 s)
- Card edited before committing: median 0% (most sessions never); going manual before the cards appear: median 50% of hand-made decisions (0–75%)
- Drones vs the Conservative card: single-type −0.5 (−1.25 to 0); multi-type +0.6 (−0.1 to +1.3)

Where each person's own session ranks [pass 5: for the paper]

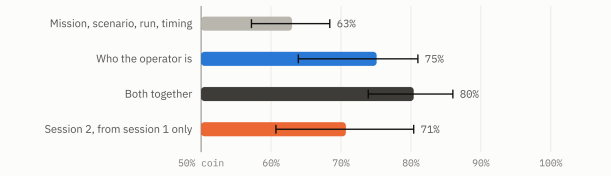


Other view: With the reliance axis removed.

Key numbers:

- Own session ranked 1st: 12 of 66 look-ups vs 2 expected; in the closest three: 27 vs 6 (41% vs 9%); permutation $p < .001$
- Average rank of own session 8.2 of 33 (chance 17.0); with the reliance axis removed 10.1 ($p < .001$)
- Profile correlation with own other session: median r = .48; with other people's: median r = −.01; own closer for 29 of 33

Who decides matters more than what is being decided [pass 5: for the paper]



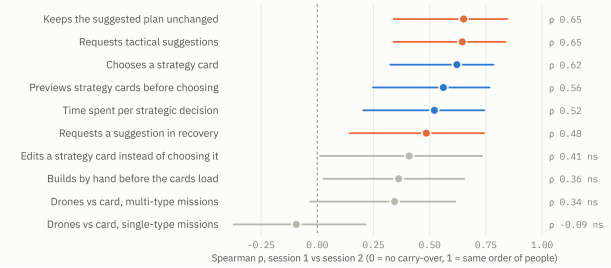
Other view: The same comparison as a ranking test (AUC); whiskers are 95% intervals (people resampled).

Key numbers:

- Pseudo-R^2: mission + timing 4%; person 24%; both 30%
- Strategy card choice per person 8%–100%, median 61%; 6 people $\geq$ 90%, 3 people $\leq$ 25%
- Telling a taken from a not-taken decision (AUC): mission and timing 63% [57%, 68%]; the person, from their other decisions, 75% [64%, 81%]; both 80% [74%, 86%]; session-2 decisions from the session-1 rate alone 71% [61%, 80%] (321 decisions).
- Right-or-wrong guessing: always 'card' 66%; mission 65%; person 70%; both 77%.
- Person-only pseudo-R^2 with the person labels shuffled: 4% on average (95th percentile 6%), against 24% observed.

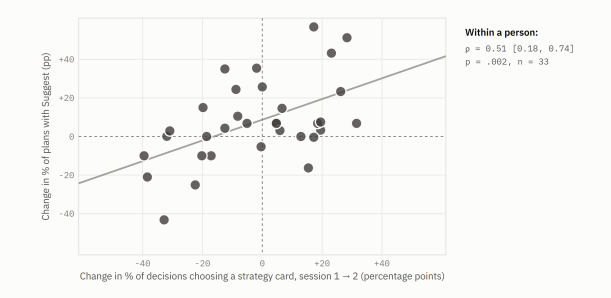
How far each habit holds from one session to the next [pass 5: for the paper; not used in the prose above]

One line per habit: how strongly a person's place on that habit in session 1 predicts their place in session 2 (ρ; 0 = no carry-over, 1 = same order of people). Six of the ten carry over clearly and survive correction for testing ten (coloured). Editing cards, going manual before the cards load and topping up drones on multi-type missions carry over weakly, and topping up drones on single-type missions does not carry over at all (ρ −.09). "Keeps the suggested plan" is high partly because 24 of 33 people kept every suggested plan in both sessions.



main

Strategic and tactical reliance go together only loosely between people [pass 5: for the paper]

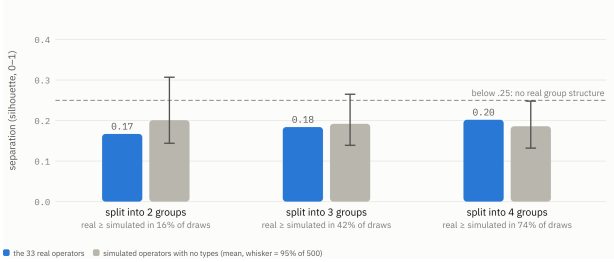


Other view: Within a person: change from session 1 to session 2.

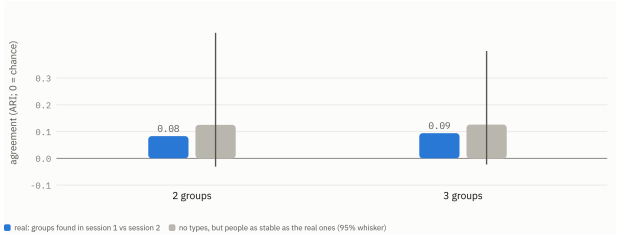
Key numbers:

- Between people, strategy card choice vs tactical suggestion use: $\rho = .37$ [.01, .65], p = .033 (raw); $\rho = .33$ [−.03, .62], p = .058 (game-adjusted)
- Within a person, change in strategy card choice vs change in tactical suggestion use, session 1 → 2: $\rho = .51$ [.18, .74], p = .002
- Strategy card choice vs Suggest in recovery: $\rho = .42$, p = .014; vs keeping the suggested plan unchanged: $\rho = .20$, p = .26
- Card taking and previewing: $\rho = .79$ (largely mechanical: choosing a card means clicking it)

Individual, not types [pass 5: numbers only]



Other view: Cluster separation (silhouette) against no-types simulations.



Other view: Recurrence of groupings across sessions (adjusted Rand index).

Key numbers:

- Separation (silhouette), real vs no-types simulations: 2 groups .17 vs .20 [.14, .31]; 3 groups .18 vs .19 [.14, .27]; 4 groups .20 vs .19 [.13, .25]; a no-types sample separates at least as well 84%, 58% and 26% of the time
- Groups in session 1 vs session 2 (ARI): 2 groups .08, 3 groups .09; a stable no-types cloud gives .13 on average (95% of simulations −.03 to .47 for 2 groups, −.02 to .40 for 3), at least as high as the real value about 55% of the time; shuffled labels give up to .12–.13
- Round 1: two groups vs one, ΔBIC 1.9 (needed > 10; 42% of no-type simulations did at least as well)
- Archetypes (extras2): three explain 46% vs 43–59% in the no-types continuum (p = .83); same archetype both sessions 61% vs 30–79% (p = .32)

Nothing measured beforehand predicts the style [pass 5: for the paper]

Key numbers:

- 0 of 40 tests survive Holm; 3 nominal (verification × score ρ −.41; delegation × Suggest consulted −.38 and × plan kept −.40)

- Chance alone gives 2.0 nominal hits on average; 3 or more in 34% of 2000 shuffles of who-goes-with-whom; a p as small as the smallest one (.021) turns up somewhere in the grid in 51% of shuffles
- AIAS-4 $\times$ strategy card choice ρ $-.15$; “A human should make the final decision” $\times$ strategy card choice ρ $-.08$
- AIAS-4 odds ratio for card taking 0.80 per SD [0.50, 1.27]; detectable only at about $\times 2$
- AIAS-4 mean 8.0 / 10 (SD 1.2); 32 of 33 from computing / engineering; detectable $|\rho| \geq .48$

7.3 Small jobs by hand, big jobs to an assistant

The bigger the job, the more likely an operator was to hand it to an assistant, on both tiers. Measured the same way on both (the drones the mission needs), a strategy card was chosen on 56% of the smallest missions and 75% of the largest, and a tactical suggestion was requested on 54% and 91%. Comparing each person only with themselves, every doubling of mission size multiplied the odds of handing over by about 1.6 on the strategic tier ($\times 1.62$ [1.23, 2.15]) and 2.9 on the tactical tier ($\times 2.91$ [2.03, 4.17], both $p < .001$), and 22 of 33 people handed over more on their biggest missions than on their smallest on each tier. The rule is a sensible one: a strategy card is much quicker than building by hand on three-type missions and no quicker on one-type missions, and a tactical suggestion is not slower than a hand-built plan.

Figure 13 shows the share of decisions handed over, on each tier, by the number of drones the mission needs. Both rise; the tactical tier rises more steeply and keeps rising. The strategic tier dips at five drones: these are seven particular one-type missions in the two Strategic Heavy games, six of which arrived in bursts, so operators reached them with three or four other missions waiting and often decided before the cards had loaded. Participants described the rule in their own words, for example: "I think if it's just one colour, I'm gonna do it, like, assign it myself, but if it's multiple colours, I think I'm gonna let, um, the AI choose for me, for the, the strategic side" (P29), and "When they only needed a single colour of drone I tended to do it manually" (P34).

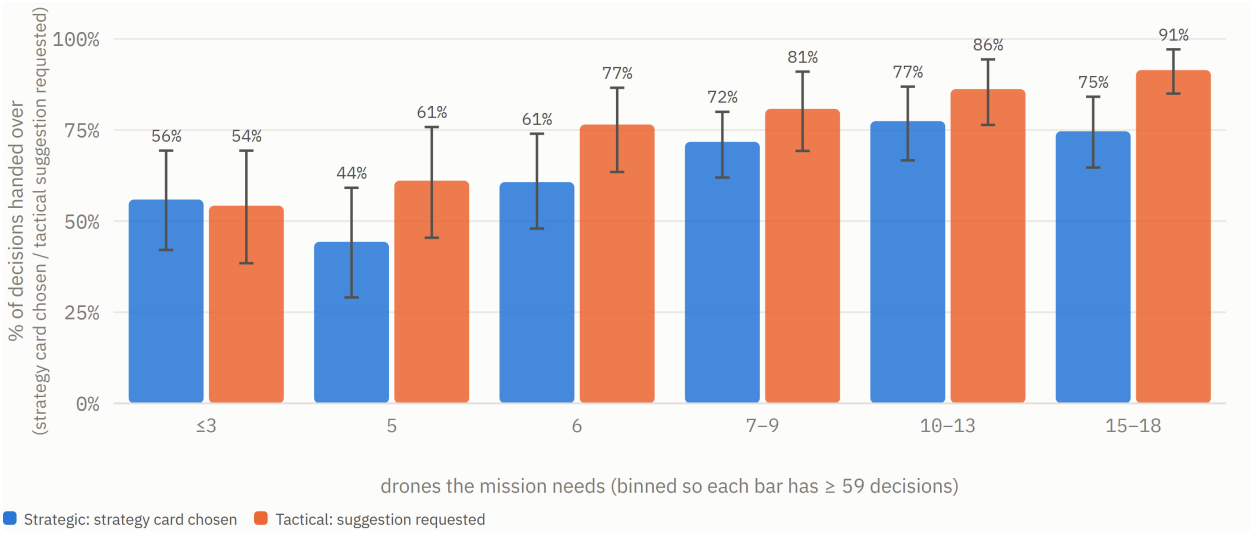


Figure 13: The bigger the mission, the more it is handed to an assistant, on both tiers. Share of decisions where a strategy card was chosen (blue) or a tactical suggestion requested (orange), by the drones the mission needs; whiskers are 95% intervals.

The effect holds when each person is compared only with themselves, for every way of measuring size and on both tiers (Figure 14). Dropping every illogical Conservative card leaves it in place (strategic $\times 1.46$ per doubling, $p = .006$), as does allowing for every operator having played the same fixed missions (strategic $\times 1.60$, tactical $\times 2.89$, both $p \leq .001$). With each mission as a single point, the share handed over still rises with its size ($\rho = .63$ on the strategic tier and .69 on the tactical, 41 missions each).

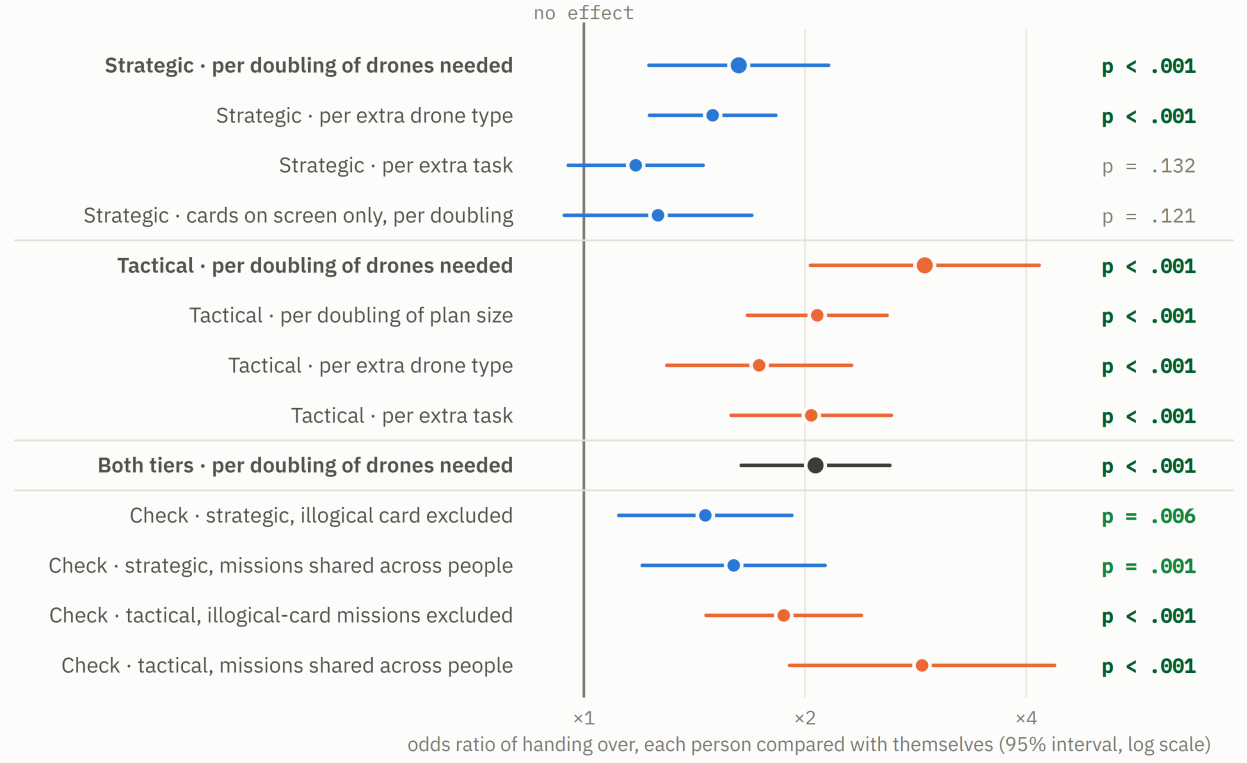


Figure 14: Bigger jobs are handed over more, for every size measure and on both tiers. Each row is a within-person odds ratio per unit of size with its 95% interval; the bottom block holds the checks for artefacts.

The two tiers are also linked. A mission whose drones came from a strategy card was more likely to get a tactical suggestion than one allocated by hand, at every size (Figure 15): for the same person and mission size, about twice the odds ($\times 2.10$ [1.30, 3.40], $p = .003$). The gap is wide on small missions (70% against 35% at three drones or fewer) and narrow on the largest (93% against 88%), but that narrowing is both rates approaching 100% rather than a reliable change. It is a general tendency rather than the habit of a few people: of the 26 who did both often enough to compare, 17 leaned this way and 8 the other, although the three largest gaps make up about half of the total. Participants described the same continuity: "Now I'm using the suggestion because I took the aggressive approach which means I know that I have enough drones" (P10). Size also mattered in plain practical terms: "If I look at a mission with 15 drones and I need to move two or three chains, that's just going to take me about a minute, and it's just no point" (P11).

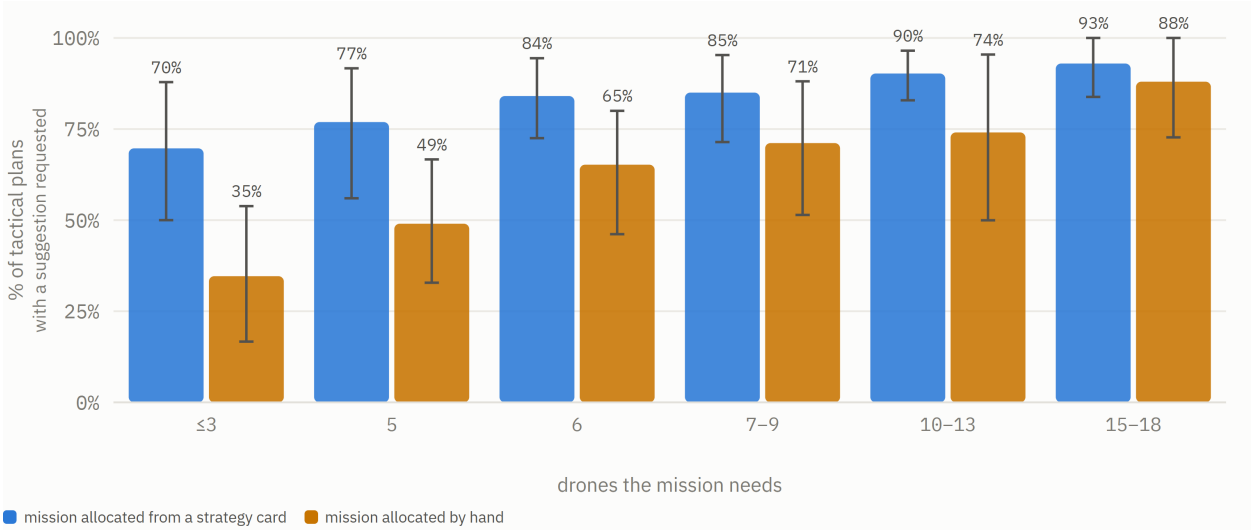
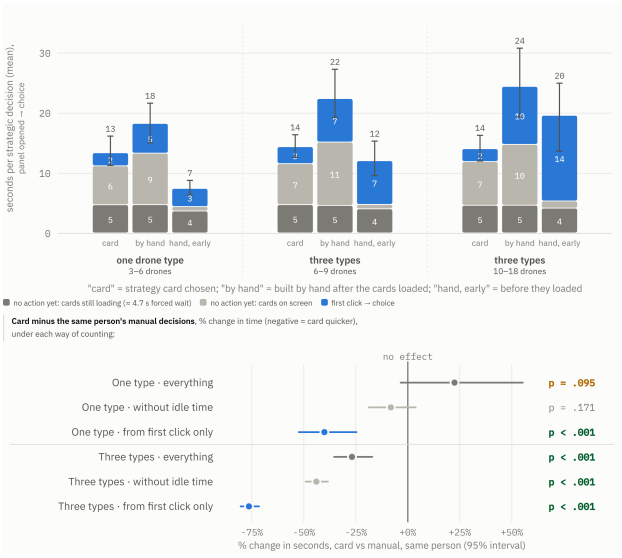
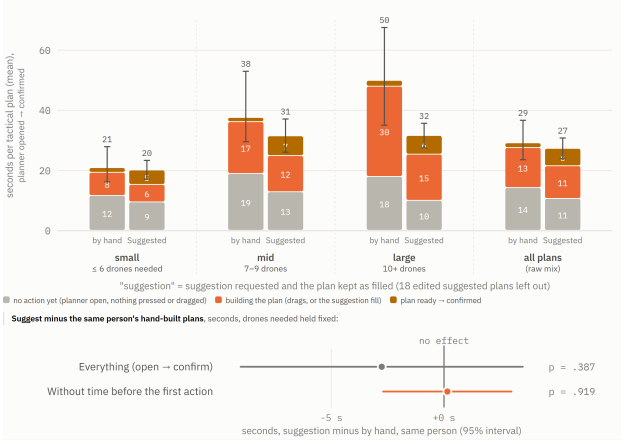


Figure 15: A mission allocated from a strategy card is more likely to get a tactical suggestion. Share of tactical plans with a suggestion requested, by the drones the mission needs, split by how the mission was allocated.

The size rule also fits the time each route takes (Figure 16). On the strategic tier, choosing a strategy card is much quicker than building by hand on three-type missions (27% less time for the same person, $p < .001$) and no quicker on one-type missions (+22%, $p = .09$); with the idle time before the first click removed, the card is 44% quicker on three-type missions and level on one-type ones. On the tactical tier, hand-built plans and tactical suggestions take about the same planner time (29 against 27 s on average), within a second of each other for the same person and mission size. Speed was the most common reason participants gave for using the assistants: "The top advantage in this situation was the speed of response" (P31).



(a) Strategic tier, all time counted.



(b) Tactical tier, all time counted.

Figure 16: A strategy card saves time only on big strategic jobs; a tactical suggestion saves no planner time. Each decision's time split into the time before the operator's first action and after it, by mission size: (a) strategic tier, (b) tactical tier.

Where a tactical suggestion does gain is in carrying the plan out (Figure 17). Splitting each tactical plan into four phases, the operator's own time is the same either way: a suggestion saves about 3 s in building the plan and gives

it back between the plan being ready and being confirmed. The only reliable difference is that a suggested plan is carried out about 11 s sooner ($p = .006$), roughly a tenth of a mission. The common feeling that the assistant is faster is therefore only weakly supported: it does not save the operator time, and its plans finish a little sooner. Nor does relying on the assistants more make an operator do better: within a person, missions allocated from a strategy card were completed in full no more often than their own hand-built ones (60% against 62%, $p = .27$).

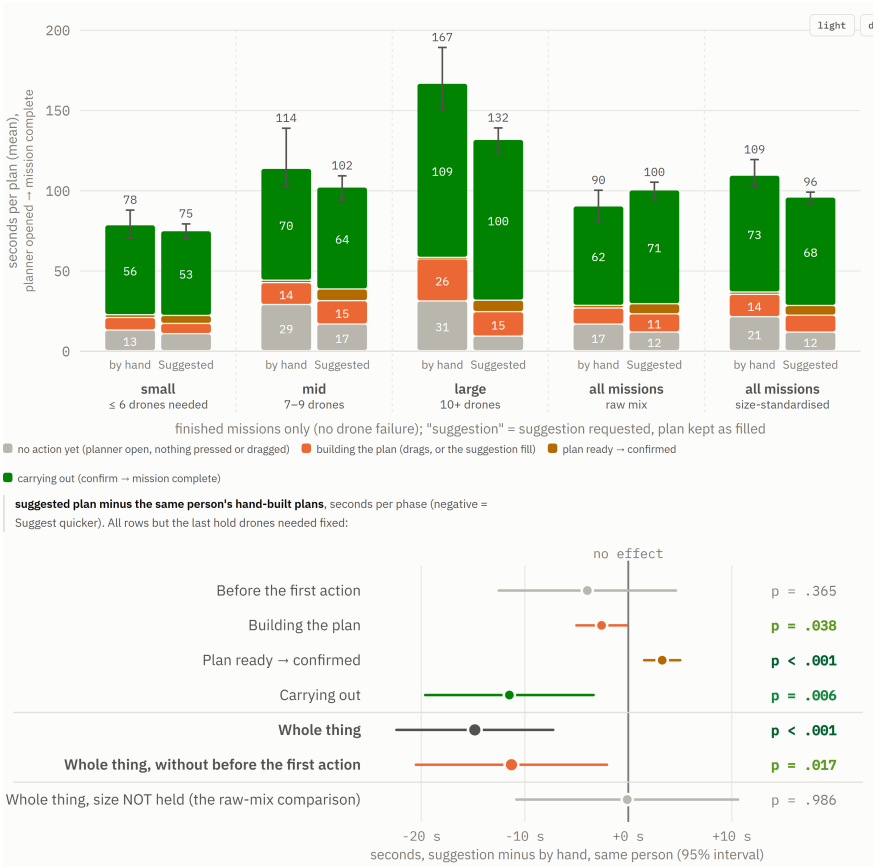


Figure 17: Hand-built and suggested tactical plans take the operator about the same time; suggested plans are carried out a little sooner. Mean time per tactical plan in four phases, by mission size.

Ten of 33 participants stated the size rule without being asked, and 15 valued the assistants for speed rather than accuracy. Several also described using the wait: while a suggestion filled in, they turned to the next mission. The logs agree: 45% of tactical plans with a suggestion requested had another mission's strategic panel opened while the planner was open, against 19% of hand-built plans.

Options for this finding

Both tiers on one axis: the bigger the mission, the more it is handed over [pass 5: for the paper]

Key numbers:

- Strategic, per doubling of drones needed: odds ×1.62 [1.23, 2.15], p < .001 (606 decisions, 33 people)
- Tactical, per doubling of drones needed: odds ×2.91 [2.03, 4.17], p < .001 (627 plans)
- Both tiers, one shared slope: ×2.07 [1.64, 2.61], p < .001 (1,233 decisions); the tactical slope is steeper than the strategic one by ×1.38 [0.99, 1.91], p = .056 — suggestive, not shown
- Raw rates, smallest to largest bin: strategic 56% → 75%, tactical 54% → 91%

- Arrival bursts (Strategic Heavy games, fixed per game): the seven 5-drone missions had on average 4.4 other missions arrive within 60 s of them, against 3.6 for the 3- and 6-drone one-type missions and 3.1 for every other mission; 6 of 7 arrived inside or straight after a burst; the seventh arrived 10 s into a game and got the card 69% of the time

Highlighted quotes not used above:

> *"either it's a really big mission, and it's like, oh, I can trust it with a large mission. [. . .] Or it's a simple mission, and I'll be like, oh, I can just bang it up myself." — P11 (debrief 0:28:41)*

> *"I did feel like some of them I should do myself and I trusted it more with the complicated tasks" — P29 (debrief 0:24:54)*

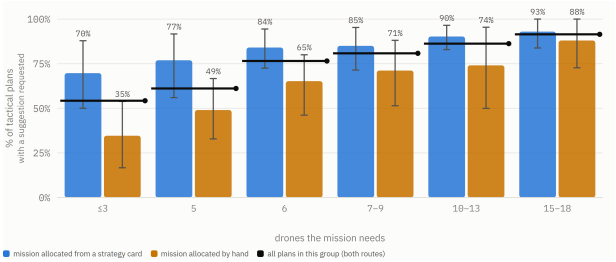
> *"I'll use the suggest, because there's a lot of drones here, so because there's a lot of drones to allocate" — P05 (think-aloud s1 0:09:36)*

Person-adjusted: every size measure, both tiers, one picture [pass 5: for the paper]

Key numbers:

- Strategic, illogical Conservative card excluded: $\times 1.46$ [1.12, 1.92], p = .006 (474 decisions)
- Missions shared across people (interval clustered on person and mission): strategic $\times 1.60$ [1.20, 2.13], p = .001; tactical $\times 2.89$ [1.91, 4.37], p < .001 (65 missions)
- Each mission as one point (missions with ≥ 5 decisions): strategic $\rho = .63$ over 41 missions, tactical $\rho = .69$ over 41 (both p < .001)

Tactical tier: Suggest rises with mission size, and goes with the strategic choice [pass 5: for the paper]



Other view: With a marker on each pair at the rate for all plans in that size group.

Key numbers:

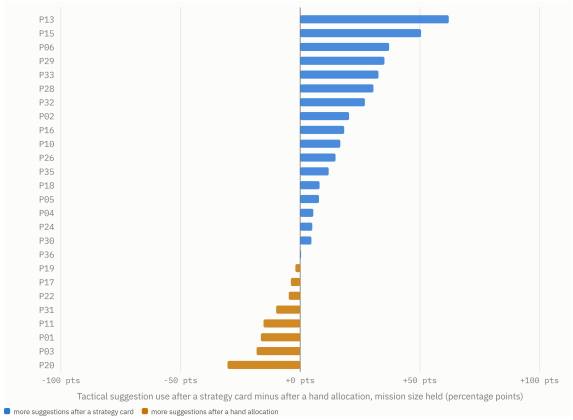
- Within person, per doubling of drones needed: odds $\times 2.91$ [2.03, 4.17]; per extra task $\times 2.04$ [1.59, 2.62]; per extra drone type $\times 1.73$ [1.30, 2.31]; per doubling of drones committed $\times 2.51$ [1.83, 3.45]; per doubling of plan size $\times 2.08$ [1.67, 2.59] (all p < .001)
- Allocated from a card vs by hand, same person, size held: $\times 2.10$ [1.30, 3.40], p = .003 (drones needed); $\times 1.78$ to $\times 2.12$ on the other four measures, all p < .05
- Aggressive card vs Conservative card, same person, drones needed held: $\times 0.88$ [0.45, 1.75], p = .72 (400 plans); raw Suggest rate after Aggressive 81% (200), after Conservative 89% (203), after a hand allocation 63% (227)
- Does the gap shrink with size? Route $\times$ size: $\times 0.76$ [0.30, 1.92], p = .56 (drones needed); p = .28 to .98 on the other measures: no reliable narrowing
- suggestion requested, card-allocated vs hand-allocated: ≤ 3 drones 70% vs 35%; 7–9 drones 85% vs 71%; 15–18 drones 93% vs 88%
- Same mission, card vs hand: $\times 1.11$ more drones committed and $\times 1.12$ larger plan (both p < .001); drones needed, tasks and types identical by construction
- Person level: people who choose strategy cards more often also request a tactical suggestion more (Spearman $\rho = .37$, p = .033, 33 people)
- P10 in the log: Suggest on 8 of 9 Aggressive-allocated missions (never took the Conservative card) vs 6 of 9 hand-allocated

Highlighted quotes not used above:

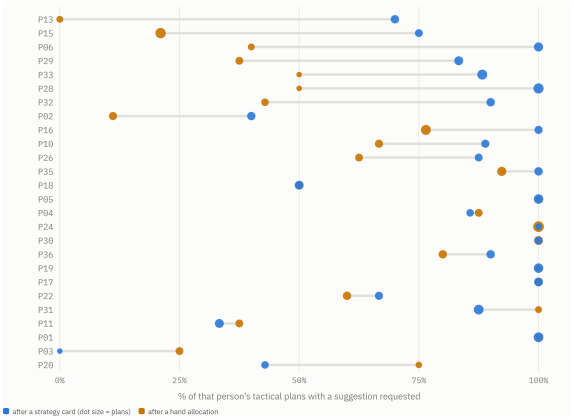
> *"Better to use tac agent when using the aggressive strat agent" — P10 (written interview) (paraphrase)*

> *"Dragging 14 drones is going to take way more time than hitting the assign and waiting for eight seconds." — P10 (think-aloud s2 0:13:43)*

Card then Suggest, person by person: a broad tilt, but a few people carry much of it [pass 5: numbers only]



Other view: Each person's gap, mission size held.

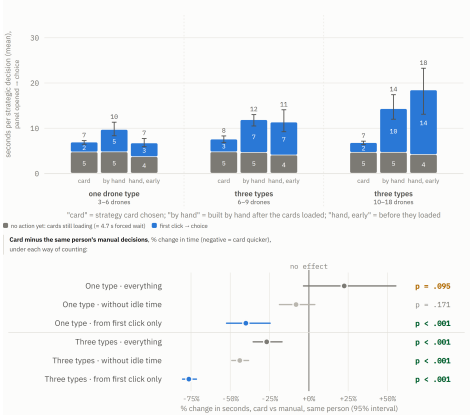


Other view: Each person's two raw rates.

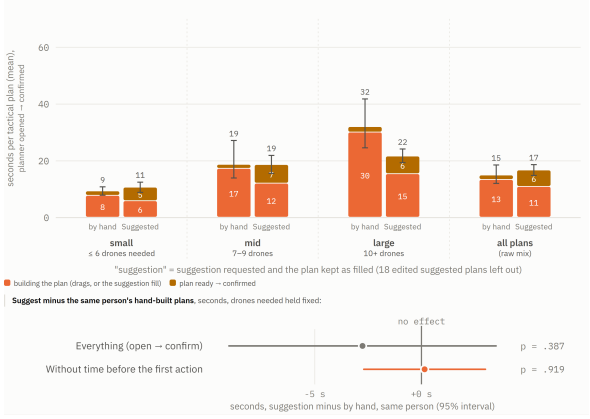
Key numbers:

- People with ≥ 2 plans each way: 26. Gap (size held) positive 17, negative 8, none 1; sign test p = .11; mean gap +11 points.
- Card vs hand odds ratio, everyone $\times 2.10$; leaving out any one person $\times 1.90$ (without P15) to $\times 2.40$ (without P11), always $p < .05$.
- Without the largest gaps: top 1 (P13) $\times 1.97$, p = .006; top 3 (P06, P13, P15) $\times 1.63$ [1.05, 2.52], p = .03; top 5 $\times 1.46$ [0.96, 2.22], p = .08; top 8 $\times 1.23$, p = .34.
- The three largest gaps are 52% of the summed gap. Not comparable: 6 people almost only planned after a card, 1 almost only after a hand allocation.
- Also holding fixed whether the mission needs one drone type: $\times 2.13$ [1.30, 3.49]; and the number of tasks too: $\times 2.06$ [1.24, 3.43], p = .005.

Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower [pass 5: extra: may make it in]



Other view: Strategic tier, idle time removed.



Other view: Tactical tier, idle time removed.

Key numbers:

- Strategic, card vs manual, same person, % change in time: everything: one-type +22% [−3, +55], p = .09; three-type −27% [−35, −17], p < .001
- Without idle time: one-type −8% [−19, +4], p = .17; three-type −44% [−49, −38], p < .001. From the first click only: one-type −40%, three-type −76% (both p < .001)
- Strategic means (s), loading + idle + acting: one-type card 4.7 + 6.5 + 2.2 = 13.3; by hand after cards 4.7 + 8.6 + 5.0 = 18.2; before cards 3.7 + 0.7 + 3.0 = 7.4. Three-type 10–18 drones: card 4.6 + 7.3 + 2.1 = 14.0; by hand 4.6 + 10.1 + 9.7 = 24.4
- Tactical means (s), idle + building + ready → confirmed: all plans by hand 14.2 + 13.3 + 1.6 = 29.1 (144) vs Suggest 10.6 + 10.9 + 5.8 = 27.2 (465); large missions 17.9 + 30.0 + 1.9 = 49.8 (22) vs 10.0 + 15.3 + 6.2 = 31.5 (159)
- Tactical, Suggest minus by hand, same person, drones needed held: everything −2.8 s [−9.0, +3.5], p = .39; without idle time +0.2 s [−2.7, +3.0], p = .92; Suggest × size (without idle) −1.3 s per doubling, p = .47
- Tactical plans idle for more than 10 s: 54% had another mission's strategic panel opened before the first action (176 plans); all plans 22%

Highlighted quotes not used above:

> *"I think the key strength when I [am] using the agent is it can help me to save the time and energy to those specific deployments" — P28 (debrief 0:32:10)*

> *"the point is, it takes me so much time to untangle that, that maybe it probably will not be faster if I do it myself manually. So as long as the AI makes sure that everything is covered once." — P17 (think-aloud s1 0:07:07)*

> *"Manually, [I] can't wait for the processing, to be honest." — P21 (think-aloud s1 0:09:32)*

> *"I think even if the tactical agent gave me a dodgy suggestion and the mission failed and I lost points, I probably would have gained more points from the time saved..." — P11 (debrief 0:30:11)*

> *"I think I was definitely a bit slower than the, than the algorithm there." — P22 (think-aloud s1 0:07:11)*

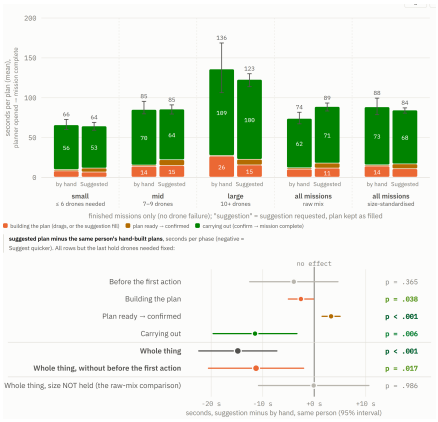
> *"You know, it's quicker to get a suggestion, but it is less clicking, and it seems to do what I would do." — P32 (think-aloud s1 0:10:07)*

> *"So, it just seems quicker to just send some myself than wait for the recommendation. I might just be impatient." — P32 (think-aloud s1 0:03:09)*

> *"So, when we are trying to do with the tactical screen, I find the agent is quite slow." — P27 (debrief 0:29:10)*

> *"Yeah, when I located, I think sometimes the AI will take a little longer. And I just mess up my strategy" — P26 (debrief 0:28:37)*

Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out [pass 5: for the paper]



Other view: Without the time before the first action.

Key numbers:

- Suggest minus hand-built, same person, drones needed held (315 finished plans): before the first action −4.0 s [−12.5, +4.6], p = .37; building −2.6 s [−5.0, −0.1], p = .038; ready → confirmed +3.3 s [+1.5, +5.0], p < .001; carrying out −11.5 s [−19.6, −3.4], p = .006; whole −14.8 s [−22.4, −7.3], p < .001
- Whole thing without the time before the first action: −11.3 s [−20.5, −2.1], p = .017; whole thing with size not held: −0.1 s [−10.8, +10.6], p = .99
- Means (s), hand vs kept Suggest: small 78.5 vs 74.8 (63 vs 97 plans); mid 113.6 vs 102.0 (12 vs 59); large 166.6 vs 131.6 (6 vs 75); all, raw 90.2 vs 100.2; all, size-standardised 109.3 [102.7, 119.4] vs 95.7 [91.9, 99.1]
- Size mix of finished plans: hand-built 78% small, 15% mid, 7% large (mean 5.7 drones needed); Suggest 42%, 26%, 32% (mean 8.8)
- All 627 plans, first three phases: before −2.9 s, p = .29; building −3.2 s [−6.2, −0.1], p = .04; ready → confirmed +3.3 s [+2.0, +4.6], p < .001
- Carrying out, including missions with a drone failure (378): −8.1 s [−17.1, +0.9], p = .08: same direction, less certain
- Projected completion vs the assistant's own plan at open, median: hand-built +10.3 s [3.7, 28.3] (144 plans), kept Suggest +1.6 s [0.8, 4.0] (468), edited Suggest +38.7 s (18)
- Interleaving, round-1 specification re-run: odds ×4.83 [2.24, 10.40], p < .001 (630 plans, 33 people); raw 45% vs 19%; 28% vs 11% actually allocated another mission

- Operator time, building + ready → confirmed, Suggest minus hand-built, same person, size held: -2.6 s + 3.3 s $\approx$ +0.7 s (negligible)
- From the retired "who does better" finding: within a person, card missions vs own manual missions, completed in full odds $\times 0.81$ [0.55, 1.18], p = .27 (raw 60% vs 62%); session score per +25 points of strategy card choice -0.10 SD [-0.47, 0.27], p = .61. Between people, heavier card users complete more ($\times 1.32$ per 25 points, p = .004): the difference is in the people, not the advice

What operators say [pass 5: for the paper]

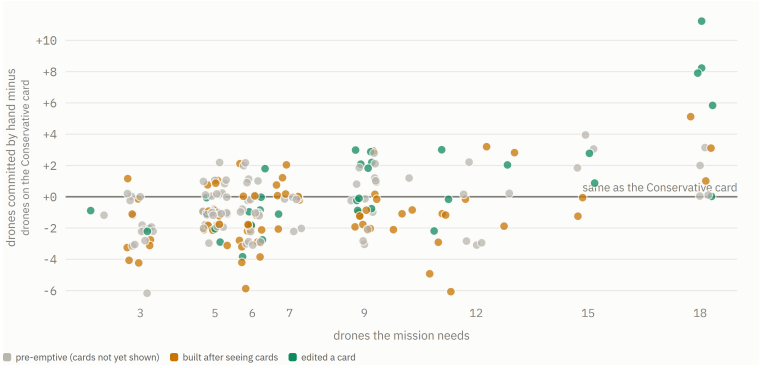
Key numbers:

- C03 10 of 33; C04 15 of 33; C05 8 of 33
- C03 vs others, size slope ratio: strategic $\times 1.28$ [0.69, 2.38], p = .43; tactical $\times 1.43$ [0.93, 2.20], p = .10

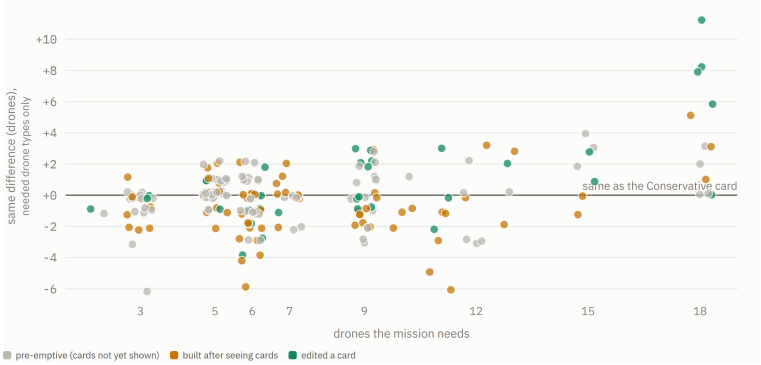
7.4 Hand-built allocations are anchored on the strategy cards

When operators allocated by hand after seeing the strategy cards, their numbers tracked the cards: one more drone on the Conservative card meant about half a drone more in the hand-built allocation, beyond what the mission needed. Operators who decided before the cards appeared showed no such link, which makes this anchoring rather than both following the mission. Hand-built allocations mostly sat at or below the cards (44% below both, 33% equal to one, 14% between and 8% above).

Figure 18 shows every hand-built allocation (209) against the Conservative card. Counting all drone types, allocations on small one-type missions sit about one drone below the card; counting only the types the mission needs, they match it (+0.05 drones). The apparent trimming of small missions is almost entirely the card's spare drones of unneeded types being left off. On big three-type missions hand-built allocations sit about one drone above the card. When short of drones, some sent everything they had: "All right. Let's just send everything I have. Edit, give me five red ones. Five green ones. Hit it." (P10).



(a) All drone types.



(b) Needed drone types only.

Figure 18: The trim on small missions is the card's unneeded spare. Drones committed by hand minus drones on the Conservative card, by the drones the mission needs: (a) all drone types, (b) needed types only.

Allocations built before the cards appeared landed exactly on a card total more often than those built after seeing them (39% against 27%, Figure 19), so matching a card is partly just the obvious allocation rather than copying; balanced for the mission, the difference is too small to tell apart (37% against 29%). Some wanted something the two cards did not offer: "But whenever I clicked manual, it's either been for the cause of 'they are sending too much', or I want it to be somewhere between those two levels" (P17).

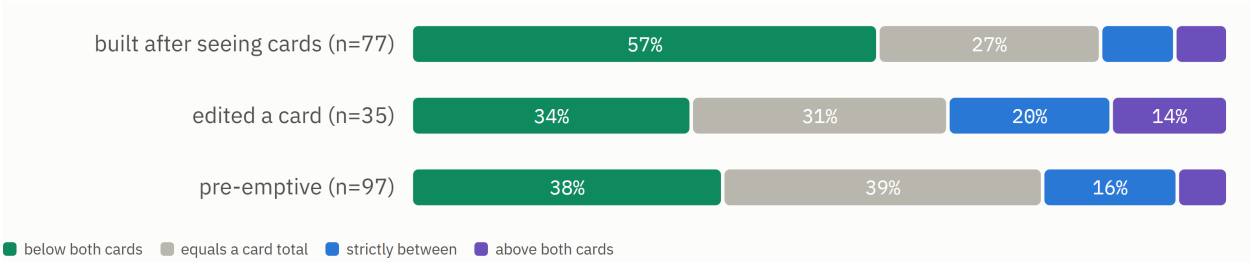


Figure 19: Hand-built allocations mostly sit below both strategy cards or equal to one, rarely above.

Options for this finding

Each manual decision against the Conservative card [pass 5: for the paper]

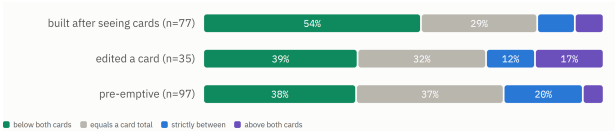
Key numbers:

- Mean manual − Conservative: −0.55 drones overall = −0.51 on unneeded types + −0.04 on needed types
- One-type missions: −1.17 overall, +0.05 on needed types, −1.22 on unneeded; three-type with 10+ drones: +0.95
- Per doubling of mission size the difference grows by +1.51 drones [0.53, 2.49] (after seeing the cards) and +1.09 [0.47, 1.72] (pre-emptive): the size pattern is just as strong in people who never saw the cards, so it is not a reaction to the card
- Needed types only, after seeing the cards: +0.93 per doubling [−0.18, 2.04], p = .10

Highlighted quotes not used above:

> *"I'm trying to reserve more and more drones in the base, rather than send them out to finish the task quickly, because I cannot predict when the next task will appear." — P28 (debrief 0:28:05)*

Where manual builds sit relative to the two cards [pass 5: for the paper]



Other view: Balanced for mission size, drone types, the gap between the cards, game, session and minute.

Key numbers:

- All 209: below both 44%, equal to a card 33%, strictly between 14%, above both 8%
- Built after seeing (77): 57% below, 27% equal, 9% between, 6% above; pre-emptive (97): 38%, 39%, 16%, 6%
- Mission mix: pre-emptive builds needed 6.9 drones on average (after seeing 7.6, edited 9.3) and 52% were one drone type (38%, 23%).
- Balanced: after seeing 54% below, 29% equal, 9% between, 7% above; pre-emptive 38%, 37%, 20%, 5%; edited a card 39%, 32%, 12%, 17%.
- Pre-emptive minus after seeing, balanced: equal to a card +8 points [−10, +23] (raw +12); below both −16 [−33, +4] (raw −19).

Highlighted quotes not used above:

> *"I felt like sometimes I could find like a middle ground that had like maybe a little bit of gambling in there, essentially, that you wouldn't have the necessary coverage." — P22 (debrief 0:28:57)*

What operators say about building by hand

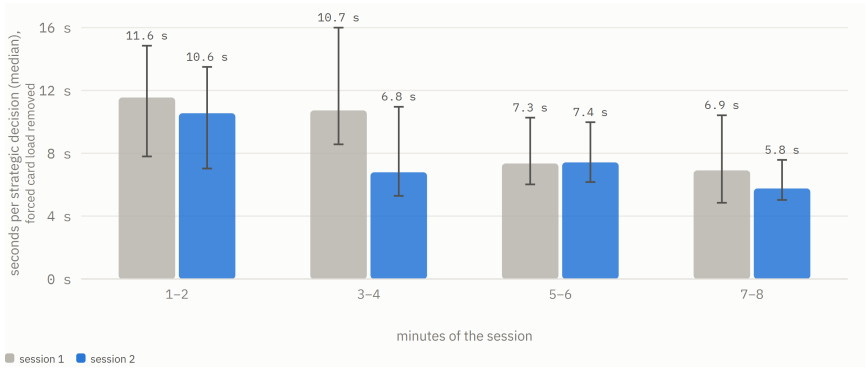
Key numbers:

- C07: 55 manual decisions from 6 people, mean +0.36 vs Conservative on needed types, 21 edited cards
- C08: 40 from 5 people, 10% strictly between the cards
- C09: 33 from 4 people, 16 edited cards
- C10: 40 from 5 people

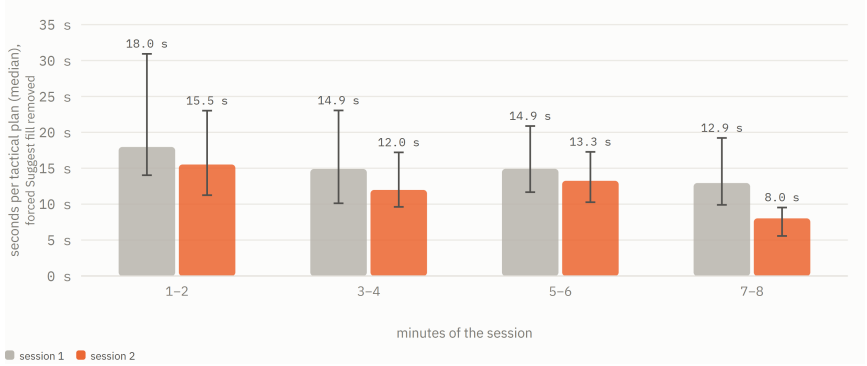
7.5 Practice makes operators faster, not better at choosing when to hand over

Decisions sped up within each session, by about 6% per minute on both tiers, and not because the queue grew. On the strategic tier most of that gain was gone by the start of the next session, so it reads as a warm-up within a session more than lasting learning; on the tactical tier the second session started about where the first ended. The size rule did not sharpen with practice.

Figure 20 shows the median decision time through each session. Strategic decisions fell from 11.6 s in the first two minutes of session 1 to 6.9 s at its end, started session 2 nearly as slow again (10.6 s) and fell once more (5.8 s). Tactical plans fell from 18.0 to 12.9 s in session 1 and from 15.5 to 8.0 s in session 2. Holding the number of waiting missions fixed does not remove the speed-up. Participants described finding a rhythm: "there was definitely an improvement further throughout session one, and in session two, you really find your stride" (P10).



(a) Strategic tier, sessions side by side.



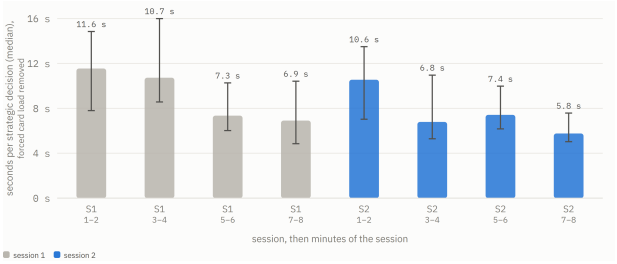
(b) Tactical tier, sessions side by side.

Figure 20: Faster within a session; on the strategic tier the gain mostly resets by the next. Median time per decision by pair of minutes: (a) strategic tier, (b) tactical tier.

The six participants who said they got better did improve their scores (all six, against 10 of the other 27), but they did not come to use the assistants more, and across everyone a bigger score gain did not go with more use of either assistant ($\rho = -.07$ for strategy cards, $-.18$ for tactical suggestions).

Options for this finding

Faster within a session; on the strategic tier the gain mostly resets by the next [pass 5: numbers only]



Other view: Strategic tier, all of session 1 then all of session 2.

Key numbers:

- Strategic, medians: session 1 minutes 1–2 11.6 s [7.8, 14.5] → minutes 7–8 6.9 s [4.9, 10.2]; session 2 10.6 s [7.0, 13.5] → 5.8 s [5.0, 7.6]
- Strategic, model: end of session 1 −39% [−51, −24]; start of session 2 −20% [−37, +2], p = .07; start of session 2 vs end of session 1 +31% [0, +72], p = .053 (slower again)
- Tactical, medians: 18.0 s [14.0, 30.9] → 12.9 s [10.1, 19.2]; session 2 15.5 s [11.3, 21.9] → 8.0 s [5.6, 9.6]
- Tactical, model: end of session 1 −24% [−45, +4], p = .09; start of session 2 vs end of session 1 +2% [−33, +56], p = .92; session 2 slope −7% per minute vs −4% in session 1 (difference p = .26)
- Within-session slope with the queue held fixed: strategic −6% per minute, tactical −8% per minute (both p < .001)
- Overall (unchanged): strategic −5.7% per minute [−7.8, −3.6], session 2 vs 1 −19% [−30, −6]; tactical −5.7% per minute [−9.1, −2.2], session 2 vs 1 −24% [−36, −9]

Highlighted quotes not used above:

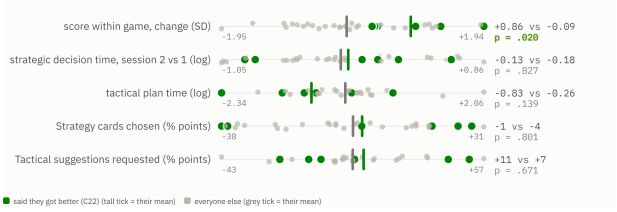
> *"i think the second scenario i used more personalised, i think it was easier because i had to i sort of got into the rhythm of having to anticipate other problems" — P30 (debrief 0:23:52)*

> *"Right. So now we also know from the last one, don't send everything straight away. Cause otherwise we end up, we're waiting for drones." — P22 (think-aloud s2 0:17:11)*

> *"And I could definitely feel that there is absolutely a learning effect." — P10 (debrief 0:22:39)*

> *"I think with having some experience, yeah, I was able to quickly understand this" — P05 (think-aloud s1 0:11:14)*

Does 'I got better with practice' match the logs? For score yes; not for speed or assistant use [pass 5: numbers only]



Other view: main

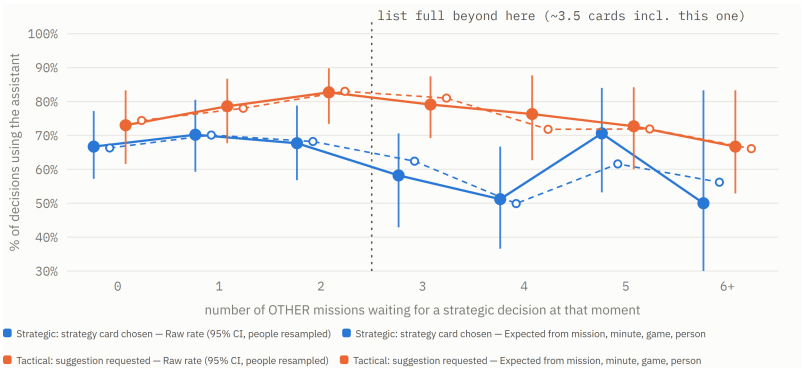
Key numbers:

- Score change: C22 +0.86 SD vs others −0.09, p = .020
- Tactical time change (log): −0.83 vs −0.26, p = .14; strategic: −0.13 vs −0.18, p = .83
- Change in strategy cards chosen (game-adjusted): C22 −1 point vs others −4, p = .80; suggestion requested: +11 vs +7 points, p = .67
- Used more in session 2: cards 3 of 6 C22 vs 12 of 27 others; Suggest 3 of 6 vs 19 of 27
- Everyone: score change vs change in strategy card choice $\rho = -.07$ (p = .71); vs change in Suggest $\rho = -.18$ (p = .32)

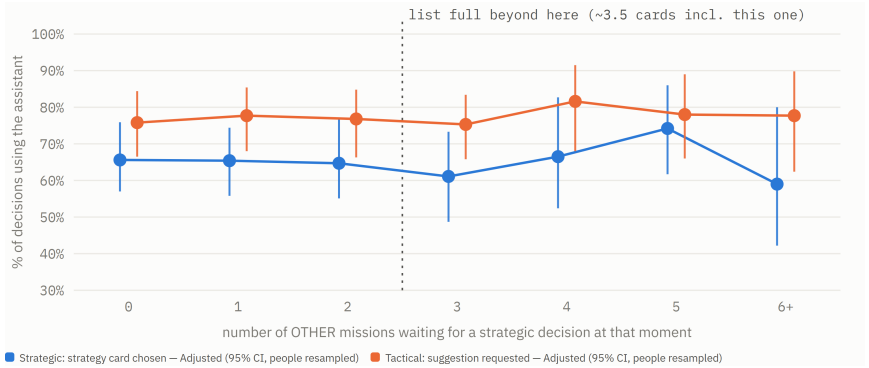
7.6 A busier moment does not push operators onto the assistants

Comparing each operator with themselves, having more missions waiting did not change how often they chose a strategy card or requested a tactical suggestion, once we account for what each decision was, when it came in the session, which game it was and who made it. The data rule out large effects (on the strategic tier, at most about 15 points more strategy card choice going from zero to five missions waiting). Nine of 33 participants said they handed over more when overloaded; their own behaviour moved with the queue no more than anyone else's.

Figure 21 shows the share of decisions handed over by the number of other missions waiting. The raw strategic rate dips at three to four missions waiting, but that dip is where the decisions themselves predicted it (dashed lines): deep queues happened almost only in the Strategic Heavy games and late in a session. Adjusted for the mission, the minute, the game and the person, every point sits between 59% and 74% on the strategic tier and between 75% and 82% on the tactical tier, with no trend, including beyond the point where the queue list runs out of room. This runs against what participants said, for example: "So I think if, at first glance, I felt really overwhelmed by the amount of drones to be assigned. I would use the assistant" (P18).



(a) Both tiers together, raw with their expected rates (dashed).



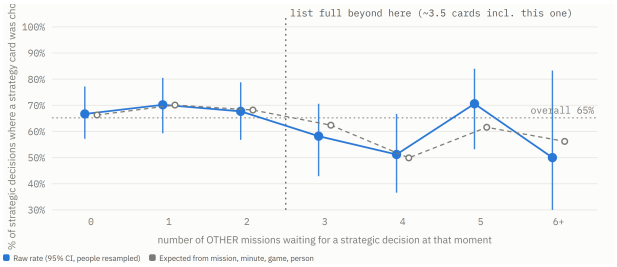
(b) Both tiers together, adjusted.

Figure 21: A busier moment does not push operators onto the assistants. Share of decisions handed over by the number of other missions waiting, both tiers together: (a) raw, with the rate the decisions themselves predict (dashed); (b) adjusted.

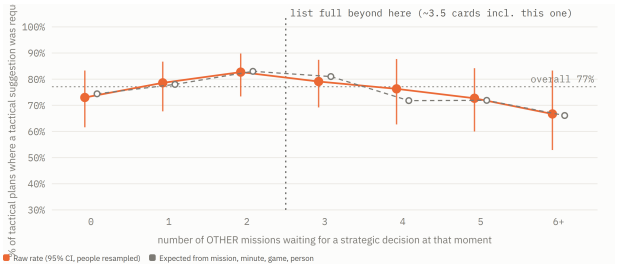
Nor were missions handed over more often after being left waiting: strategy card choice was 63–68% (adjusted) whether a mission had waited ten seconds or more than two minutes, and tactical suggestions, if anything, became a little less likely the longer a plan had been left open.

Options for this finding

Queue depth against reliance — raw, and adjusted [pass 5: for the paper]



Other view: Strategic tier, raw with the expected rate (dashed).



Other view: Tactical tier, raw with the expected rate (dashed).

Key numbers:

- Strategic raw: 67% (0 waiting), 70%, 68%, 58%, 51%, 71%, 50% (6+); adjusted: 66%, 65%, 65%, 61%, 67%, 74%, 59% — all intervals overlap.
- Tactical raw: 73%, 79%, 83%, 79%, 76%, 73%, 67%; adjusted: 76%, 78%, 77%, 75%, 82%, 78%, 78%.
- The queue rises with the minute (Spearman $\rho = .49$) but hardly with mission size ($\rho = -.08$): the adjustment mostly removes WHO reaches deep queues and WHICH GAME they are in.
- Deep queues are a Strategic Heavy phenomenon: 90–97% of decisions at 4+ waiting are in those two games.

Highlighted quotes not used above:

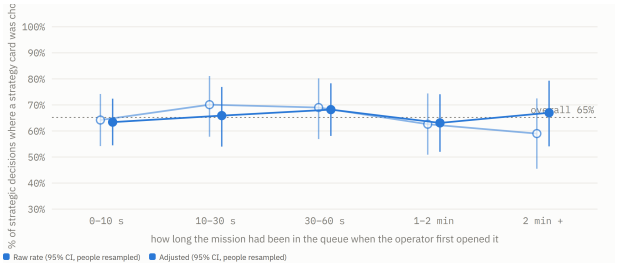
> *"You can have a lot of tasks to be overloaded and try and offload some of that work to AI and then for the very complex stuff, maybe you don't." — P20 (debrief 0:28:46)*

> *"there was no more active missions coming in, so therefore I could take more time and ensure that it was correct. But perhaps if I was more time pressured, then I'd ask it to suggest." — P05 (think-aloud s1 7:31)*

> *"But in an urgent situation and the number of the swarm is not enough, so I need to consider about the strategies and I think I need the help of AI assistant." — P35 (debrief 0:28:19)*

> *"I think the first one, I feel the pace is very, very fast. So, I can't do it very well in a very short time. So, I used a lot of AI suggestions." — P33 (debrief 0:27:33)*

A mission left waiting is not handed over more often [pass 5: numbers only]



Other view: Strategic tier: how long the mission had waited in the queue.

Other view: Tactical tier: how long the plan had been left open.

Key numbers:

- Strategic (mission left in the queue): adjusted 63% at 0–10 s, 66% 10–30 s, 68% 30–60 s, 63% 1–2 min, 67% at 2 min+ (n 173, 117, 100, 91, 83). Within person, odds $\times 1.10$ per doubling of the wait [0.96, 1.28], p = .17; holding the queue fixed $\times 1.10$, p = .19.
- Tactical (plan left open): adjusted 80% at 0–5 s, 76% 5–10 s, 76% 10–20 s, 73% 20–40 s, 70% at 40 s+ (n 270, 182, 93, 41, 44). Within person, odds $\times 0.80$ per doubling [0.60, 1.06], p = .12; holding the queue fixed $\times 0.76$, p = .10.

7.7 A long queue does not change what operators open

With a long queue, operators did not abandon order or pick smaller missions: compared with what was actually waiting, they kept the same priorities at any queue length.

The share of decisions that opened the oldest waiting mission fell as the queue grew (59% with one other waiting, about 30% with five), but so did the chance of picking the oldest at random (50% to 17%); measured against chance, the pull towards the oldest stayed the same (Figure 22). In a model of the choice among what was waiting, being the oldest roughly tripled a mission's chance of being opened, at short and long queues alike. Criticality pulls a little too: each category above the waiting missions' average raised the odds of being opened by about 1.45 (Section 8.5).

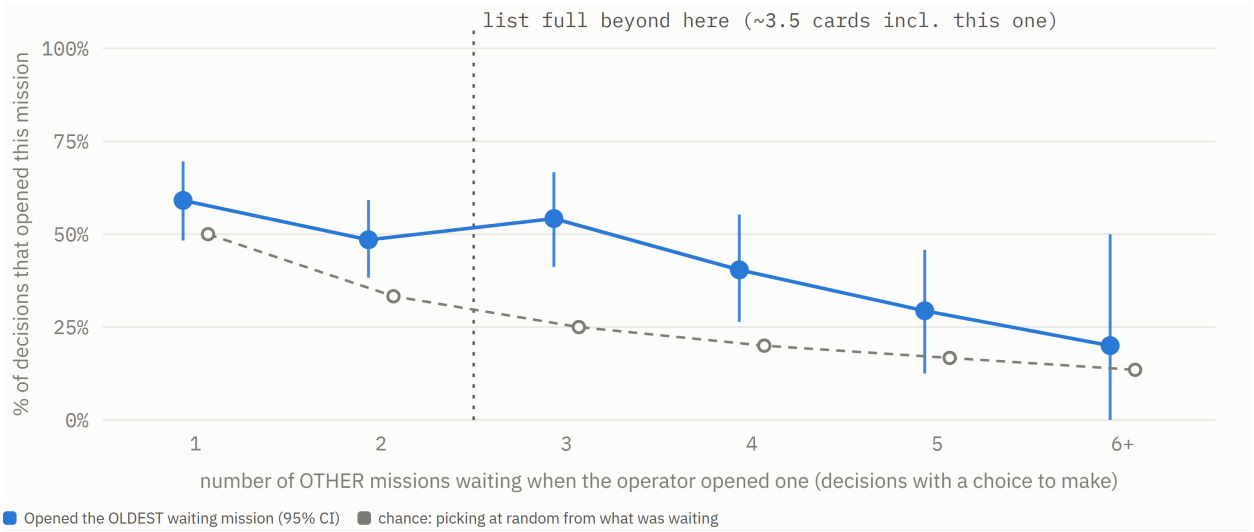


Figure 22: Operators open the oldest waiting mission well above chance at every queue length.

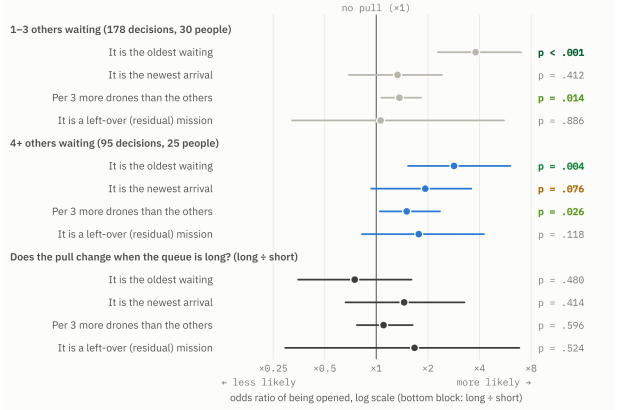
Options for this finding

Order: still oldest-first well above chance [pass 5: for the paper]

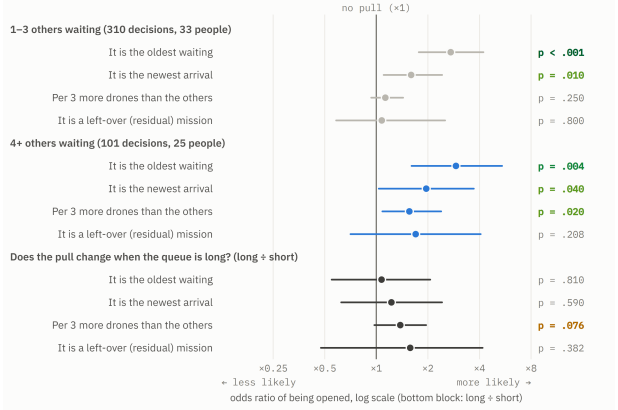
Key numbers:

- Opened the oldest: 59% (1 other waiting; chance 50%), 48% (33%), 54% (25%), 40% (20%), 29% (17%), 20% (14%).
- Within person, the excess over chance does not shrink with depth: +2.4 points per extra mission [−1.0, +5.7], p = .16.

What makes a waiting mission get opened, short vs long queues [pass 5: numbers only]



Other view: Strategic Heavy games only.



Other view: All four games.

Key numbers:

- Strategic Heavy games, short queues (178 decisions): oldest $\times 3.80$ [2.29, 6.99], $p < .001$; newest $\times 1.33$ [0.69, 2.41], $p = .41$; +3 drones $\times 1.36$ [1.07, 1.82], $p = .014$; residual $\times 1.06$, $p = .89$
- Strategic Heavy games, long queues (95 decisions): oldest $\times 2.84$ [1.53, 6.06], $p = .004$; newest $\times 1.93$ [0.94, 3.58], $p = .076$; +3 drones $\times 1.50$ [1.05, 2.35], $p = .026$; residual $\times 1.77$, $p = .12$
- Long $\div$ short, Strategic Heavy games: oldest $\times 0.75$ [0.35, 1.60], $p = .48$; newest $\times 1.45$, $p = .41$; size $\times 1.10$, $p = .60$; residual $\times 1.67$, $p = .52$ (none differs)
- All four games: oldest $\times 2.72$ (short) and $\times 2.92$ (long), both $p \leq .004$; newest $\times 1.60$, $p = .010$ and $\times 1.96$, $p = .040$; size $\times 1.13$, $p = .25$ and $\times 1.56$, $p = .020$; long $\div$ short for size $\times 1.38$ [0.97, 1.95], $p = .076$, all others $p \geq .38$

7.8 The illogical strategy card, and who stopped choosing the cards

The Conservative strategy card sometimes added a drone type the mission did not use (22% of decisions, nearly all on one-type missions), and all 33 people met it, usually on their first or second decision. The 15 who later said they noticed it chose strategy cards less afterwards, and on the flawed one-type missions they learned to skip the card. But the dip in their use is not tied to the illogical card itself: it appears early in the first session whether or not they had met that card yet, and in the same stretch the people who never mentioned it chose the cards more. Our best reading is that "noticed" marks people who scrutinise the cards, and the dip is that scrutiny at the first busy stretch of the study. The evidence is suggestive (15 against 18 people).

Lined up on each person's first illogical card (Figure 23), noticers went from choosing a strategy card on 76% of decisions before it to 43–52% for their next ten decisions, recovering to 67–68% only from the eleventh on; people who never mentioned it stayed at 67–75% throughout. On three-type missions, which never carry the flaw, noticers dipped for a short stretch and recovered (74% before, 43% for the next two decisions, then 59–63% and 75%). Many first met the card with puzzlement rather than a verdict: "Oh, this just sends a bunch of extra of these even though it doesn't need it?" (P19); "But in conservative, you don't need camera or lifters, yet it adds it as reserve. Even for tasks it doesn't do. That doesn't make sense." (P13).

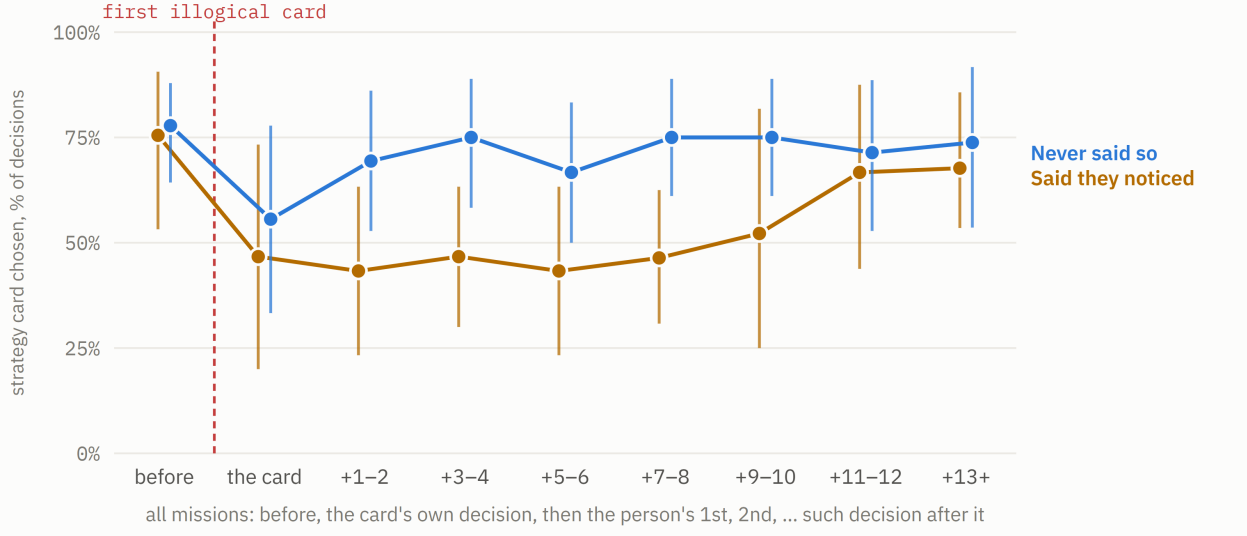
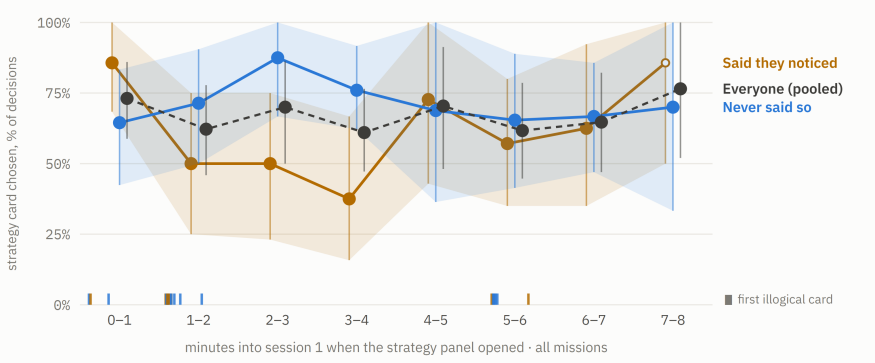
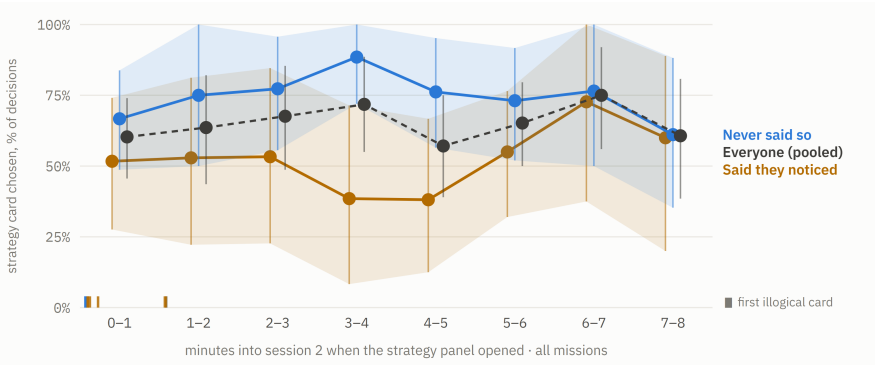


Figure 23: After the first illogical strategy card, those who later said they noticed it choose the cards less for about ten decisions. Share of decisions where a strategy card was chosen, by decision number after the first illogical card, in pairs.

Not lined up on the card but placed by the minute of the session (Figure 24), the dip is still there. In session 1 noticers started at 86% in the first minute, fell to 38–50% over minutes one to four and came back to 73%, while the others rose over the same minutes (65% to 88%). Pooled over everyone the line stays at 61–77%, because the two groups move in opposite directions. Several kept using the cards anyway for the time they saved: "I think it's less reliable than before, but I still use its suggestion, and that's the case. Yeah, because at least it saves time." (P28).



(a) Session 1: each group and everyone.



(b) Session 2: each group and everyone.

Figure 24: Early in the session the two groups move in opposite directions, so pooling them hides the dip. Share of strategic decisions with a strategy card chosen, by minute of the session, for noticers, the others and everyone (dashed): (a) session 1, (b) session 2.

Counted simply by decision number (Figure 25), noticers chose a strategy card on 80% of their first decisions of session 1 and 46–50% of decisions three to five, then 83% at the sixth, while the others rose to 89–94% in the same stretch. People who played the Tactical Heavy game first had not yet met an illogical card by decisions three to five (95% of those decisions came before it), and their noticers dipped just as much (-21 points per person, against -32 for those who had met it; $p = .79$). Noticers recovered within session 1 ($+24$ points per person) and started session 2 lower again, but so did the others, so a weaker carry-over for noticers cannot be shown (difference at the break -6 points $[-32, +20]$).

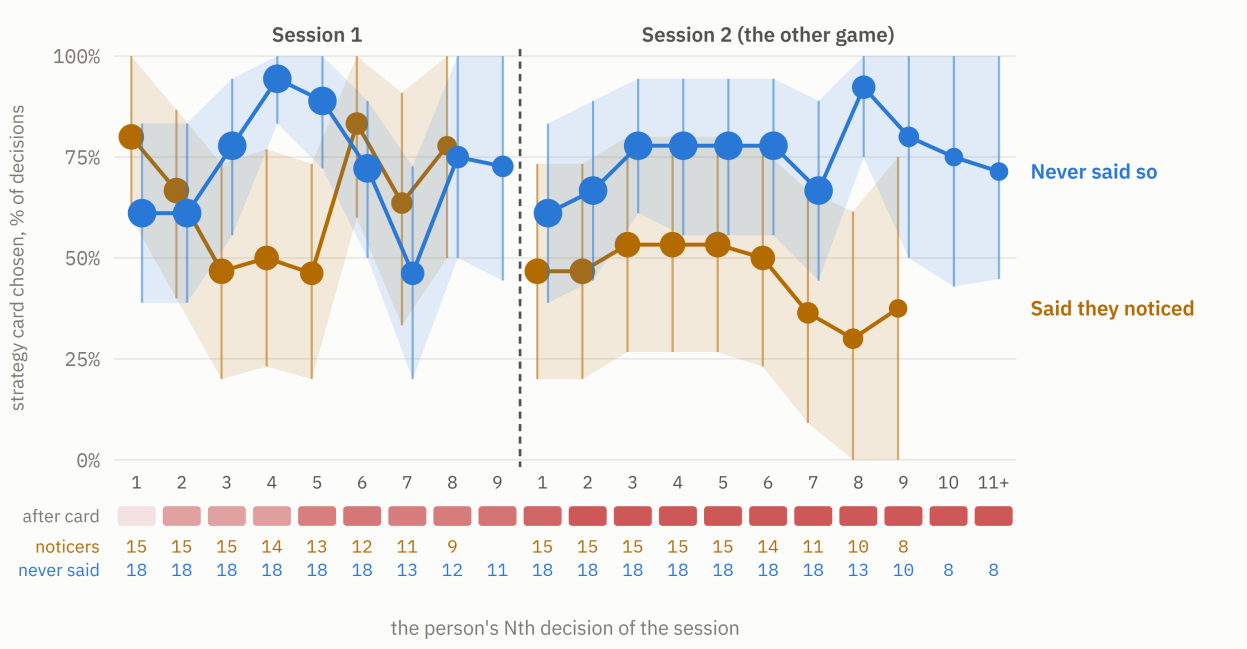


Figure 25: Counted by decision number, the noticers' dip comes before most of them have met an illogical card. Share of decisions with a strategy card chosen, session 1 then session 2.

Lining people up instead on the moment they said aloud that a card was wrong gives the same direction (Figure 26). Seven people made such a remark; they went from choosing a strategy card on 80% of decisions before it to 49% after (adjusted -21 points, $p = .11$), and on the one- and two-type missions that carry the flaw from 83% to 35%, although that 'before' rests on only six decisions. Seven people cannot settle it, but the direction matches. A failure of the other assistant had no such effect: after the Tactical Assistant first answered 'No workable plan' when asked for a recovery plan (18 people), neither strategy card choice ($+2$ points, $p = .62$) nor requests for tactical suggestions (-1 point) changed, and nobody said a refusal made them doubt either assistant.

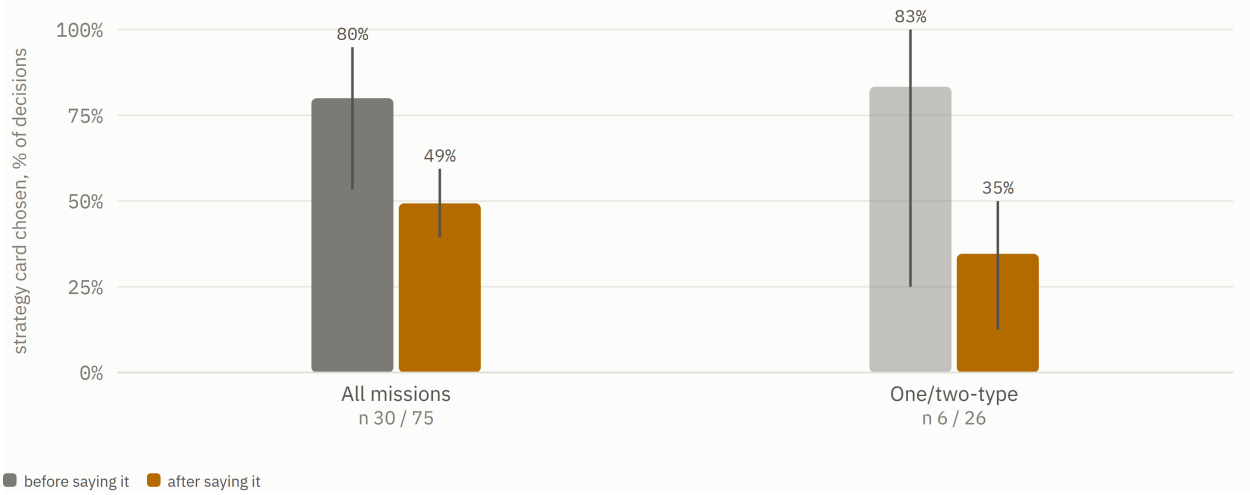
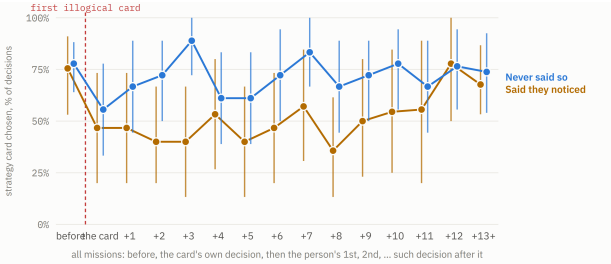


Figure 26: After saying aloud that a strategy card was wrong, card choice falls (seven people; a direction, not a measured size). Share of strategic decisions with a strategy card chosen, before and after the remark: all missions, and one- and two-type missions.

Options for this finding

The event study, aligned on the first illogical card (every step, pairs or threes) [pass 5: for the paper]



Other view: All missions, every step.

Key numbers:

- All missions, noticers (15): 76% before (n 53) → card 47% (n 15) → pairs +1–2 to +9–10: 43%, 47%, 43%, 46%, 52% → 67% (+11–12) → 68% (13th+, n 31)
- All missions, never said (18): 78% before (n 54) → 56% at the card → 67–75% for every later pair
- All missions, before → after, adjusted: noticers −14 points, p = .08; never said +9, p = .14; difference −14 [−30, +1], p = .06
- Three-type, noticers by pair: 74% before (n 51), 43% (n 30), 59% (n 29), 62% (n 24), 63% (n 19), 75% (n 12)
- Why expected is lower for noticers: their own-rate term (take rate on all their other decisions) is 56% vs 70%; 79% of a noticer's decisions come at or after the card, where they took 50% per person vs 72% (p = .007); before the card the groups were alike (68% vs 63%, p = .71)
- Mission-only expected (own-rate term set to the average): after the card 68% noticers vs 69% never-said; before 80% vs 81%
- Steps before the card, all missions: noticers ≤ -4 87% (n 23), −3 88% (n 8), −2 50% (n 8), −1 64% (n 14); never-said 82%, 100%, 88%, 56% (n 22, 8, 8, 16)
- −1 against −3/−2, the 16 people with three steps (all tactical-heavy-first): adjusted −13 points [−40, +13], p = .29
- Matched view (8 noticers with 3+ steps, 6 of whom met the card at the start of session 2): 87%, 88%, 50%, 63% before → 38% at the card → 31–44% for the next ten decisions

Highlighted quotes not used above:

> *"Why is it that the green ones, there's no green or red, unless I'm missing something." — P17 (think-aloud s1 5:25)*
>
> *"And then when it came to assigning them, I noticed, oh, okay, it's made a mistake." — P20 (debrief 0:26:11)*
>
> *"where we can go conservative, but we don't need any, in this one we don't need any lifter or cameras." — P30 (think-aloud s1 1:01)*
>
> *"But I find that the first task it allocated too much Lifter and Camera. So I think I'm less confident for it." — P35 (think-aloud s1 1:22)*
>
> *"there was one example in the first round where it was required all fast drones. And I think the AI assistant recommended I add some red and green drones." — P20 (debrief 0:25:51)*
>
> *"I mean, it was giving me the wrong type of drones. It required only blue drones and kept giving me red and green ones." — P21 (debrief 0:35:03)*
>
> *"deploy a conservative strategy for that one doesn't make sense, I don't think. Because I don't need any red or blue drones." — P20 (think-aloud s1 3:41)*

Card use through the session, noticers and never-said [pass 5: for the paper]

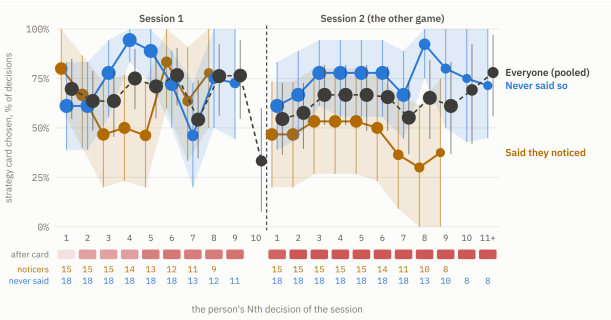
Key numbers:

- Session 1, noticers by minute 0–1 to 4–5: 86% (n 21), 50% (n 16), 50% (n 14), 38% (n 16), 73% (n 11)
- Session 1, never said: 65% (n 31), 71%, 88%, 76%, 69%
- Session 2, noticers: 52% (n 29), 53%, 53%, 38% (n 13), 38% (n 21), 55%, 73%, 60%; never said 61–89%

Highlighted quotes not used above:

> *"I know there is a better choice, but I have no time to tune everything." — P28 (debrief 0:30:33)*

Card use by decision number, session 1 then session 2, not aligned to the card [pass 5: for the paper]

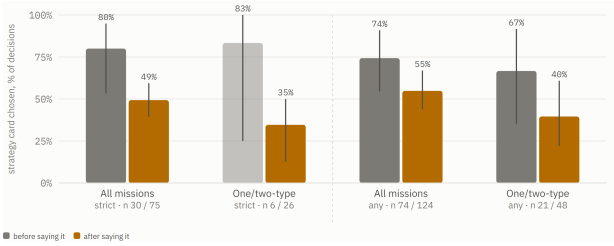


Other view: Each group and everyone.

Key numbers:

- Everyone, session 1, decisions 1–6: 70%, 64%, 64%, 75%, 71%, 77% (n 33 → 30)
- Noticers, session 1: 80%, 67%, 47%, 50%, 46%, 83%, 64%, 78% (decisions 1–8, 15 → 9 people); never said: 61%, 61%, 78%, 94%, 89%, 72%, 46%, 75%, 73% (decisions 1–9, 18 → 11 people)
- Session 1 fixed panel (9 noticers, 12 never-said, decisions 1–8): noticers 78, 67, 56, 44, 44, 89, 56, 78%; never said 50, 42, 67, 92, 83, 67, 50, 75%
- Session 1 tail (hidden): decision 9, 6 noticers 83%; decision 10, 5 noticers 1 of 5 took (20% [4, 62]), 7 never-said 43% [16, 75]; 11+, 2 noticers 3 of 7 (one took 3 of 4, the other 0 of 3), 5 never-said 7 of 12. All strategic-heavy-game players; 57–67% of the 11+ decisions in the last 90 s
- Session 2 tail (hidden for noticers): decision 10, 5 noticers 60%; 11+, 4 noticers (all tactical-heavy first, so playing the strategic-heavy game) 12 of 13 = 92% [67, 99], against 8 never-said 20 of 28 = 71%. These 4 took cards on 42% of their decisions 1–6, below their group's 51%: not keen takers surviving
- Strategic-heavy game: share of decisions with an illogical card at decisions 7, 8, 9, 10, 11+: 48%, 55%, 30%, 4%, 5%. At decisions 7+, noticers took 17% of illogical-card missions (n 18, 9 people) and 62% of the rest (n 53); never-said 68% and 69%
- Session 2, no illogical card: noticers 71% at decisions 7 and 8 (7 people) against never-said 73% and 70%; earlier the gap is there (decisions 2–3: 33–47% against 67–89%)
- Session 2 fixed panel (8 noticers, 10 never-said, decisions 1–9): gap over the last 3 decisions −47 points [−81, −12] against −19 [−50, +12] before; on missions with no illogical card (panel to decision 6) −17 [−45, +12] against −30 [−61, +1]
- At the session break (last 3 of session 1 → first 3 of session 2): noticers 64% → 49%, −16 points [−38, +4], p = .17; adjusted −9 [−30, +12]; never-said 76% → 69%, −7 [−24, +7]; difference −6 [−32, +20], p = .66
- Recovery within session 1 (decisions 3–5 → 6+): noticers 48% (n 42) → 66% (n 50), +24 points per person [+5, +43], p = .04 (window read off the curve, so descriptive)

Lined up on the moment they said a strategy card was wrong: card use falls (7 people, a direction only) [pass 5: for the paper]

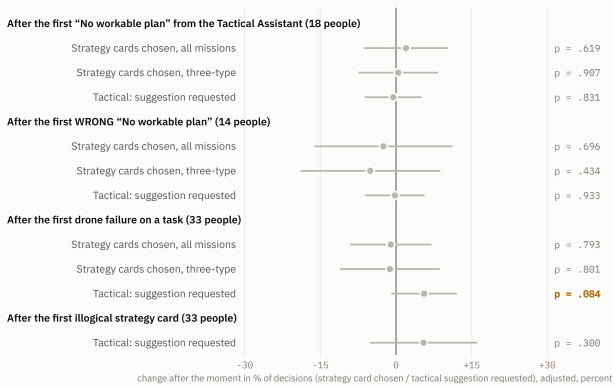


Other view: Strict and any remark, side by side.

Key numbers:

- Strict remarks (7 people), all missions: 80% (n 30) → 49% (n 75); adjusted −21 points [−49, +7], p = .11; 5 fell, 2 rose; placebo p = .51
- Strict, one/two-type: 83% (n 6, 4 people) → 35% (n 26, 7 people); three-type: 79% (n 24) → 57% (n 49)
- Any remark (12 people), all missions: 74% (n 74) → 55% (n 124), adjusted −4 points, p = .60; one/two-type 67% (n 21) → 40% (n 48)

A different event: after the Tactical Assistant's first 'No workable plan', nothing changes on either tier [pass 5: numbers only]



Other view: main

Key numbers:

- First refusal (18): strategy cards, all missions, adjusted +2.0 points [−6, +10], p = .62; three-type +0.4, p = .91; tactical Suggest −0.6 [−6, +5], p = .83
- First WRONG refusal (14): strategy cards −2.5 [−16, +11], p = .70; three-type −5.1, p = .43; tactical −0.2, p = .93
- First drone failure (33): strategy cards −1.0, p = .79; tactical +5.6, p = .08
- After the first illogical card, tactical Suggest +5.4 points, p = .30 (no spill-down either)

Why the pre-registered test (H8) was dropped [not used in the prose above]

H8 asked whether uptake falls after an operator 'realises' the card can be illogical. Its marker for that realisation was the first time a person went manual on an illogical-card mission and left out the unneeded type. The trouble is that, by that definition, every illogical-card decision *before* the marker was not such a manual choice. So the 'before' window was mostly forced to be card takes, near 100%. Any later manual choice then shows up as a drop, even if the person never changed at all. A simulation confirmed it. In simulated people who keep the same habits all study, the same marker produced a large 'drop' (average odds ratio 0.37, against the real 0.12). Measured against that, the real drop was ordinary (calibrated p = .30, about −10 points after correction). Defining the marker as the first manual choice on a *clean* card produced the same kind of drop. H8 was replaced by tests whose anchor is not defined by the outcome (the first illogical card shown; the moment someone said it), each checked against a placebo. Those are the tests in this finding.

Key numbers:

- Pre-registered estimate OR 0.12; the same test on no-change simulations averages OR 0.37; calibrated p = .30 (risk difference)
- Placebo marker on clean cards: OR 0.22, p = .03, the same kind of 'drop'

7.9 The Tactical Assistant is trusted more

On the questionnaire after each session, operators trusted the Tactical Assistant more than the Strategic Assistant (5.44 against 4.61 on a 1–7 scale; 26 of 33 people higher for the tactical, $p = .001$). Most of the gap opened in the second session and in the Strategic Heavy games. Tactical plans got only a few seconds of checking, whatever their size.

The gap is a medium-sized effect (mean $+0.83$ points, $d_z = .62$), and every one of the six questionnaire items points the same way (Figure 27). The usual reason given is that the strategy cards visibly wasted drones while the tactical plans matched what the operator would have done: "So the tactical one seemed to be doing what I was doing. So quite reliably. And obviously the strategic one wasn't." (P32). Another put it in terms of blame: "You can't blame the tactical for drone failures but you can blame the strategic for not having redundancy" (P03).

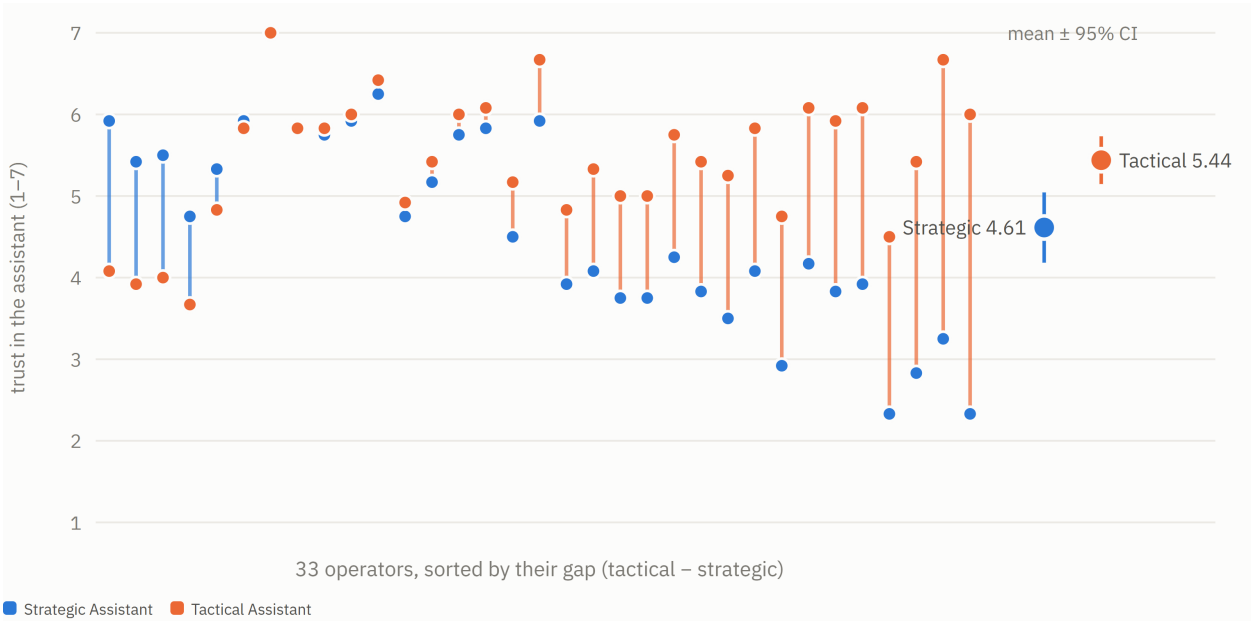
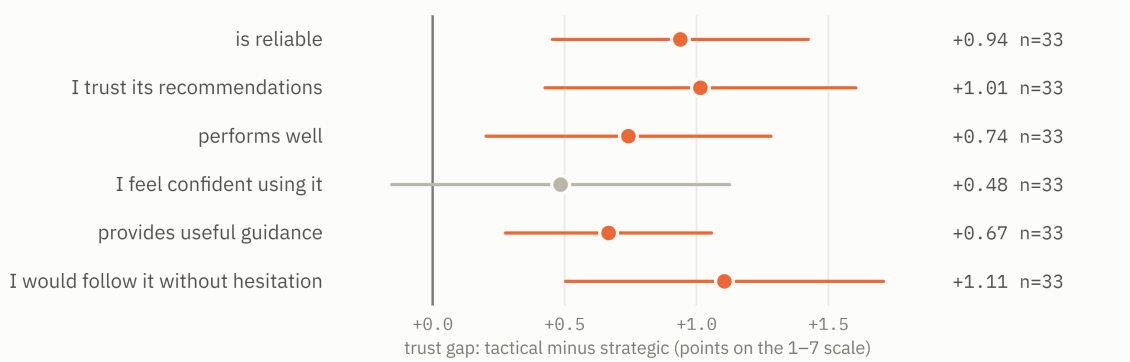
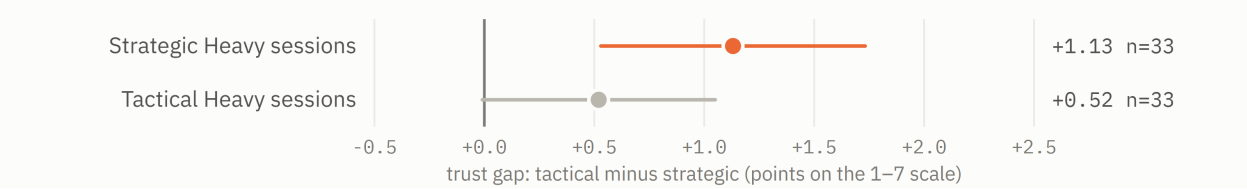


Figure 27: The Tactical Assistant is trusted more. Each person's mean trust in each assistant over their two sessions (1–7).

The gap was larger in Strategic Heavy sessions ($+1.13$) than in Tactical Heavy ones ($+0.52$), and small after the first session ($+0.37$) but large after the second ($+1.28$); with both in one model, Strategic Heavy adds about 0.6 to the gap and the second session about 0.9 (Figure 28). Trust in the Tactical Assistant hardly moved (5.36 to 5.51 across sessions): it is trust in the Strategic Assistant that fell, and most in the scenario where its cards are used most.



(a) By questionnaire item.



(b) By scenario played.

Figure 28: Where the gap comes from. Tactical minus strategic trust: (a) by questionnaire item, (b) by scenario played.

Checking a tactical plan was brief and did not grow with the plan (Figure 29). As plans grew, operators requested a suggestion more often (63% of plans with three drones or fewer, 93% with ten or more) and kept the plan unmodified more often (63% to 86%), and once the plan could be deployed they spent a median of 2–4 s on it whatever its size, against about 9 s on a strategy card. Almost every suggested plan was deployed as it came: 468 of 486. Participants were candid about why: "as soon as it got to any sort of non-linear, difficult, high dimensional problem, I just took the AI's word for it, essentially, probably because I couldn't assess in the time" (P22). The test of this mechanism is fragile, though: it does not survive correction for the round's several tests ($p = .065$).

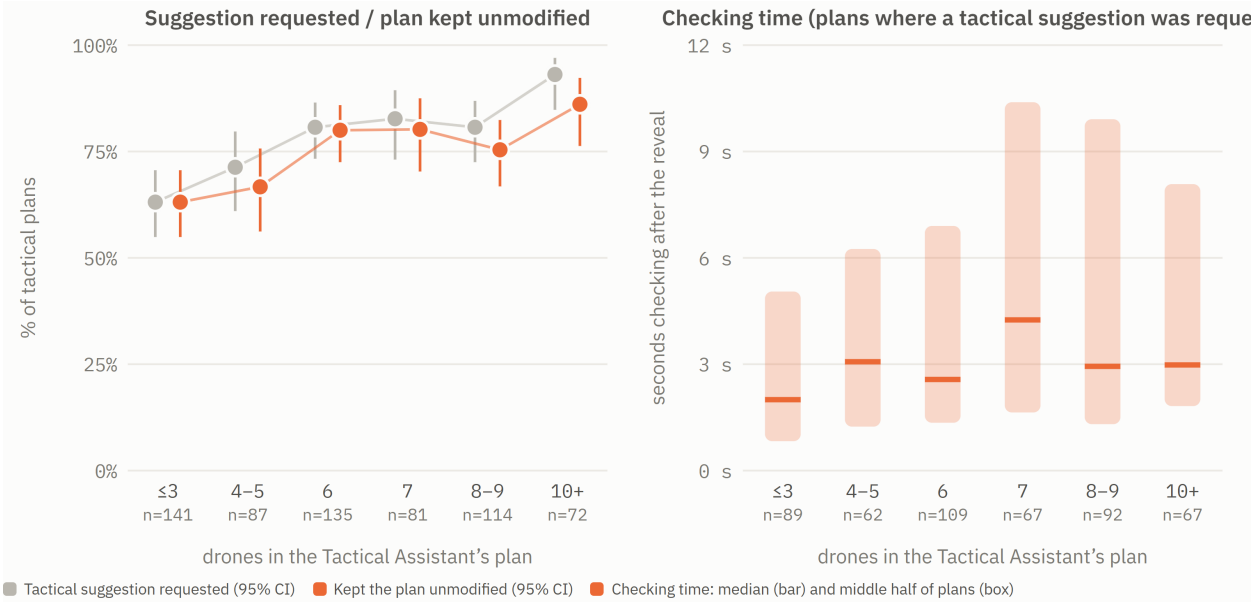


Figure 29: Checking a tactical plan takes a few seconds, whatever its size. By plan size: the share of plans with a suggestion requested, the share kept unmodified, and the median checking time once Deploy unlocks.

Trust and use line up when people are ranked against each other (the trust gap correlates $\rho = .47$ with the gap in use), but not in size and not over time: a change in a person's trust from one session to the next does not go with a change in their use ($\rho = .16$), and trust after the first session does not predict how their use changes in the second ($\rho = .07$).

Options for this finding

Trusted more: the basic result [pass 5: for the paper]

Key numbers:

- Strategic 4.61 [4.18, 5.04], Tactical 5.44 [5.15, 5.73] (1–7, person means)
- Gap +0.83 [+0.35, +1.30]; paired t p = .0012; Wilcoxon p = .0010; dz = .62; n = 33
- 26 higher for tactical, 6 for strategic, 1 tied
- Item gaps: follow its recommendations +1.11, I trust it +1.01, reliable +0.94, performs well +0.74, useful +0.67 (all p < .01); confident in it +0.48 (p = .13)

Highlighted quotes not used above:

> *"But I think in the [tactical level], it works very well." — P33 (debrief 0:28:24)*

Where the gap comes from: the second session and the Strategic Heavy scenario [pass 5: for the paper]

Other view: By session order as well.

Key numbers:

- Session 1 gap +0.37 [−0.15, +0.88]; session 2 +1.28 [+0.69, +1.87] (n = 33 each)
- Strategic Heavy sessions +1.13 [+0.53, +1.73]; Tactical Heavy sessions +0.52 [−0.01, +1.05] (n = 33 each)
- Same people, Strategic Heavy minus Tactical Heavy: +0.61 [−0.01, +1.23], paired t p = .054, Wilcoxon p = .038; strategic trust −0.46 (p = .087), tactical +0.15 (p = .25)
- Both in one model: Strategic Heavy +0.64 [+0.11, +1.17], p = .019; second session +0.93 [+0.40, +1.47], p < .001
- Strategic trust 5.00 → 4.23 from session 1 to 2; tactical 5.36 → 5.51

Why the tactical plan is kept: checking it is brief and does not grow with the plan [pass 5: for the paper]

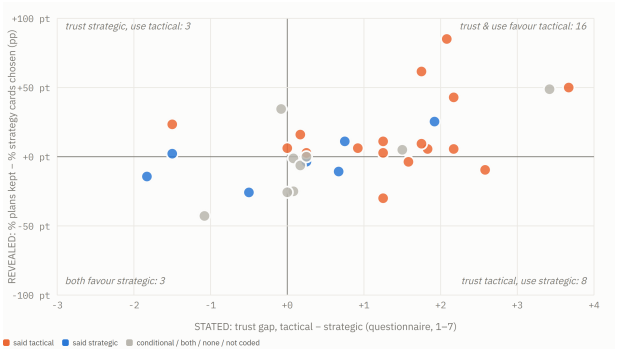
Key numbers:

- Kept unmodified rises with plan size: OR 1.76 per SD [1.25, 2.47], p = .001 (all 630 plans)
- Consulted rises: OR 2.52 [1.87, 3.38]; kept once consulted FALLS: OR 0.68 [0.50, 0.94], p = .017
- By bin (≤3, 4–5, 6, 7, 8–9, 10+ drones; n = 141, 87, 135, 81, 114, 72 plans): suggestion requested 63%, 71%, 81%, 83%, 81%, 93%; kept unmodified 63%, 67%, 80%, 80%, 75%, 86%
- 10+ drones vs ≤3: suggestion requested 93% vs 63%, OR 10.8 [3.2, 36.0]; kept 86% vs 63%, OR 5.2 [2.1, 12.7], both p < .001
- Median checking time by bin: 2.0, 3.1, 2.6, 4.2, 2.9, 3.0 s; per SD +0.60 s [−0.15, +1.35], p = .12; per drone 0.68 s at ≤3 drones → 0.23 s at 10+
- 468 of 486 consulted plans deployed unmodified (96%)
- Person-median check: tactical 2.85 s vs strategy card 8.80 s (n = 33, Wilcoxon p < .0001)
- H2 primary p = .019 uncorrected, .065 after Holm correction across round 1

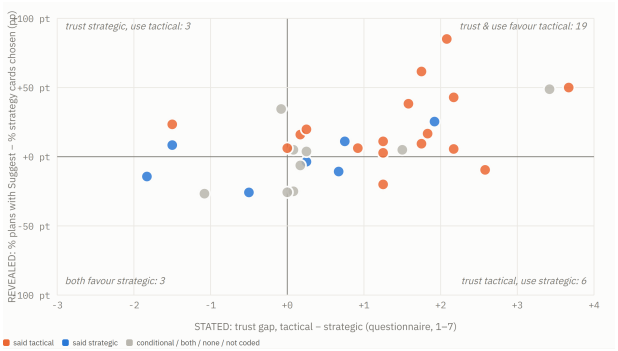
Highlighted quotes not used above:

> *"My reason is there are so many drones, so I don't have enough energy and attention to just the check them." — P27 (think-aloud s1 0:56)*

Stated vs revealed preference: they line up in rank, not in size or over time [pass 5: numbers only]



Other view: Use measured as keeping the plan unmodified.

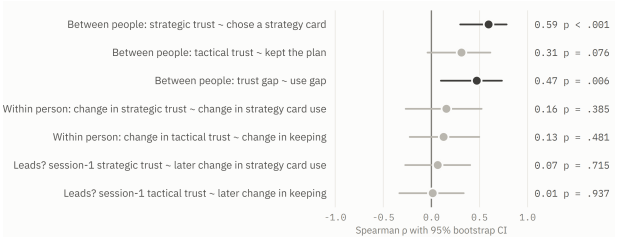


Other view: Use measured as pressing Suggest at all.

Key numbers:

- Trust gap vs usage gap (kept − took): ρ = .47 [.10, .73], p = .006, n = 33
- Quadrants: both favour tactical 16, both strategic 3, trust tactical/use strategic 9, trust strategic/use tactical 3 (2 on a zero line)
- Group use: chose a strategy card 64%, consulted the plan 76%, kept it 72%; usage gap +0.08 [−0.02, +0.17]
- Within person, change in strategic trust vs change in strategy card use: ρ = .16 [−.27, .52], p = .38
- Session-1 strategic trust vs change in strategy card use s1→s2: ρ = .07 [−.27, .40], p = .71; tactical .01
- Between people, strategic trust vs strategy cards chosen ρ = .60 [.30, .78]; tactical trust vs plan kept ρ = .31, p = .08

All the trust–use correlations in one view [pass 5: numbers only]



Other view: main

7.10 What operators asked for

Asked about real search and rescue, nobody rejected assistants like these, and 9 of 33 said, mostly unprompted, that a human must keep the final decision. What they asked for was mostly information: how the assistants decide, and how likely drones were to fail.

Figure 30 counts the requests. Beyond a human final say, the most common were failure probabilities or performance data (6), options between the two strategy cards (6) and the assistants' reasoning (5); only four wanted the assistants to do more. Several credited the assistants with more than they had, assuming they learned from the operator or that the two tiers shared one plan, while others saw through them to simple rules. These themes are taken up in Section 8.

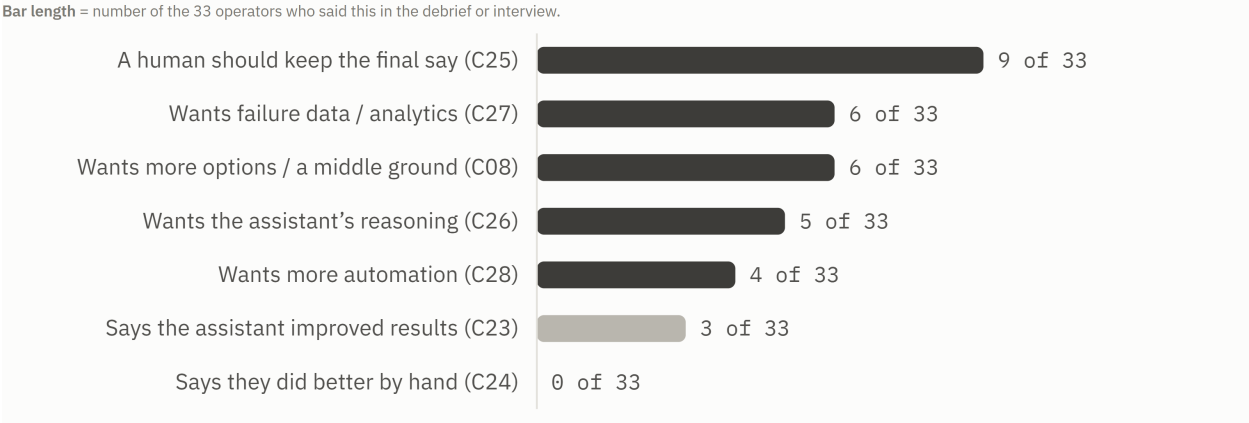


Figure 30: What operators asked for. Number of participants raising each request (of 33).

Options for this finding

What they asked for [pass 5: numbers only]

Key numbers:

- Human final say 9 of 33 (8 unprompted); failure data 6; more options 6; reasoning 5; more automation 4 (all unprompted)

8 In Their Own Words

[Generated (`qual.tex`): the qualitative findings built from long passages of the think-aloud, the questionnaire talk and the debriefs. Two findings you pruned this pass are left out ("Trust in the process, not the answer" and "What a real operation would need"; your note on the second: the limitations are implied and can be mentioned near the end). Participant P36 is quoted only in Section 9.]

8.1 You trust what you cannot check

The most articulate operators described trusting the Tactical Assistant *because* they could not assess its plan, while a strategy card was simple enough to judge at a glance, and so was judged, often harshly. Under time pressure checking shrank to a glance, and the minority who did check earned their trust by comparing the assistant's plan with their own.

A strategy card shows a few numbers per drone type beside the mission's requirements, so its oddities are visible; a tactical plan is a web of routes on a map. Participants drew the consequence themselves: they took the tactical plan on trust because they could not see how far it was from optimal, and suspected that was part of why it seemed so good. This is their own account of two of the quantitative results, the trust gap between the tiers (Section 7.9) and the visible and invisible flaws (Section 7.8), and it suggests the gap is partly one of checkability rather than quality.

> "The tactical assistant, I never really saw it do anything wrong, but that's basically because I think the problem is so complex [that] seeing a more optimal solution is really hard. So I wonder if it's one of those things where these are based on the fact that I actually can't perceive the difference between its performance and the optimal."
>
> — P22 (talk during the questionnaire survey1 0:14:45)

> "I will be honest with you. I don't think I was paying as much attention to the suggestions of the tactical assistant as I could have been. Just purely because the tasks were often very multi-drone. I think I would have committed a lot of extra thought-space to assigning all of the drones individually to those. So I definitely inadvertently trusted the tactical [agent]. I think it's worth saying any assessment of the tactical system would probably be disingenuous just because of my lack of engagement with it. But the strategic assistant, I think on the most part, I have absolutely no problems with. There was that one issue, but that's okay because I noticed it because the display is straightforward and it's clear enough that you can as a human spot discrepancies if they arise."
>
> — P19 (talk during the questionnaire survey1 0:17:06)

> "i felt like i took a bigger pinch of salt for the strategic one because i felt like i could assess the quality of it more immediately whereas with the tactical assistant, I couldn't accurately and rapidly assess how well it was performing. So I just kind of said, OK, I have to take your word for it, essentially."
>
> — P22 (talk during the questionnaire survey2 0:26:53)

Nobody described careful checking under pressure. They described a quick look, and accepted the trade because penalty accrued while they checked; one participant set out the expected-value argument directly.

> *Researcher:* Am I correct in saying you should be using the agents a little bit more often now?
>
> "Whatever lack of optimality that comes from using the AI agents is more than compensated for in their speed. Or at least in their speed of decision making. I also feel like the longer that this session has gone on, the less oversight that I've been giving it, and the more I've been focusing on the strategic layer and the incoming missions."
>
> — P31 (think-aloud s2 0:20:58)
>
> *Researcher:* So you say you especially trust it with a large mission. You mean you're happy to rely on it because of the speed?
>
> "Yeah. I think even if the tactical agent gave me a dodgy suggestion and the mission failed and I lost points, I probably would have gained more points from the time saved"
>
> — P11 (debrief 0:29:11; includes the researcher's words)

Where trust was earned, it was by comparison with what the operator would have done themselves.

> *Researcher:* So I would start with just how did you find that? Anything immediately that you would comment on?

> "I think I developed a stronger opinion on how much to trust each of them. By, I guess, comparing them to what I would have done. And I mean, I understood the task more as I went. So the tactical one seemed to be doing what I was doing. So quite reliably. And obviously the strategic one wasn't. So I guess it was very much comparing them to what I'd do. And then letting the tactical one get on with it."
>
> — P32 (debrief 0:23:45)

> *Researcher:* The degree to which you trusted the decisions the AI was making, uh, and how that progressed throughout the study?
>
> "I mostly trusted it, although, uh, I did kind of want to test it more at the beginning, like comparing it, like doing a few manual ones myself to see, like, whether it would be better, faster. I guess it was kind of hard to keep track of which tasks were being finished when, but I noticed that the ones that failed were normally cause I'd just done a manual one. So I guess that made me want to use the AI more often."
>
> — P29 (debrief 0:25:59)

Options for this finding

Trusted because it cannot be checked [pass 5: for the paper]

Your note: A nice few quotes that we can use to make a point.

Checking shrinks to a glance [pass 5: extra: may make it in]

Your note: Less significant but might be worth a mention

Highlighted quotes not used above:

> *"Speed is the name of the game. So sometimes in these ones, I guess making a decision earlier is beneficial because at some point the other ones are going to pop up." — P22 (think-aloud s1 0:04:20, includes the researcher's words)*

Trust earned by comparison, and doubts that came with practice [pass 5: for the paper]

Your note: Minimal but nice

8.2 Who decides, and who answers for it

Almost everyone wanted a human to keep the decision, and the reasons were moral as much as practical: accountability for a life not saved, a human connected to the operation, and the plan held in someone's head. People were candid that the human they wanted in the loop might not be able to check much.

The clearest statement of the moral reason is about accountability: even if an AI saves more lives on average, when it fails to save one there is no one to answer for it. Against it sits the plain value of speed in an emergency.

> *Researcher:* But one of my questions is if in five years' time this was much more sophisticated, far more complicated, how would you feel about search and rescue scenarios being conducted with AI systems of this type in this kind of setup?
>
> "Yeah, I feel like that's a very interesting question for more of a moral reason, because if you look at it entirely from the perspective of your option A, option B, and option A saves more lives, that always seems like a good thing, but taking it ... in a slightly less simplified way, you've got to make sort of moral judgments about how these resources are being allocated, and if someone's life was like failed to be saved, under a completely human system, you'd have some sort of like, not responsibility, what's the word I'm looking for? [interviewer suggests the word 'accountability']. Accountability, thank you! Of who was on the job that failed to save your life, whereas if it's an AI system, even if it performs on average better than a human, that accountability just isn't there."
>
> — P31 (debrief 0:27:53; includes the researcher's words)

> *Researcher:* How do you feel about people using systems like this in, um, high risk scenarios?
>
> "I think definitely with emergency scenarios, it's best to just get something going, even if it's not the most perfect plan. And AI lets you do that quite good, like quite quickly."
>
> — P29 (debrief 0:27:05; includes the researcher's words)

Others wanted a human in the loop whatever that human could check, and one participant framed it organisationally: measure where people and assistants each add value, and assign the tiers accordingly.

> *Researcher:* If we took this five years in the future, and systems of this type were being used in real search and rescue scenarios with more complicated AIs and operators being assisted, how do you feel about that? And do you see any potential for good and for bad of doing that sort of thing?
>
> "I think if I were being organisationally strategic about that, it seems like it's really helpful. Assuming that you're human constrained (if you've got loads of spare humans, I'd want to see if the AI was actually better) but assuming that you've got less humans than you ideally would have, it feels like a good accelerator. I'd want the analytics on, I guess, like drone utilisation and mission success. I feel like that would be a good place to be making these decisions. And should I be, for example, in this case where I think the strategic one was less good than the tactical, if I were being really strategic, I'd be saying: 'okay, people are going to do the strategic stuff, and we're just going to trust it for the tactical'. So I guess very much doing that optimisation around: 'is it the agents or the humans who are adding the most value, and where do they add the most value', and then have them allocate them... I'd want to understand, well, just because I like to understand everything, I'd want to understand empirically actually how it was performing, who should be doing what."
>
> — P32 (debrief 0:26:10; includes the researcher's words)

For some the reason to keep the work was not safety but connection to the task, and pride in their own solution.

> "I don't want to be reduced to pressing allocate, suggest, deploy, because otherwise I might as well be in the next room. And I think the small missions give me the chance to do that more. And I'm like, 'oh, I can just do that. Don't worry. I'll do that'. Done. And I feel satisfaction."
>
> — P11 (debrief 0:30:55)

> "it would be a dreadful task without any of the tools, especially on the tactical side. It's just a lot of laborious dragging, holding shift at the right time.... And I guess I do feel I want to use (empahsis)my(/emphasis) solution, and I don't want to like, oh, you know, of course the computer thinks it's better. Or like personify the computer and feel some sort of grudge to hold it."
>
> — P11 (debrief 0:26:55; includes the researcher's words)

Options for this finding

Ownership, and the opposite view [pass 5: for the paper]

Your note: This is one of the better ones

8.3 Failure without feedback

Drone failures were random and uniform, and nothing on screen said so or said how likely they were. Operators felt the lack, built their own theories about where failures happened, and blamed themselves.

Several wanted failure rates built into the allocation, so that the reserve could be sized to the risk rather than padded.

> "I think in a in a real world scenario where if I imagine I'm a drone swarm operator you will have some numbers presumably about 'so what's the likelihood for drone failure' right? Based on historic data, and that will presumably bias (or inform, not bias) but inform your selection if I'm like okay the lifter drone is much heavier therefore there's more wear and tear on it or whatever it's more likely to fail down that means that my redundancy strategy might change from 'send one additional drone for the lifter' towards 'send 20 percent more more lifters than you need"
>
> — P10 (debrief 0:28:53; includes the researcher's words)

8.4 The hierarchy as one plan

Operators did not experience two independent assistants. They described one plan that starts at the strategic step: allocate too little at the top and the shortage cascades down; allocate generously and there is only one sensible way to assign the drones; build the allocation yourself and the tactical plan is already in your head.

Several placed the responsibility for failures with the strategic tier, since only drones sent at the start can absorb them; others weighed the same step from the other side, holding drones back for missions not yet arrived.

> "Basically, it did everything. If failure rate of drones is unrelated to my assignment, so rate of failure as opposed to how many drones I send out, then all things considered, the tactical assistant is probably the best. The tactical assistant completed all the jobs and we didn't end up in any kind of a situation where we didn't have enough drones if it hadn't been for random drone failures. And the only way to remedy drone failures is to have sent enough drones at the beginning, which I think is the strategic assistant's prerogative."
>
> — P19 (talk during the questionnaire survey1 0:18:53)

> *Researcher:* You found that one more demanding?
>
> "I'm trying to be more defensive in the strategy, so I'm trying to reserve more and more drones in the base, rather than send them out to finish the task quickly, because I cannot predict when the next task will appear. So luckily, there's only one task is holding for a long time, only one there, and another problem is [that] such a choice will make the drones in the case becomes not sufficient for that case. So if I only send out one drone for one specific color, and it fails, then I have to abort that mission, and I only abort one of them, because that happened, and the rest is still able to finish, so I think this, such a deployment is still all right."
>
> — P28 (talk during the questionnaire survey2 0:28:05)

The link between the tiers in the logs (Section 7.3) was described from both sides: a generous strategy card makes a tactical suggestion the obvious next step, while a hand-built allocation already lays out a plan the operator does not want disrupted. Whether one plan held across both tiers is the right ideal is a question the practitioner's account takes up, and answers differently (Section 9.1).

> "it's better to use the autonomy on the tactical level on top of the strategic"
>
> — P10 (debrief 0:25:46; includes the researcher's words)

> "[Having manually set a strategic allocation] I prefer to [tactically] set the drones myself. Because I'm not sure how exactly it will be allocated [if I use the agent]. Because I have something in mind when I consider the allocation... When I make the selection for the drones, I kind of lay out a plan in my mind initially. So I feel like it's going to disrupt my thought process if I try to go around that."
>
> — P21 (think-aloud s1 0:14:16; includes the researcher's words)

> "Actually, they might work better if I use the AI agent for the selection and that as well, because. Yeah, because when I assume when it makes a selection, when it tries to make the drone allocation, it follows certain patterns, which could then be adopted by the tactical agent afterwards. And if I try to think of something else, so I'm going to split them like that and change them in a certain way. I'm not sure if the tactical agent will be able to match my thought process."
>
> — P21 (talk during the questionnaire between1 0:20:36)

A strategy card considers one mission. Operators overrode the cards to act on what the cards could not see: keeping drones back for other missions, and weighing which missions could afford to wait.

> *Researcher:* Could you just summarize again, just in a sentence or two, how you were approaching each one, what scenarios you felt you were using each one?
>
> "I felt like sometimes I could find like a middle ground that had like maybe a little bit of gambling in there, essentially, that you wouldn't have the necessary coverage. But then also then that meant that you'd keep some more in reserve. So it wasn't just fully because it's cautious, it was also because it was actually to keep some in reserve for other tasks, because, yes, you're covering this task specifically very well. But that doesn't mean that then you actually have enough coverage for the, for the rest of your tasks. So I felt that often I could find, I could strike a middle ground. So I felt comfortable assessing and then overriding the strategic assistant."

— P22 (debrief 0:28:28)

> *Researcher:* Now that I've had a little bit more time, I'm just thinking if my opinions have changed on this slightly. And for what reason?
>
> "If you wanted to send as few drones as possible, you could send four fast drones, one lifter, one camera. Whereas even the conservative recommendation recommended sending two of the lifters, and the cameras in case of drone failure. When I was in that situation, I had a lot of other pending tasks, so I went more conservative than the conservative recommendation."
>
> — P31 (talk during the questionnaire survey2 0:23:49; includes the researcher's words)

Options for this finding

The cross-mission view the cards lack

Highlighted quotes not used above:

> *"And the problem is we don't know whether a extremely critical job is going to show up later." — P19 (think-aloud s1 0:09:07)*

8.5 An assistant for what to open next

Asked what the assistants should do, participants rarely asked for more automation of the same kind. Two asked for something neither assistant does: help deciding which waiting mission to open next.

Neither assistant helps with the first decision of every allocation, which waiting mission to open. With two or more missions waiting (411 decisions), operators opened the oldest well above chance (49%, against 34% by chance) and leaned towards the more critical (odds $\times 1.45$ per category above the waiting missions' average, $p = .02$), but not towards the missions they could actually cover: whether the reserve could supply a mission's drones made no difference ($\times 1.04$, $p = .79$), and at 41% of these decisions the oldest waiting mission could not be covered. An ordering of the kind one participant proposed (missions that can be done first, then by criticality) would have picked a different mission at 59% of decisions.

Options for this finding

An assistant for what to open next [pass 5: for the paper]

Key numbers:

- Decisions with 2+ missions waiting: 411 (33 people). Opened the oldest 49% (chance 34%); when one waiting mission was uniquely the most critical, opened it 49% (chance 33%)
- Choice among the waiting missions (conditional logit, person bootstrap, 400 draws): oldest ×2.75 [1.97, 4.12], p < .001; newest ×1.64 [1.17, 2.32], p < .001; criticality, per category above the set's mean ×1.45 [1.09, 2.15], p = .02; coverable from the reserve ×1.04 [0.72, 1.58], p = .79; size per 3 drones ×0.95 [0.78, 1.11], p = .61
- The oldest waiting mission could not be covered from the reserve at 41% of these decisions; some waiting mission could not be covered at 57%. When some could not, the one opened was coverable 32% of the time (chance 39%)
- P27's order (coverable, then most critical, then oldest) picks a different mission from oldest-first at 59% of decisions; there operators opened P27's pick 35% of the time (chance 30%)

9 A Practitioner's Reading

[Generated (`practitioner.tex`). One participant with operational military experience, one point per passage you picked, each with its links back to the results above (agrees / qualifies / goes beyond) as cross-references; the last subsection gathers every link by kind. In the paper, describe them only as "a participant with operational military experience".]

9.1 A practitioner's reading

One participant had operational military experience. Their play was typical of the cohort; they differed in who they are. Their sessions and debrief are the richest material in the study, and nearly every result above has a passage of theirs that agrees with it, qualifies it or goes beyond it. This is one person's expert commentary, not evidence about the cohort. Each point below gives the passage, as trimmed, and the results it bears on.

How typical they were. Our practitioner was a good performer (0.7 SD above the game average, the 80th percentile) with a slightly higher work rate than most: quicker on the strategic step (9.6 s per decision against a median of 11.4 s) and more often deciding before the strategy cards had loaded (32% of decisions against a median of 10%). On every other habit the findings measure they were typical, sitting in the middle half of the cohort (Figure 31). Where they differed was in who they are: the oldest participant (44; median 27), the only one with extensive command experience, and at the top of the attitude-to-AI scale.

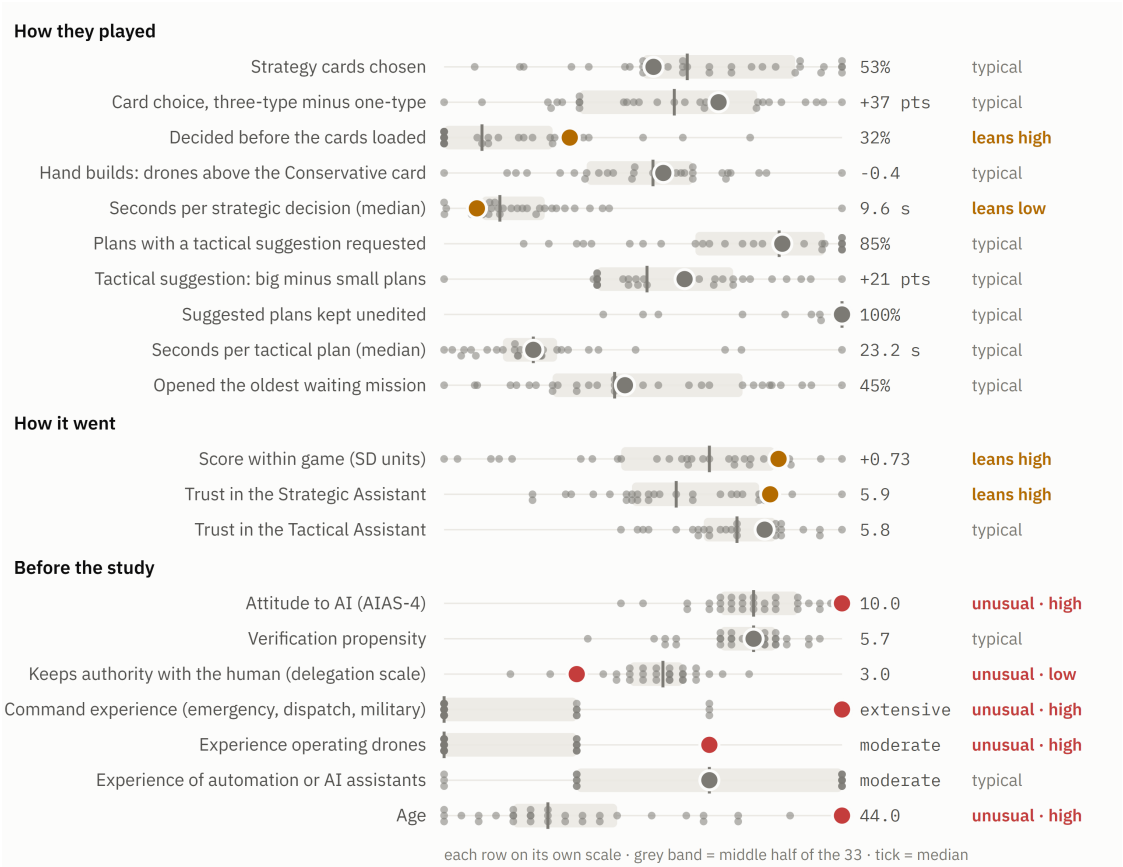


Figure 31: Typical play, unusual background. One row per measure, each on its own scale: grey dots are the 33 participants, the shaded band the middle half, the tick the median and the large dot our practitioner.

For the whole board, not the mission. The practitioner played for the whole board rather than the mission in front of them: the cross-mission view that other operators said the strategy cards lack (Section 8.4).

> "so when I think it's going to be quicker because it's not necessarily about just that task it's about all of the tasks and success is generated across all of the tasks rather than just an individual task."
>
> — P36 (think-aloud s1 0:05:58)

Your comment on this quote: Interesting

Bears on:

- *Agrees with* Section 8.4, *The cross-mission view the cards lack*: Operators overrode the cards to act on what one mission's card cannot see. P36 says why in one line: "success is generated across all of the tasks rather than just an individual task".
- *Agrees with* Section 7.4, *Each manual decision against the Conservative card*: Playing for every task shows in the cohort's hand builds: on 44% of decisions they sit below both cards, mostly by leaving off the card's spare, which keeps drones for the next mission.

The card can be slower than doing it yourself. Three minutes into the first session, after choosing a strategy card on a small mission, they decided it would have been quicker by hand: the speed result (Section 7.3) in miniature.

> "Do you know what? Probably quicker just to hit three on there. But we'll go three on that one there. [. . .] I probably should have done that one manually. It would have been quicker."
>
> — P36 (think-aloud s1 0:03:21)

Bears on:

- *Agrees with* Section 7.3, *Why the rule makes sense: the card pays only on big strategic jobs; a tactical suggestion is not slower*: The card saves time only on big three-type jobs. Three minutes into their first session P36 finds it out: "Probably quicker just to hit three", then "I probably should have done that one manually. It would have been quicker."

Simple by hand, complex to the assistant, and why. They gave the fullest statement of the size rule in the study, with both halves of the reason: on a simple mission they already know the answer by eye and build in redundancy, so a failure needs no re-plan; on a complex one the options branch, so they let the assistant start and step in when needed. This points to something the study did not measure: the more complex the mission, the more branching its space of possible failures, and the more a plan built in advance must change when one occurs. They also named the limit of the quick check that goes with handing over.

> "sometimes it's going to be a little bit uh quicker to do these myself only because i know the answer based on what i'm seeing visually in front of me so i don't need to do that much [...] simple task you probably want to do yourself if you're going to get the solution quicker and the more complex tasks you're going to want somebody to give you that solution in front of you so you can give it a quick once over and go that looks about right um whether or not it is right of course it's another question entirely [...] the solutions it is coming up with uh i would have to go and do a much more thorough scan of the solution to say have you put redundancy into that plan or do i care like i've put redundancy into that plan so i don't have to come back to it i've put every turn on that last blue one every single drone is going to every single task so that if one of my drones fails my assumption is that my other drones are picking up the task whereas the more complex ones just literally looking have they allocated enough to that task started going because if it fails um think about like hypertextual branching paths so as soon as i get one failure that's going to change the entire story how much of that story do we need to change on the blue ones you don't really need to change the story you're just going to flick through whereas the other ones it's like okay no i've got more options so my option scope is broader and so it's going to need more human intervention to go into it so there's no point in me trying to think about all of that ahead of time i may as well let them start on that task and then intervene when i need to"
>
> — P36 (think-aloud s1 0:09:12; includes the researcher's words)

Bears on:

- *Agrees with* Section 7.3, *Both tiers on one axis: the bigger the mission, the more it is handed over*: Small jobs by hand, big jobs to the assistant (cohort: card 56% to 75%, Suggest 54% to 91%). P36 gives both halves of the reason: on a simple job "i know the answer based on what i'm seeing visually", and a failure needs no re-plan ("you don't really need to change the story"); on

a complex one "my option scope is broader", so "let them start on that task and then intervene when i need to". Their own log follows it (card 33% to 70%, Suggest 79% to 100%).

- *Agrees with* Section 7.9, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: "Give it a quick once over and go that looks about right ... whether or not it is right of course it's another question entirely": the brief check the timing data show, with its limit named.
- *Agrees with* Section 8.4, *One plan across both decisions*: P21 chained drones through each other's tasks for "redundancy for free". P36 does the same on simple missions ("every single drone is going to every single task so that if one of my drones fails ... my other drones are picking up the task"), which is why those plans need no checking afterwards.

The rule bends when the reserve is full. On a simple mission, with the reserve full and nothing else coming in, they requested a tactical suggestion anyway: the size rule is a default, and slack loosens it.

> "so again this one here that's relatively simple and I'm resource rich at the moment so we'll go nuts I'll start suggesting some stuff through because there's nothing else that's coming in"
>
> — P36 (think-aloud s2 0:20:33)

Bears on:

- *Qualifies* Section 7.3, *Both tiers on one axis: the bigger the mission, the more it is handed over*: The size rule bends with slack: this mission is "relatively simple", but "I'm resource rich at the moment so we'll go nuts I'll start suggesting".
- *Agrees with* Section 7.6, *Queue depth against reliance — raw, and adjusted*: A busier moment does not push people onto the assistants. P36 requests a tactical suggestion at a quiet moment, "because there's nothing else that's coming in".

Never without a reserve. They stated, and named, their own rule for hand-built allocations, "never deploy without a reserve", alongside opening the mission that has waited longest. A named rule of this kind is one way the consistency of individual styles (Section 7.2) could be built through training.

> "this one's been there for the longest i've got nothing in there that's going to change my prioritization so i'm going to go into here and probably send manually i'm going to give you minimum two we're going to go three just never deploy without a reserve"
>
> — P36 (think-aloud s1 0:07:50)

Bears on:

- *Agrees with* Section 7.7, *Order: still oldest-first well above chance*: Operators open the oldest waiting mission well above chance. P36: "this one's been there for the longest i've got nothing in there that's going to change my prioritization".
- *Qualifies* Section 7.4, *Each manual decision against the Conservative card*: Most of the cohort's hand builds sit at or below the cards (44% below both). P36 adds a spare on purpose ("minimum two we're going to go three just never deploy without a reserve"), yet their builds land where the cohort's do: on average 0.4 of a drone under the Conservative card, against a median of 0.6 under.

Critical tasks over score. Out of Fast drones, they accepted a lower score to finish the critical tasks, framing it as what a real operation would demand. The cohort leans the same way when choosing what to open next (Section 8.5).

> "i've run out of blue so that's gonna hurt my score but at the same time i've got the critical tasks done so i'd rather do the critical tasks if this is a real life scenario i'd rather be focused on the critical tasks than getting those done to a higher value rather than moderate tasks of getting them done to a low"
>
> — P36 (think-aloud s1 0:07:50)

Bears on:

- *Agrees with* Section 8.5, *An assistant for what to open next*: Operators lean towards the more critical waiting mission ($\times 1.45$ per category, on top of oldest-first). P36 goes further and spends score on it: "i've run out of blue so that's gonna hurt my score but ... i'd rather do the critical tasks if this is a real life scenario".

Easy to check is not the same as low stakes. One of the few who trusted the Strategic Assistant more than the Tactical, they explained it by how easy a strategy card is to check, then reversed themselves mid-sentence: a bad allocation at the start leaves the tactical step with no way back.

> "the strategic one i was less concerned about the solutions it presented so i hesitated less on the strategic than the tactical but only because all i was looking at there was have i got a reserve allocated whereas a tactical one it's 'have you covered all the tasks' and there is a reserve [...] I wouldn't trust any of them without checking but I trust the strategic slightly more because the consequences are slightly - less they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical"
>
> — P36 (talk during the questionnaire survey1 0:13:46)

Bears on:

- *Qualifies* Section 7.9, *Trusted more: the basic result*: The cohort trusts the Tactical Assistant more; P36 is one of the exceptions, and their ratings match what they say ("I trust the strategic slightly more": 5.9 against 5.8 on the 1–7 scale, where the cohort's median strategic rating is 4.5). They hesitated less over the cards only because "all i was looking at there was have i got a reserve allocated".
- *Agrees with* Section 8.4, *The strategic step carries the risk*: P19 put the only remedy for failures with the strategic tier. P36 reaches the same point in mid-sentence, and it reverses their reason for trusting the cards: "they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical".

Physics sets the pace. The pace of the game is not realistic: in a real operation physics bounds the pace, so decisions get faster but the operation does not, and the operator's load moves to waiting and contingency planning.

> *Researcher:* so I would ask in your professional opinion obviously this is clearly we're trying to get at something that we think is of interest in this kind of setting and stuff in reality I'd imagine the strategic and tactical thing would probably be done by two people but how relevant would you say this kind of set up is
>
> "so in terms of delivery absolutely in terms of the pace probably not so much [...]if you think so the biggest thing for us that is always hard to convey to decision makers is: physics remains the bounding factor I've got to get from this point here to that point there and if the adversary can see me from 20k away then I'm doing that activity from 20k and if you're going at 3k an hour then that's going to take you a long time to get from here to over there so there's a level of that acceleration in terms of decision however on the flip side of it when the shit hits the fan you are probably going to be very frustrated; As I move this thing to here I've now got to wait for this thing to get into place to go and do this thing here hurry up and get there but at the same time as a commander you're sat there thinking okay what am I going to do if this thing fails what am I then going to do what am I going to do so your mental load is still loaded it's just loaded in a slightly different way so you're always trying to balance cognitive capacity against the time pressure effectively"
>
> — P36 (debrief 0:26:31)

Bears on:

- *Qualifies* Section 7.6, *Queue depth against reliance — raw, and adjusted*: A busier queue did not push anyone onto the assistants. P36 puts the real pressure elsewhere: "physics remains the bounding factor", and the load becomes waiting and asking "what am I going to do if this thing fails". "Your mental load is still loaded it's just loaded in a slightly different way."
- *Agrees with* Section 7.3, *Where the tactical time goes: hand-built and suggested plans are close, with a small edge to Suggest in carrying the plan out*: The time a tactical suggestion frees is spent elsewhere while a plan runs. P36 describes that dead time from the commander's side: "I've now got to wait for this thing to get into place", and uses it to plan for failure.
- *Agrees with* Section 7.10, *It's a game: what real stakes, pace and training would change*: Others said the pace was unreal; P36 agrees ("in terms of the pace probably not so much") and says what sets the real pace: "if you're going at 3k an hour then that's going to take you a long time".

The pressure lands on the lower tier. In a real hierarchy the time pressure falls on the lower tier: higher up, a new item is routine; lower down, everything arrives at once and has to be sequenced.

> *Researcher:* and I suppose I think it would be fair to say that a lot of traditional ways this kind of stuff is working I know people do incredibly long shifts and we're talking about watchkeeping over a wide area with increasing use of systems like this in a more

> I suppose the pace you can talk about that the distance context those distances are scaled down quite a lot within a tactical kind of small scale
>
> "so it's how you're talking about when does the effect need to be delivered and coordinated in time and space then it's the higher up you are the further out and then the less so when you see something, 'cool I've now got to pass that information down but that's just like a routine routine task I'm going to move on to the next task' whereas if I see 10 things in front of me as it's all come down the chain I've got to do this thing now then that thing then that thing then that thing and then for it to stack back up the chain so yes there is a level of that is getting both faster"
>
> — P36 (debrief 0:28:21)

Bears on:

- *Qualifies* Section C.3.2, *Said vs did: the people who said they hand over when busy*: Nine people said they lean on the assistant when busy, yet their reliance did not rise with the queue. P36 puts the busy in a real hierarchy: "the higher up you are the further out", where a new item is "just like a routine routine task"; at the lower tier "10 things in front of me as it's all come down the chain". The game gave one person both tiers.
- *Goes beyond* Section 8.5, *An assistant for what to open next*: P27 and P32 asked for an assistant that says what to open next. P36's lower tier is where that job would sit: "I've got to do this thing now then that thing then that thing then that thing".

Trust in the process, as with staff. Above battlegroup level, which the participant mapped onto the study's strategic tier, a commander takes plans worked up by staff, and trusts them because they have done those jobs themselves and know how the answers are made. No operator has done the assistant's job, so that basis for trust is missing.

> *Researcher:* speaking specifically about the relationship between you and these two assistants these being autonomous systems yeah obviously that's not a million miles away from that being a human assistant who's helping you there that it's gonna be you and two people right do you have any feeling about the the difference there you spoke a lot about socializing socializing the decision-maker something like that how does that compare do you think to if you had two subordinate people here
>
> "well it's kind of this how it works in military planning anyway so in anything above battlegroup level [our 'strategic'] there would be a staff there that would work up plans socialise those plans amongst themselves and back brief me as a commander. I would then go okay yeah I want you a bit of that plan and a bit of that plan give me this new plan. They'll go and write up by then sort of basically finish off that way. So it's a it is always a collective input into it but I'm just getting the synthesis of them work because I can't as an individual I couldn't work through the ground the enemy, the enemy's plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it, so at that level you're already in a position where somebody's just coming back to you with the answers as long as you have trust in how they've come up with those answers then if it looks right: because you've gone through the process yourself [...] you'd have worked up in those levels so you kind of understand how those answers are generated so there's a level of trust in that process"
>
> — P36 (debrief 0:35:27; includes the researcher's words)

Bears on:

- *Agrees with* Section 7.9, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: Plans are accepted after a brief look. P36 explains why that can be rational: answers from staff are taken "as long as you have trust in how they've come up with those answers", not re-derived.
- *Agrees with* Section 7.10, *Wanting to know how the assistants decide*: Several operators wanted to know how the assistants decide. For P36 that is the basis of trust: a commander can take the staff's answers because "you'd have worked up in those levels so you kind of understand how those answers are generated". No operator has done the assistant's job.
- *Agrees with* Section 7.3, *Why the rule makes sense: the card pays only on big strategic jobs; a tactical suggestion is not slower*: The card pays where a job is too big to do quickly by hand. P36 gives the same reason for having staff at all: "I couldn't work through the ground the enemy, the enemy's plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it".

Verify by socialising the plan. Verification, for the practitioner, is a process around the plan, socialising it with those who will carry it out and keeping it open to challenge, together with trust that the machine has done the maths.

Researcher: so you thought they say I might be worse

> "I write a plan I will go and take it to the lowest tactical level which is something like this where you're moving different units around different people you'd probably speak to the platoon commander you'd speak to the platoon sergeant: 'I'm thinking of doing this', you'd bring your section commanders in you'd go through yours and that's the process 'guys this is what I'm thinking and this is what your likely mission is, go away and work out the plan'. [...] As humans we should be used to socialising our decision making process to an extent. How you do that is entirely up to the commander but at the same time I'm still going through a process where somebody is checking my thinking and has the option to challenge me so with AI it should always be open to challenge [...] but also trust in the fact it's done the maths behind it and has allocated enough platforms to do the task you're after"
>
> — P36 (think-aloud s2 0:17:40; includes the researcher's words)

Bears on:

- *Qualifies* Section 7.9, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: This is from someone who kept every consulted tactical plan unedited. In P36's account verifying happens around the plan, not in a long look at it: take it to the people who will carry it out ("go away and work out the plan") and keep it "open to challenge". Beside that sits "trust in the fact it's done the maths behind it and has allocated enough platforms", which is about all the cohort's 2–4 seconds of checking can cover.
- *Goes beyond* Section 8.2, *A human in the loop, even one who cannot check*: Some operators want a human in the loop even one who cannot check. P36's check is not one person looking harder but a plan socialised so that "somebody is checking my thinking and has the option to challenge me".
- *Agrees with* Section 7.10, *Wanting to know how the assistants decide*: "With AI it should always be open to challenge": the same ask as the operators who wanted to know how the assistants decide.

Hand over the routine, keep the critical. Handing over lets them work elsewhere, but it has a price: a plan they did not build is not in their head when it fails. For them the key is therefore not trusting the AI but the scarce skill of exploiting it.

Researcher: so the last scenario you said something about sort of speed accuracy and the main advantage of the agents being speed

> "If I've sent that off it allows me to scan over here and then go back and check the answer over there. [...] That's what's going to be required if you're doing a drone swarm the whole point of it is to make sure that you have got, but the utility of it is that point of presence that point of intervention where I need something to check that and know I'm going to take over that critical thing myself because it's the most important thing whereas for the routine sort of activity of getting from A to B do you know what whatever you want to do as long as the resources are in place and I can see what resource I'll have to play with if something goes wrong whereas if I did all of that manually I would know all of that because I've built the plan I've got it in my head if I just said 'go nuts' when something goes wrong I've got to play catch up I've got to start thinking how do I now have to do that differently so the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this [...] that's what air traffic controllers do and they're actually testing it for it and their pay reflects that because it's not a particularly common skill if you're going to start saying search and rescue teams you now need this skill set it's going to cost you this much to pay these people because they're a niche skill set"
>
> — P36 (think-aloud s2 0:20:56)

Bears on:

- *Agrees with* Section 7.3, *Where the tactical time goes: hand-built and suggested plans are close, with a small edge to Suggest in carrying the plan out*: Handing over frees the operator to work elsewhere: "If I've sent that off it allows me to scan over here and then go back and check the answer over there", which is where the cohort's Suggest time goes.
- *Agrees with* Section 7.9, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: Brief checking, whatever the plan size. P36 names its price: "if I just said 'go nuts' when something goes wrong I've got to play catch up", because the plan is not in your head.
- *Goes beyond* Section 7.2, *Who decides matters more than what is being decided*: Who decides predicts handover far better than the mission does. P36 draws the operational conclusion: "the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this", a niche skill set like air traffic control, paid accordingly.

The failure rate of doing nothing is 100%. Failure rates should be shown with context, or people will refuse any risk: doing nothing fails every time.

> *Researcher:* you would appreciate having a sort of a prior estimation of the failure chance and of what's coming in kind of thing
>
> "but the interesting thing so that that always messes with people's heads doesn't it because we look at the failure 40% failure chance fucking no chance it's like okay but if you put a person into that scenario and it was all people doing it your failure chance would be this as well so there's got to be something where it shows a bit of context to it so just to understand that when you see that failure rate because everyone will try to push it up and up and up and up and up no one likes gambling particularly gambling in people's lives but particularly in the military environment or even in the search and rescue environment you're always gambling with someone's life like the failure rate of doing nothing is 100% so you've got to do something you're just trying to chip away at it and prioritise accordingly so having that information presented again great to see the information but it's then down to that person that you're training for them to be used to there's always going to be a failure rate I'm just trying to make it as low as reasonable"
>
> — P36 (debrief 0:32:25)

Bears on:

- *Qualifies* Section 7.10, *Failure risk and feedback: wanting the numbers*: Several operators asked for failure numbers. P36 wants them with context ("there's got to be something where it shows a bit of context to it"), because a bare "40% failure chance" reads as "no chance" when "the failure rate of doing nothing is 100%".
- *Agrees with* Section 8.3, *Blind decisions, and failure data built into the plan*: Operators facing random failures with no numbers felt they were deciding blind. P36 agrees the numbers should be shown ("great to see the information") and adds that people then need training to act on them: "there's always going to be a failure rate I'm just trying to make it as low as reasonable".
- *Goes beyond* (not in the paper), *Lower-reliance operators attempt fewer recoveries*: Lower-reliance operators attempted fewer recoveries. P36's rule cuts the other way: in search and rescue "you're always gambling with someone's life", so you act and "chip away at it".

Real stakes and shared risk. Their own decisiveness came from experience and from knowing it was a simulation. With real lives at stake an inexperienced commander might be reckless or paralysed, and remote autonomy removes the shared risk that used to temper an order.

> *Researcher:* I noticed that you're you were very decisive you were you didn't really wait around you just you just did stuff
>
> "yes well and that's kind of a challenge within this is because I know it's a simulation there isn't someone's life on the line at the end of it and I've done enough planning activities to know and I'm very visual so looking at a glance yes it's allocated what I needed to allocate it cool let's start running with it whereas I can imagine if you've not had that experience that I've generated through non-use of drone swarming technology if you've not had that if you're you know young platoon commander who's literally just come out of Sandhurst and all is only a very experienced perhaps this is a simulation at Sandhurst and all of a sudden it's now you know the bullets are flying it's a two-way range they're gonna be much either they're gonna be foolhardy and just fucking go for it and people are gonna die or they'll be almost paralysed because it's like oh shit no if I do that someone is gonna die here's 40% failure rate on this task I'm gonna lose people but if I don't do that task this whole battle falls apart that's always been true regardless of whether or not technology was in there but before there was a level of interpersonal sort of contact I'm out there with my people I might die alongside them but I'm so I'm when I'm putting them under that risk I'm because I'm taking the same risk myself whereas something like this where you're not necessarily taking the risk yourself and you've put humans albeit they might be one removed from your autonomy but you've put humans on the way then that's going to then come back and impact on your decision-making so I imagine you're going to see me less decisive if someone's gonna die at the end"
>
> — P36 (debrief 0:33:43)

Bears on:

- *Goes beyond* Section 7.10, *It's a game: what real stakes, pace and training would change*: "It's a game" was a common caveat. P36 says what changes with a life at stake ("I imagine you're going to see me less decisive if someone's gonna die at the end"), and that a commander who no longer shares the risk ("I might die alongside them") loses what used to temper the order.
- *Qualifies* Section 7.2, *Nothing measured beforehand predicts the style*: Prior experience did not predict style here. P36 expects it to matter most when the stakes are real: their own "looking at a glance yes it's allocated what I needed ... cool let's start

running with it" comes from experience, where a new platoon commander would be "foolhardy" or "almost paralysed". A game with no lives on the line hides that difference.

The top allocation sets the risk below. Allocate too little at the top and the lower tier carries the risk; allocate too much and it is wasted elsewhere.

> *Researcher:* one thing that you did clearly is you very much immediately were happy to sort of automate the low level of detail the tactical level and spend more of your attention on the high level which makes a lot of sense to me and that's how probably about half my participants have behaved half the participants have said well I want to automate the high level and then I can fix it
>
> "I would argue that if you've not got the resources available to you at the low level then or you've over allocated resources at the high level then you're running out of resources cascade down so you're it's a balance it's never perfectly weighted one way or the other so if you're let's say you split the task this dude here is doing this task and all of a sudden has got no resource left look at the task I need more resource I told you need more resource when I told you this is my plan and this is I've got no redundancy here you're carrying risk whereas if you've done both it's your risk to carry on the flip side of it if you've over allocated a resource that's just wasted resource now I can't complete you might want more resource but I've got five tasks that I need to do so I'm balancing these five do what you can with what you've got let me know where it all falls down because it doesn't matter if yours falls down because yours is a moderate task I've got a critical one over here that I'm allocating more to"
>
> — P36 (debrief 0:29:43)

Bears on:

- *Agrees with* Section 7.3, *Tactical tier: tactical suggestions rise with mission size, and goes with the strategic choice*: Handing over at one tier goes with handing over at the other (Suggest $\times 2.1$ after a card, same person, same size). P36 explains the coupling: the top allocation sets the lower tier's risk ("I told you need more resource when I told you this is my plan ... you're carrying risk"). Their own card-then-Suggest gap was close to zero, so the link is the cohort's, not theirs.
- *Agrees with* Section 7.4, *Each manual decision against the Conservative card*: Over-allocating at the top: "that's just wasted resource". In their own builds the cohort mostly trimmed the card's spare drone.
- *Agrees with* Section 8.4, *The cross-mission view the cards lack*: Operators overrode the cards to weigh other missions. P36 says it as a commander: "I've got five tasks that I need to do so I'm balancing these five", and "I've got a critical one over here that I'm allocating more to".

Narrow context is normal. Narrow context is normal in staff planning, and fine as long as each part has been through a sound analytical process. This sits directly against the operators who wanted one plan held across both tiers (Section 8.4).

> *Researcher:* both of them are myopic in terms of what they see so they see only the current thing so the strategic agent all it sees is the list of drones and it has a simple rule that says one drone for each dot plus one of each or something like that the fact that they see only that context how would you say that compares to the scenario you're talking about with humans planning across multiple layers this might be a six or seven layer command structure
>
> "You kind of work in your own silo so if you're the g2'er [intelligence branch of general's staff planning team] you just focus on the enemy and you're delivering what the enemy is likely to do and you have a team working to you to do that so as a g2 officer you'll have like a one person will work on the enemy capability, one person will look at on the ground, one person will look on the sort of the human terrain and they'll work through their own analytical processes to come up with an answer then gets aggregated up and you go into it so going into that planning brief you as a g2'er are will only see your part of the plan and then the g3 [operations branch] guys will see their part of the plan and then that's kind of the first time you come together but generally speaking you're you're not getting all of the context okay so but you don't that you need that because as a person you need to be able to work through the context solution part of it yeah and this type of scenario here you wouldn't need that because of the simplicity of the task it's i've got to go and do these things and for those things i need these allocations so it's hard to sort of translate across the two as long as in each of them it's gone through the analytical process and spat out an answer"
>
> — P36 (debrief 0:38:36)

Bears on:

- *Qualifies* Section 8.4, *One plan across both decisions*: Several operators wanted one plan held in mind across the tiers. P36 sees narrow context as normal ("you as a g2'er are will only see your part of the plan") and fine "as long as in each of them it's gone through the analytical process", and adds that this game's tasks are simple enough not to need more: a direct tension with those operators.
- *Goes beyond* Section 7.10, *How the two tiers fit together*: Operators wondered how the two assistants fit together. P36's model is staff cells that each see their part and first "come together" at the planning brief.

One up, two down. Each level should be given the part of the level above that its task needs, and no more; an agent that reaches past its remit gives worse answers.

> *Researcher:* there's a philosophy of understanding what's going on one level above you and one level below you to have that or one or two levels above you or something for the context right so you you get told to go and i don't know um recce that road you need to know why you're wrecking that road so there's something above right and the idea that actually you need to have each of your agents in this in this scenario understanding the context a little above them i understand what you're saying that you don't want them to be thinking about the super wide context but is there an advantage to that linking up of understanding what's a little way up above you
>
> "so it depends yeah so you could have your agent that's checking the colors might be the one that understands two up [the intent of those above you, and one about that] so you understand two up in the same way as you plan two down issue orders one down so you think about i'm going to be sure to report to my section commanders but my i'm thinking about what each of those individual soldiers doing with that scenario and that might be your agent that corrals all your thing but in a when we go through that staff process the person who's just doing a scenario they will understand the one up they will understand or part of the one up that's given to them as their task is there's an area of responsibility there's an area of interest there's a thing that the enemy is doing within this space i just need to kind of understand that that so it might be you might understand that your mission that's come to you within your organisation so that's technically your organisation's mission but you might consider that your one up so it's depending on on as long as the person doing the task is given the understanding they're required to do that task then it doesn't matter how it docks into everything else that's not their job that's the person above them to do that so for your agents yes it might be useful but no it might not be useful it depends and particularly it's not useful if they're giving shit answers like if they're coming out and going no i've allocated five over here because there's a thing over there it's like cool that's not your job your job is to do this thing over here and then there's other agents to do this thing over here and they don't need them to do both because that's where this agent is going to do both for them and that's quite often even humans doing it humans start thinking or g2 again they'll start thinking blue plan but your job is not to do the blue plan your job is to do the red plan stop thinking about or the blue is probably going to do this so i'm going to do this no no but i need to know that sir because i need to understand the context of what the red is going to do so you need to have an idea of what blue capabilities have but you don't have control over the blue plan and you're not there to anticipate the blue plan you're there to put it within this context just stick to your lane and do that output because someone else will do that for you"
>
> — P36 (debrief 0:42:47)

Bears on:

- *Goes beyond* Section 7.10, *How the two tiers fit together*: How should the tiers fit? From doctrine: "you understand two up in the same way as you plan two down", each task comes with the part of the level above that it needs, and an agent reaching outside its lane is told "that's not your job".
- *Qualifies* Section 8.4, *One plan across both decisions*: The operators who wanted one shared plan are answered by a different design: each level gets "the understanding they're required to do that task", and "it doesn't matter how it docks into everything else".

An assistant that red-teams the others. An assistant whose only job is to check the data the others' answers rest on would catch a bad input before the answers built on it, without relying on the operator to notice (compare Section 7.8).

> "One of the good things about ai is that you can have an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly so if you've got that there is that hole in that data set and it go through and start all your job is to do is to check the data that they've based their answers on and then you can't go well this data is wrong so all these answers are now redundant we now need to look at this change this and then the answers will be right"

— P36 (debrief 0:39:26)

Bears on:

- *Goes beyond* Section C.4.1, *Strategic flaws are visible; tactical flaws are hidden*: The tactical flaw (a wrong "No workable plan") went almost unseen: only 4 people reacted to one aloud. P36's answer does not rely on the operator: "an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly", checking "the data that they've based their answers on".
- *Agrees with* (not in the paper), *Assistants that check rather than decide*: P11 wanted an assistant that checks their own plan. P36's checker tests the other agents' data instead, so a bad input shows up before the answers built on it.

Read against the results. The links on the points above, gathered by how each passage bears on the results. This could close the paper.

Agrees with the results (24).

- Section 7.3, from the point in Section 9.1: Their aim is throughput across the board ("get as many across as quickly as I can to complete all of the tasks"), the same reason the cohort gives for starting Suggest and moving on.
- Section 8.4, from the point in Section 9.1: Operators overrode the cards to act on what one mission's card cannot see. P36 says why in one line: "success is generated across all of the tasks rather than just an individual task".
- Section 7.4, from the point in Section 9.1: Playing for every task shows in the cohort's hand builds: on 44% of decisions they sit below both cards, mostly by leaving off the card's spare, which keeps drones for the next mission.
- Section 7.3, from the point in Section 9.1: The card saves time only on big three-type jobs. Three minutes into their first session P36 finds it out: "Probably quicker just to hit three", then "I probably should have done that one manually. It would have been quicker."
- Section 7.3, from the point in Section 9.1: Small jobs by hand, big jobs to the assistant (cohort: card 56% to 75%, Suggest 54% to 91%). P36 gives both halves of the reason: on a simple job "i know the answer based on what i'm seeing visually", and a failure needs no re-plan ("you don't really need to change the story"); on a complex one "my option scope is broader", so "let them start on that task and then intervene when i need to". Their own log follows it (card 33% to 70%, Suggest 79% to 100%).
- Section 7.9, from the point in Section 9.1: "Give it a quick once over and go that looks about right ... whether or not it is right of course it's another question entirely": the brief check the timing data show, with its limit named.
- Section 8.4, from the point in Section 9.1: P21 chained drones through each other's tasks for "redundancy for free". P36 does the same on simple missions ("every single drone is going to every single task so that if one of my drones fails ... my other drones are picking up the task"), which is why those plans need no checking afterwards.
- Section 7.6, from the point in Section 9.1: A busier moment does not push people onto the assistants. P36 requests a tactical suggestion at a quiet moment, "because there's nothing else that's coming in".
- Section 7.7, from the point in Section 9.1: Operators open the oldest waiting mission well above chance. P36: "this one's been there for the longest i've got nothing in there that's going to change my prioritization".
- Section 8.5, from the point in Section 9.1: Operators lean towards the more critical waiting mission ($\times 1.45$ per category, on top of oldest-first). P36 goes further and spends score on it: "i've run out of blue so that's gonna hurt my score but ... i'd rather do the critical tasks if this is a real life scenario".
- Section 8.4, from the point in Section 9.1: P19 put the only remedy for failures with the strategic tier. P36 reaches the same point in mid-sentence, and it reverses their reason for trusting the cards: "they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical".
- Section 7.3, from the point in Section 9.1: The time a tactical suggestion frees is spent elsewhere while a plan runs. P36 describes that dead time from the commander's side: "I've now got to wait for this thing to get into place", and uses it to plan for failure.
- Section 7.10, from the point in Section 9.1: Others said the pace was unreal; P36 agrees ("in terms of the pace probably not so much") and says what sets the real pace: "if you're going at 3k an hour then that's going to take you a long time".

- Section 7.9, from the point in Section 9.1: Plans are accepted after a brief look. P36 explains why that can be rational: answers from staff are taken "as long as you have trust in how they've come up with those answers", not re-derived.
- Section 7.10, from the point in Section 9.1: Several operators wanted to know how the assistants decide. For P36 that is the basis of trust: a commander can take the staff's answers because "you'd have worked up in those levels so you kind of understand how those answers are generated". No operator has done the assistant's job.
- Section 7.3, from the point in Section 9.1: The card pays where a job is too big to do quickly by hand. P36 gives the same reason for having staff at all: "I couldn't work through the ground the enemy, the enemy's plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it".
- Section 7.10, from the point in Section 9.1: "With AI it should always be open to challenge": the same ask as the operators who wanted to know how the assistants decide.
- Section 7.3, from the point in Section 9.1: Handing over frees the operator to work elsewhere: "If I've sent that off it allows me to scan over here and then go back and check the answer over there", which is where the cohort's Suggest time goes.
- Section 7.9, from the point in Section 9.1: Brief checking, whatever the plan size. P36 names its price: "if I just said 'go nuts' when something goes wrong I've got to play catch up", because the plan is not in your head.
- Section 8.3, from the point in Section 9.1: Operators facing random failures with no numbers felt they were deciding blind. P36 agrees the numbers should be shown ("great to see the information") and adds that people then need training to act on them: "there's always going to be a failure rate I'm just trying to make it as low as reasonable".
- Section 7.3, from the point in Section 9.1: Handing over at one tier goes with handing over at the other (Suggest $\times 2.1$ after a card, same person, same size). P36 explains the coupling: the top allocation sets the lower tier's risk ("I told you need more resource when I told you this is my plan ... you're carrying risk"). Their own card-then-Suggest gap was close to zero, so the link is the cohort's, not theirs.
- Section 7.4, from the point in Section 9.1: Over-allocating at the top: "that's just wasted resource". In their own builds the cohort mostly trimmed the card's spare drone.
- Section 8.4, from the point in Section 9.1: Operators overrode the cards to weigh other missions. P36 says it as a commander: "I've got five tasks that I need to do so I'm balancing these five", and "I've got a critical one over here that I'm allocating more to".
- ("Assistants that check rather than decide", not in the paper), from the point in Section 9.1: P11 wanted an assistant that checks their own plan. P36's checker tests the other agents' data instead, so a bad input shows up before the answers built on it.

Qualifies the results (10).

- Section 7.3, from the point in Section 9.1: The size rule bends with slack: this mission is "relatively simple", but "I'm resource rich at the moment so we'll go nuts I'll start suggesting".
- Section 7.4, from the point in Section 9.1: Most of the cohort's hand builds sit at or below the cards (44% below both). P36 adds a spare on purpose ("minimum two we're going to go three just never deploy without a reserve"), yet their builds land where the cohort's do: on average 0.4 of a drone under the Conservative card, against a median of 0.6 under.
- Section 7.9, from the point in Section 9.1: The cohort trusts the Tactical Assistant more; P36 is one of the exceptions, and their ratings match what they say ("I trust the strategic slightly more": 5.9 against 5.8 on the 1–7 scale, where the cohort's median strategic rating is 4.5). They hesitated less over the cards only because "all i was looking at there was have i got a reserve allocated".
- Section 7.6, from the point in Section 9.1: A busier queue did not push anyone onto the assistants. P36 puts the real pressure elsewhere: "physics remains the bounding factor", and the load becomes waiting and asking "what am I going to do if this thing fails". "Your mental load is still loaded it's just loaded in a slightly different way."
- Section C.3.2, from the point in Section 9.1: Nine people said they lean on the assistant when busy, yet their reliance did not rise with the queue. P36 puts the busy in a real hierarchy: "the higher up you are the further out", where a new item is "just like a routine routine task"; at the lower tier "10 things in front of me as it's all come down the chain". The game gave one person both tiers.
- Section 7.9, from the point in Section 9.1: This is from someone who kept every consulted tactical plan unedited. In P36's account verifying happens around the plan, not in a long look at it: take it to the people who will carry it out ("go away and work

out the plan") and keep it "open to challenge". Beside that sits "trust in the fact it's done the maths behind it and has allocated enough platforms", which is about all the cohort's 2–4 seconds of checking can cover.

- Section 7.10, from the point in Section 9.1: Several operators asked for failure numbers. P36 wants them with context ("there's got to be something where it shows a bit of context to it"), because a bare "40% failure chance" reads as "no chance" when "the failure rate of doing nothing is 100%".
- Section 7.2, from the point in Section 9.1: Prior experience did not predict style here. P36 expects it to matter most when the stakes are real: their own "looking at a glance yes it's allocated what I needed ... cool let's start running with it" comes from experience, where a new platoon commander would be "foolhardy" or "almost paralysed". A game with no lives on the line hides that difference.
- Section 8.4, from the point in Section 9.1: Several operators wanted one plan held in mind across the tiers. P36 sees narrow context as normal ("you as a g2'er are will only see your part of the plan") and fine "as long as in each of them it's gone through the analytical process", and adds that this game's tasks are simple enough not to need more: a direct tension with those operators.
- Section 8.4, from the point in Section 9.1: The operators who wanted one shared plan are answered by a different design: each level gets "the understanding they're required to do that task", and "it doesn't matter how it docks into everything else".

Goes beyond the results (8).

- Section 8.5, from the point in Section 9.1: P27 and P32 asked for an assistant that says what to open next. P36's lower tier is where that job would sit: "I've got to do this thing now then that thing then that thing then that thing".
- Section 8.2, from the point in Section 9.1: Some operators want a human in the loop even one who cannot check. P36's check is not one person looking harder but a plan socialised so that "somebody is checking my thinking and has the option to challenge me".
- Section 7.2, from the point in Section 9.1: Who decides predicts handover far better than the mission does. P36 draws the operational conclusion: "the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this", a niche skill set like air traffic control, paid accordingly.
- ("Lower-reliance operators attempt fewer recoveries", not in the paper), from the point in Section 9.1: Lower-reliance operators attempted fewer recoveries. P36's rule cuts the other way: in search and rescue "you're always gambling with someone's life", so you act and "chip away at it".
- Section 7.10, from the point in Section 9.1: "It's a game" was a common caveat. P36 says what changes with a life at stake ("I imagine you're going to see me less decisive if someone's gonna die at the end"), and that a commander who no longer shares the risk ("I might die alongside them") loses what used to temper the order.
- Section 7.10, from the point in Section 9.1: Operators wondered how the two assistants fit together. P36's model is staff cells that each see their part and first "come together" at the planning brief.
- Section 7.10, from the point in Section 9.1: How should the tiers fit? From doctrine: "you understand two up in the same way as you plan two down", each task comes with the part of the level above that it needs, and an agent reaching outside its lane is told "that's not your job".
- Section C.4.1, from the point in Section 9.1: The tactical flaw (a wrong "No workable plan") went almost unseen: only 4 people reacted to one aloud. P36's answer does not rely on the operator: "an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly", checking "the data that they've based their answers on".

Options for this finding

How typical are they? [pass 5: numbers only]

Your note: We will say that our expert was a good performer with a slightly higher workrate than others, prempting cards more, but was otherwise fairly typical (insert a couple numbers into here)

How they played: for the whole board, not the mission [pass 5: for the paper]

Your note: We'll briefly quote the second part I think

Highlighted quotes not used above:

> *"so generally speaking all I'm trying to do here is get as many across as quickly as I can to complete all of the tasks managing my resources that I've got left here so I'm running a little bit low on blue over there and then just double checking on here so I've got the redundancy in there" — P36 (think-aloud s1 0:05:32)*
>
> *Your comment: Minor*

How they played: simple by hand, complex to the assistant, and why [pass 5: for the paper]

Your note: This one is a new idea that we can talk about in more depth in the paper: More complicated tasks can have a more branching failure space. This is not something we cover in this work, but could be an important factor in future works.

How they played: the rule bends when the reserve is full [pass 5: extra: may make it in]

Your note: minor, may make it in

How they played: oldest first, and never without a reserve [pass 5: for the paper]

Your note: Short note - they identified an classified a rule "never without a reserve". Links to the consistency factor in training.

How they played: critical tasks over score [pass 5: for the paper]

Your note: good

The task: the pressure lands on the lower tier [pass 5: for the paper]

Your note: maybe include

10 Discussion

[To write. Your pass-5 notes for it: the branching failure space of complex tasks as future work (Section 9.1); a rule like "never without a reserve" and the consistency of habits, for training (Section 9.1); the clash between one plan across the tiers and narrow context per tier (Sections 8.4 and 9.1). Material for it: the f-operator-wants finding (Section 7.10), the open questions recorded in the register, and the old draft's Step-4 benchmarking note: compare against `adams2025supervise` (workload at scale), `wischnewski2023measuring`/`wildman2024trust` (across-tier and multilevel trust), the variable-autonomy cluster including `oberacker2026mosaic`, and `ramchurn2021trustworthy` for error propagation across a pipeline.]

10.1 Limitations

[Your pass-5 note: the real-operations limitations (pace, stakes, unknown task locations) are implied and can be mentioned here rather than as a finding. Draft list from the register: one sample of 33, almost all computing/engineering and strongly AI-positive (AIAS-4 mean 8.0/10); a game, not real stakes; four fixed games played in an order that depended on when a person took part; build changes during collection (Appendix B); single AI coder for the qualitative counts; the illogical card and the wrong refusal are side effects found after collection, not manipulations; trust measured only after each session, so a same-session trust–use link can run either way.]

11 Conclusion

[To write.]

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A Simulation Parameters

B Build Versions

[P09 (`v1.11`, errors on) excluded. Check the table against docs/STUDY_BUILD.md before submission; the register's sensitivity analyses on the `v1.12+` cohort are in the analysis scripts.]

Table 4: Simulation parameters (scenario parameter set v2.1). Everything except the arrival rate and the category weights is identical in the two scenarios.

Parameter	Description	Value
Fleet		
Fast (blue)	count; speed	11; 11.0 u/s
Lifter (red)	count; speed	11; 10.0 u/s
Camera (green)	count; speed	11; 9.0 u/s
Missions		
Category penalty rate	A / B / C / D / E	0.05 / 0.10 / 0.15 / 0.25 / 0.40 pts/s
Archetype weights	Fast / Lifter / Camera / Mixed	38 / 26 / 21 / 15
Zone	radius; min. distance to hub; to other zones	80; 150; 200 units
Scenario-specific		
Strategic Heavy	mean inter-arrival; category weights A–E	37 s; 40/38/16/5/1
Tactical Heavy	mean inter-arrival; category weights A–E	75 s; 5/13/28/38/16
Session, assistants, failures		
Session length	per session	480 s (8 min)
Seed	all sessions	42 (four games; Section 6.1)
Strategy card reveal	per card, independent	≈4–5 s
Conservative card	spare per used type; top-up of remaining reserve	+1; 15% of every type
Suggest fill rate	tactical plan	1 drone→task entry per second
Drone failure hazard	per deployed drone per second	$1/900$ (`v1.4`+)
Post-failure grace	no further failure on that mission	30 s after recovery (`v1.5`+)
Penalty display step		15 s
Assistant error rates	ε_S, ε_T	0, 0 (all analysed sessions)

Table 5: Platform builds used during data collection, with the changes that bear on the data. The build is recorded in every session's log; the full history is in `docs/STUDY_BUILD.md`.

Build	Participants	Relevant to the data
`v1.0`	P01	Task points half the later values (rescaled in analysis); failures from a per-mission schedule.
`v1.2`	P02	As above; abandoned work re-queued as a residual mission from here on.
`v1.3`–`v1.4`	P03	Points doubled; pre-study AI-attitude scales added; live failure hazard (1/900 per drone-second) on a fixed grid from `v1.4`; substitute compositions executable.
`v1.5`–`v1.6`	P04	Recovery Suggest produces plans; 30 s post-failure grace; a half-built recovery plan survives a task completing.
`v1.7`–`v1.9`	P05–P07	Strategy-card score bars corrected (Speed = best ETA / ETA; Reserve = share of reserve left); allocating clears the tactical screen; deploying closes the planner.
`v1.10`–`v1.11`	P10, P11	Error-injection machinery added, switched off ($\varepsilon = 0$).
`v1.12`	P12–P30	NASA-TLX performance slider carries a "runs backwards" note.
`v1.13`	P31–P36	Games swapped: the same four mission streams played in the other session position.

Table 6: The same measures by the four fixed games (mean (SD) over sessions). SH = Strategic Heavy, TH = Tactical Heavy; the number is the mission stream (plan index), not the session order. Sessions per game: SH1 17, SH2 16, TH1 16, TH2 17.

Measure	**SH1**	**SH2**	**TH1**	**TH2**
Missions arrived (non-residual)	13.0 (0.0)	14.0 (0.0)	6.0 (0.0)	9.0 (0.0)
Strategic decisions made	10.8 (2.3)	12.1 (2.3)	6.2 (1.4)	7.6 (1.2)
Tactical plans confirmed	10.6 (2.5)	11.8 (2.4)	6.1 (1.5)	7.5 (1.1)
Tasks completed	25.1 (4.4)	28.0 (7.2)	21.2 (6.0)	32.8 (7.8)
Completion points[a]	1266 (248)	1288 (384)	1169 (331)	2058 (480)
Time penalty (points)	288 (52)	264 (66)	232 (79)	549 (92)
Score (points − penalty)[a]	978 (284)	1023 (443)	940 (398)	1509 (567)
Strategy card chosen (% of decisions)	58 (23)	61 (31)	78 (29)	68 (26)
Built by hand before cards loaded (%)	15 (19)	22 (27)	11 (26)	11 (14)
Strategy card edited, then deployed (%)	10 (21)	0 (2)	1 (4)	10 (19)
Tactical suggestion requested (% of plans)	75 (22)	68 (30)	75 (35)	84 (21)
Suggested plan kept unmodified (% of plans)	72 (24)	66 (32)	74 (35)	76 (29)
Drone failures	3.2 (1.7)	4.9 (3.1)	3.1 (2.0)	4.5 (2.9)
Missions abandoned	1.0 (1.1)	1.4 (2.0)	0.7 (0.9)	0.9 (0.9)
NASA-TLX, raw mean (0–20)	9.4 (3.1)	8.2 (3.1)	6.5 (2.9)	10.6 (2.8)
Mental demand	13.1 (3.8)	10.2 (5.5)	7.0 (5.2)	14.7 (4.2)
Temporal demand	13.5 (3.5)	13.7 (4.4)	11.2 (5.4)	13.4 (4.5)
Performance (0 = perfect)	8.1 (4.2)	6.9 (5.6)	7.8 (5.8)	9.8 (3.8)
Effort	10.2 (4.7)	8.9 (4.7)	5.9 (3.4)	11.7 (4.0)
Frustration	6.9 (4.8)	5.1 (5.3)	3.6 (3.9)	8.7 (6.3)
Trust, Strategic Assistant (1–7)	4.35 (1.29)	4.42 (1.77)	4.95 (1.43)	4.75 (1.26)
Trust, Tactical Assistant (1–7)	5.45 (1.05)	5.58 (0.67)	5.43 (0.95)	5.30 (0.94)

C Supplementary Results

Sidelined analyses from the analysis register: robustness checks, alternative specifications and closer looks that support the main results but are not needed to follow them.

C.1 Each operator has a personal style — individual, not a type (supplementary analyses)

Supplement to Section 7.2.

C.1.1 After one session we can recognise a person's style in the next

[Pass 5: numbers only]

Each session is summarised as ten habits (below). Comparing everyone's session-1 profile with everyone's session-2 profile, the closest match is far more often the **same person** than chance allows. It is a "near neighbour" result, not exact identification: the own session is the single closest match for 7 of 33 people (chance: 1), but in the closest three for 41% (chance 9%), and on average ranks 8th of 33 (chance 17th). It survives taking the reliance axis out — so the style is more than "how much do you lean on the assistants". In the matrix, every row is one person's session 1, every column a session 2, darker = closer; the outlined diagonal is each person against themselves. (Figure 32)

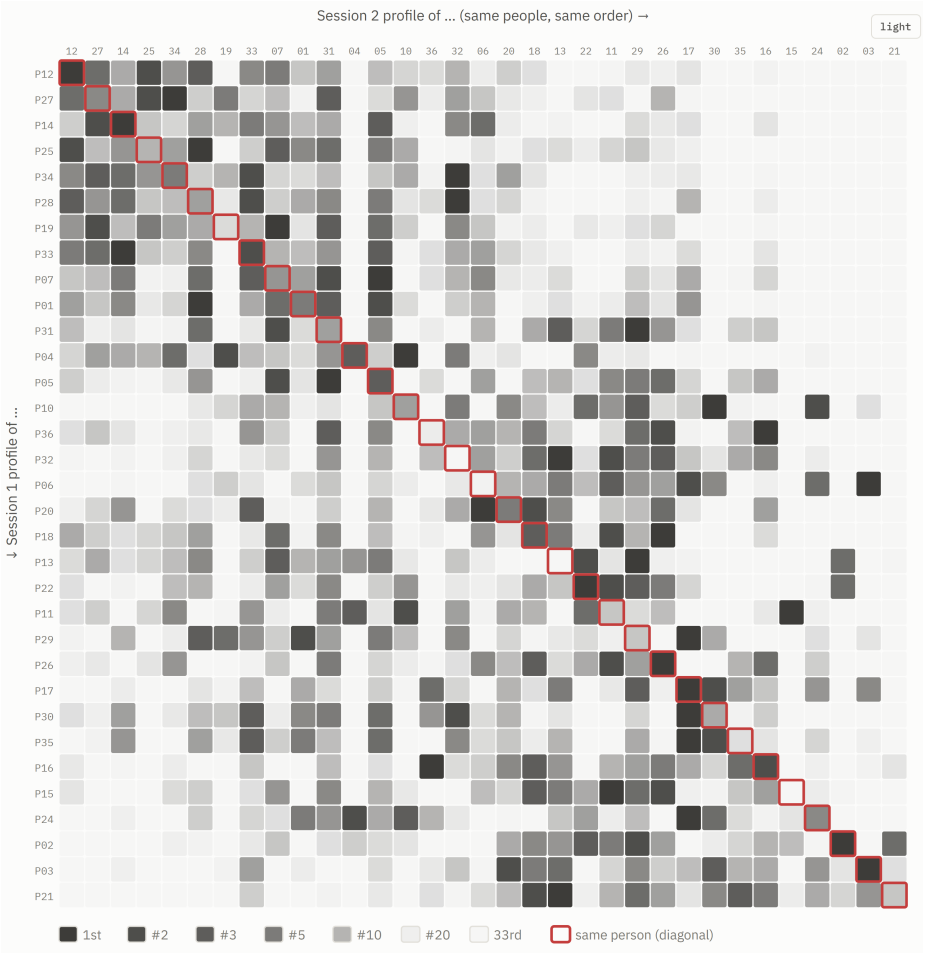


Figure 32: After one session we can recognise a person's style in the next

Key numbers.

- Own session nearest: 7 of 33 (session 1 $\rightarrow$ 2) and 5 of 33 (2 $\rightarrow$ 1); chance $\approx$ 1; permutation $p < .001$
- Own session in the closest 3: 41% vs 9% by chance; mean rank 8.2 of 33 vs 17.0 ($p < .001$)
- Reliance axis removed: 6 and 6 of 33, top-3 32%, mean rank 10.1 ($p < .001$) — the style is more than reliance
- Reliance axis alone: 4 and 4 of 33, top-3 27%, mean rank 11.1
- Profile correlation with own other session: median $r = .48$; with other people's: median $r = -.01$; own closer for 29 of 33 (sign test $p < .001$)

Method. Rows: 66 sessions (33 people $\times$ 2). Each session becomes a vector of ten habits: share of strategy strategy cards chosen; share of manual choices made before the cards loaded; share of cards edited; share of tactical plans where a tactical suggestion was requested; share of suggested plans kept unchanged; share of recovery episodes with a suggestion requested; time per strategic decision beyond the forced card delay (log); drones committed vs the Conservative card on single-type and on multi-type missions; card previews per decision. Each habit is turned into a z-score *within the game played* (so a hard game does not make everyone in it look alike; a habit with no data in a session, e.g. no recoveries, is set to that game's median). For each person's session-1 vector we measure the straight-line (Euclidean) distance to all 33 session-2 vectors and rank them; "rank 1" means their own session 2 is the nearest. Done both ways (session 1 looking among session 2s and vice versa). The p-value comes from shuffling who-is-who in session 2 10 000 times and counting how often chance does as well. "Reliance axis removed": the single strongest direction of variation (PC1, see below) is subtracted from every vector first. The profile correlation r is the Pearson correlation between a person's two ten-habit vectors (1 = same shape, 0 = unrelated). Exploratory and post hoc; the permutation test is clean. People are ordered by reliance in the matrix, so neighbours look alike — the claim is that the diagonal beats them.

C.1.2 What a personal style looks like

[Pass 5: extra: may make it in]

Every person's ten habits in session 1 (light) and session 2 (dark). The top rows are people whose two sessions have almost the same shape (r up to .89); at the bottom are the few whose style changed (P13, P29, P32). Most people repeat themselves. (Figure 33)

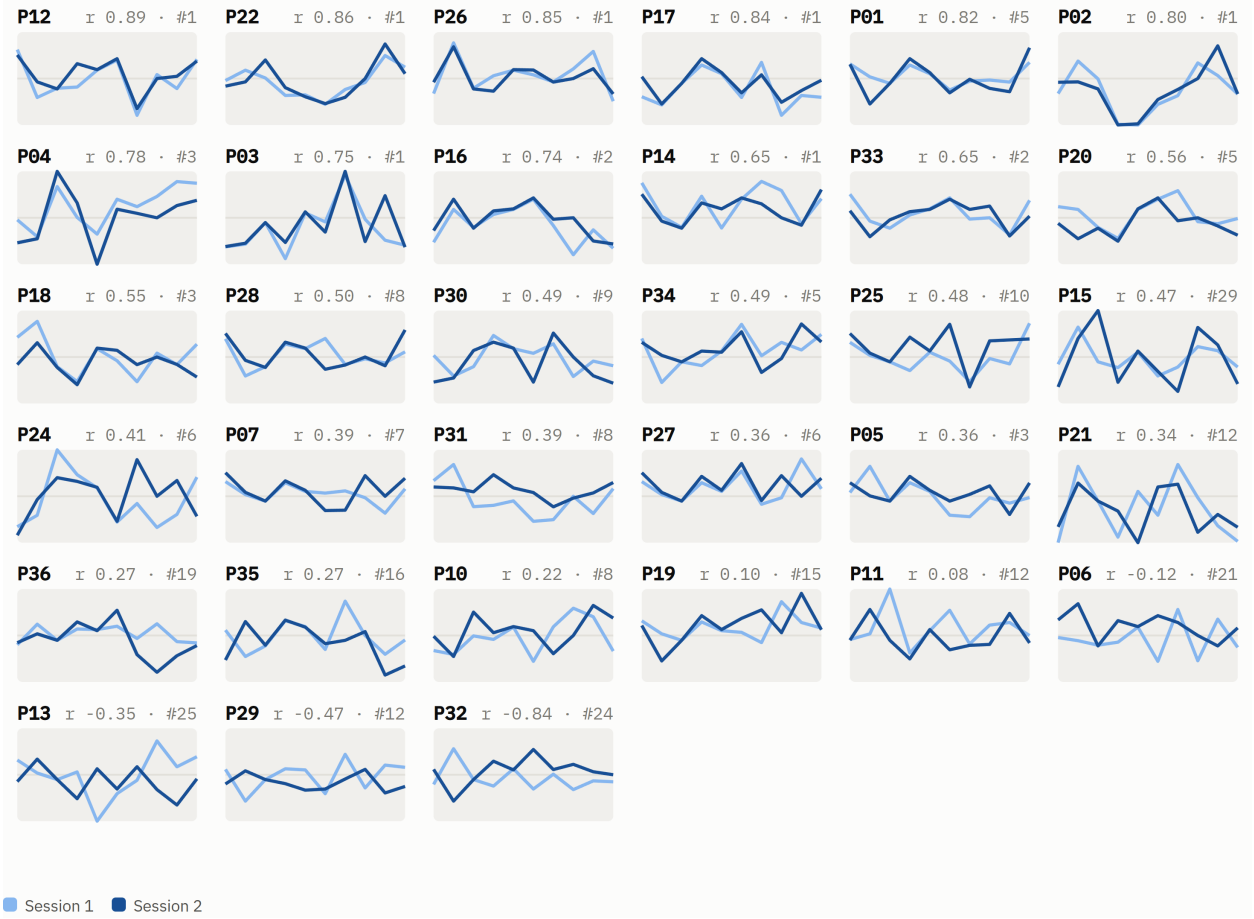


Figure 33: What a personal style looks like

Key numbers.

- 24 of 33 people have r $\geq$ .27 between their two sessions; 4 have r < 0

Method. Each line joins the ten habit z-scores (within game, clipped at $\pm$2.5 for drawing). r = correlation of the two lines; #k = where their own session 2 ranked among 33 (1 = nearest).

C.1.3 Which habits carry over — and what each correlation means

[Pass 5: extra: may make it in]

Each small scatter is one habit: a dot is a person, x = that habit in session 1, y = the same habit in session 2. If people kept their place in the queue exactly, the dots would lie on the dashed line; if session 1 told us nothing, they would be a shapeless cloud. Six of the ten habits carry over clearly: keeping suggested plans, requesting a tactical suggestion, taking cards, previewing cards, decision time, and Suggest in recovery. Topping up drones on single-type missions does not carry over at all. (Figure 34)

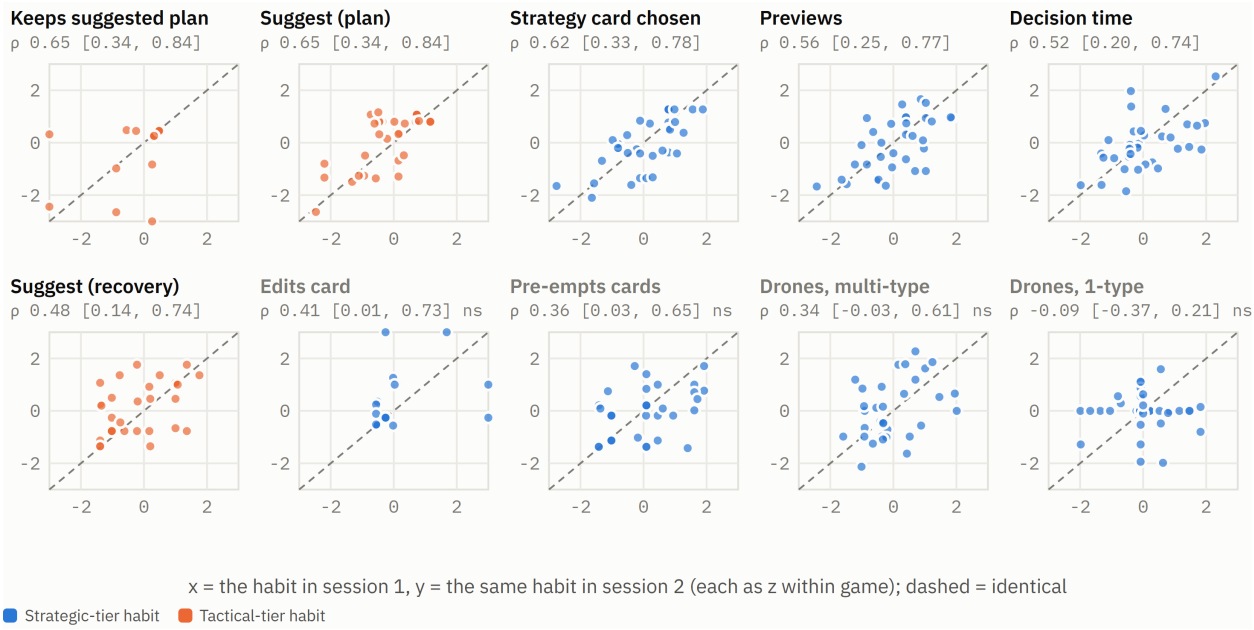


Figure 34: Which habits carry over — and what each correlation means

Key numbers.

- Survive Holm: keeps suggested plan ρ .65 [.34, .84]; requests a tactical suggestion .65 [.34, .84]; takes card .62 [.33, .78]; previews .56 [.25, .77]; decision time .52 [.20, .74]; Suggest in recovery .48 [.14, .74]
- Do not: edits a card .41; manual-before-cards (share of manual) .36; drones on multi-type .34; drones on single-type −.09
- Strategy card choice: ~2 distinct levels of people from one session, ~3 from two
- Score on the reliance axis, session 1 vs 2: ρ = .65 [.38, .82]

Method. ρ (Spearman) for habit X = the correlation, across the 33 people, between their rank on X in session 1 and their rank on X in session 2, each session first adjusted for the game played (z within game). ρ = .62 for "takes the strategy card" means people keep roughly the same order: high card-takers in session 1 are mostly high in session 2, though not exactly. 95% intervals come from resampling people 2000 times; Holm corrects for testing ten habits. "Distinct levels" (separation strata) says how many groups of people the habit can tell apart reliably: one session of strategy card choice separates about 2 (high vs low), the average of two sessions about 3. Caution on "keeps the suggested plan": 24 of 33 people kept every suggested plan unchanged in both sessions (a ceiling), so its high ρ mostly says that the nine who ever edit a plan tend to do so in both sessions. The pre-empting habit here is "share of *manual* choices made before the cards loaded" (undefined for 15 people who never chose manual in a session); the clearer per-decision version is in the next section.

C.1.4 Not waiting for the cards is one of those habits

[Pass 5: extra: may make it in]

Committing a manual allocation before the strategy cards have even loaded is something some people do and others never do. People who did it in session 1 did it on a quarter of their decisions in session 2; people who did not, on 8%. Fifteen people never did it in either session. (Figure 35)

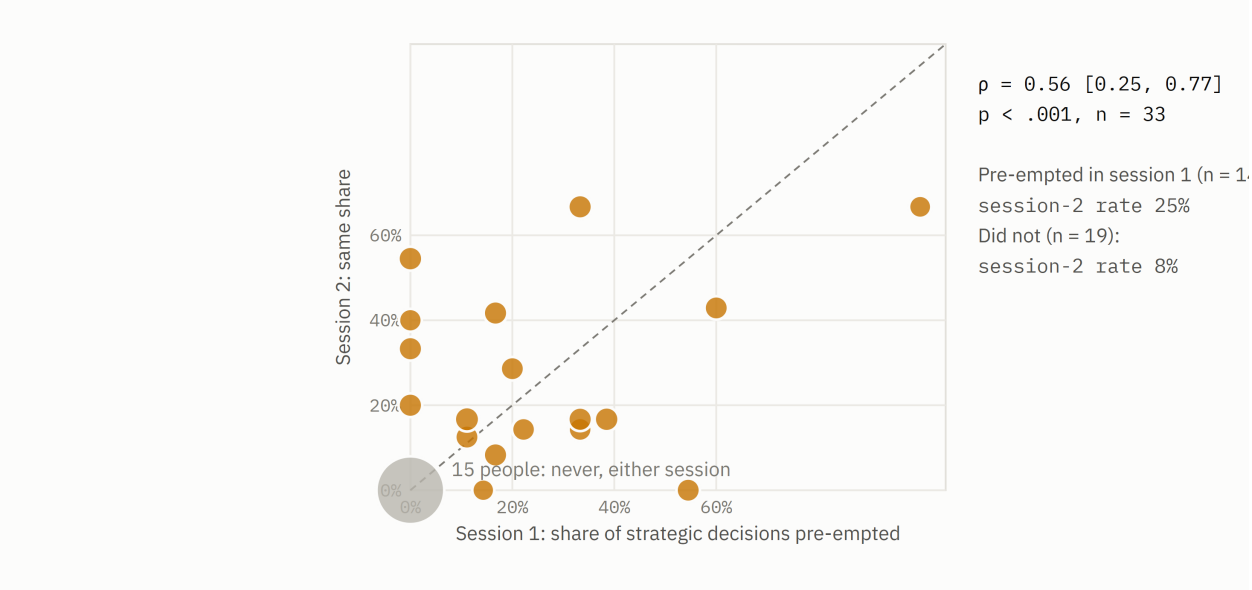


Figure 35: Not waiting for the cards is one of those habits

Key numbers.

- Session 1 vs 2: $\rho = .56$ [.25, .77], $p < .001$, $n = 33$
- Session-2 rate 25% for those who pre-empted in session 1 (n = 14) vs 8% for those who did not (n = 19)
- 86 of 564 decisions pre-empted; 46% of all manual choices; 15 of 33 never did it
- Round 2: person share of variance .24, mission .05

Method. Rows: 564 non-residual strategic decisions. Per person and session: share of decisions committed manually before the cards appeared (logged `manualBeforeCardsLoaded`). ρ = Spearman of the session-1 share against the session-2 share across 33 people. Variance shares (person ~24%, mission ~5%) are from round 2 (F4.1, crossed random intercepts for person and the exact mission, which repeats across people because games are fixed). The 15 "never" people sit together at the origin; ρ is driven both by them staying at zero and by the regular pre-empters staying high.

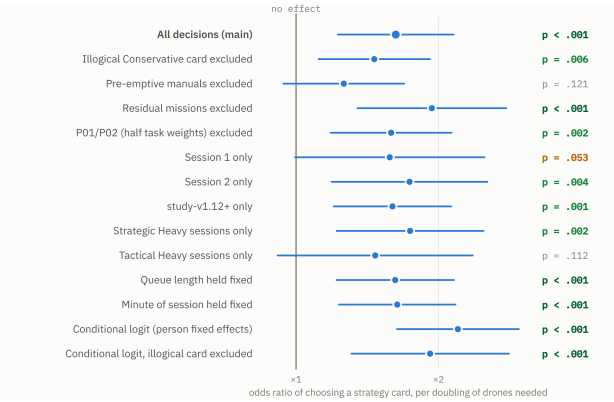
C.2 Small jobs by hand, big jobs to the assistant — on both tiers, and it is where the assistant saves time (supplementary analyses)

Supplement to Section 7.3.

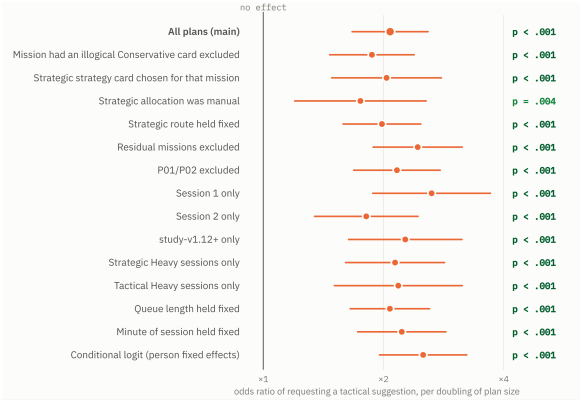
C.2.1 Is it an artefact? Checks, all reported

[Pass 5: numbers only]

The worry: the illogical Conservative card (one that adds a drone of a type the mission does not need) appears only on one- and two-type missions — 71% of one-type decisions, none of the three-type ones — so "people avoid the card on small missions" could be "people avoid a visibly odd card". It is not: on one-type missions uptake is the same with an odd card (50%, 129 decisions) and without one (56%, 52 decisions), and with every odd-card decision removed the size effect is still there ($\times$1.46 per doubling; clean one-type vs three-type $\times$1.83). No ε-corrupted (deliberately wrong) cards were shown in any session, so card quality did not otherwise vary by size. The rule also holds with queue length held fixed, with the minute of session held fixed, per session, per scenario, without the half-weight participants and in a person-fixed-effects model. Two strategic rows do not reach $p < .05$: Tactical Heavy sessions alone (few one-type missions there) and cards-on-screen only (above). The tactical rule clears every row. (Figure 36)



(a) Strategic tier.



(b) Tactical tier.

Figure 36: Is it an artefact? Checks, all reported

Key numbers.

- Strategic, illogical card excluded: $\times$1.46 [1.12, 1.92], p = .006 (474 decisions); conditional logit $\times$1.92 [1.31, 2.82]
- One-type missions, odd card vs clean card: odds $\times$0.77 [0.33, 1.78], p = .55 — no difference

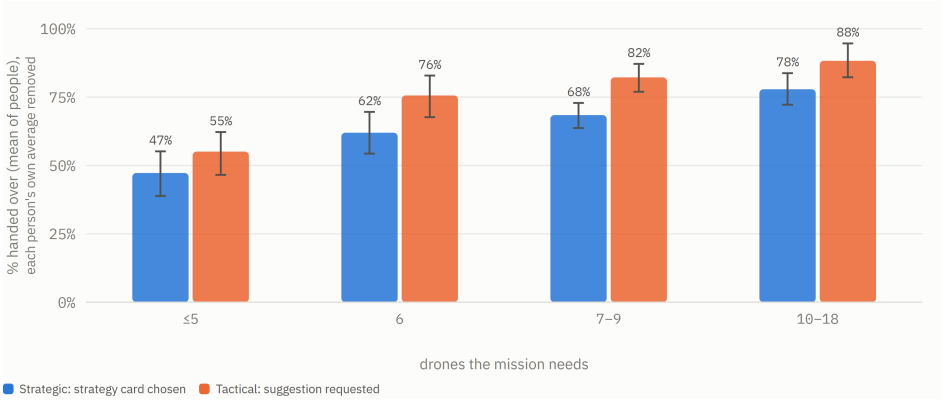
- Strategic range across 14 specifications: $\times 1.26$ to $\times 2.20$; tactical range across 15: $\times 1.75$ to $\times 2.64$
- Tactical, illogical-card missions excluded: $\times 1.87$ [1.47, 2.39]; strategic route held fixed $\times 1.98$

Method. Each row refits the main within-person model on a subset or with one more control. Illogical card: the *odd* flag from build_findings_synthesis.load_decisions (Conservative card carries a type the mission needs none of). Queue held fixed: the number of missions waiting at the decision, split within/between person (the queue story itself is Section 7.6's). Conditional logit: a different model family that compares each person only with themselves by construction (people who always or never hand over drop out, hence fewer people). The odd-card story itself is Section 7.8's — here it is only checked as a confound.

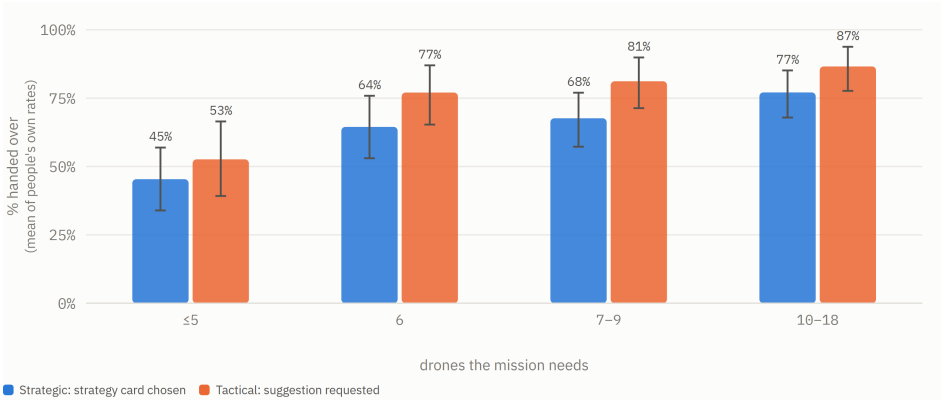
C.2.2 Person by person

[Pass 5: extra: may make it in]

The same rule with each person counted once: for each band of mission size, each person's own hand-over rate, averaged over people. The default view removes each person's overall level first, so it shows how the *same* people shift between small and big missions. Both tiers rise band by band. 22 of 33 people hand over more on their biggest band than their smallest, on each tier; only 4 (strategic) and 2 (tactical) do the reverse. (Figure 37)



(a) Each person compared with their own average.



(b) Plain average of people.

Figure 37: Person by person

Key numbers.

- Strategic, own average removed: 47% ($\leq$5 drones, 32 people) $\rightarrow$ 62% $\rightarrow$ 68% $\rightarrow$ 78% (10–18, 33 people)
- Tactical, own average removed: 55% $\rightarrow$ 76% $\rightarrow$ 82% $\rightarrow$ 88%
- Biggest band above smallest: strategic 22 people, below 4, equal 7 (sign test $p < .001$); tactical 22, 2, 9 ($p < .001$)

Method. Bands of drones needed: $\leq$5, 6, 7–9, 10–18 (both tiers; drones needed is the measure both tiers share). A person contributes to a band if they made at least 2 decisions in it. "Own average removed" = the person's rate in the band minus their rate over all bands, plus the overall rate, so bars stay on the percentage scale. Whiskers: 95% intervals from resampling people. Sign test: for each person, their biggest band against their smallest; are "more" more common than "less" than a coin flip would give (ties dropped)?

C.2.3 The speed difference as estimates

[Pass 5: numbers only]

The same comparison as model estimates: negative means the assistant route was faster for the same person. (Figure 38)

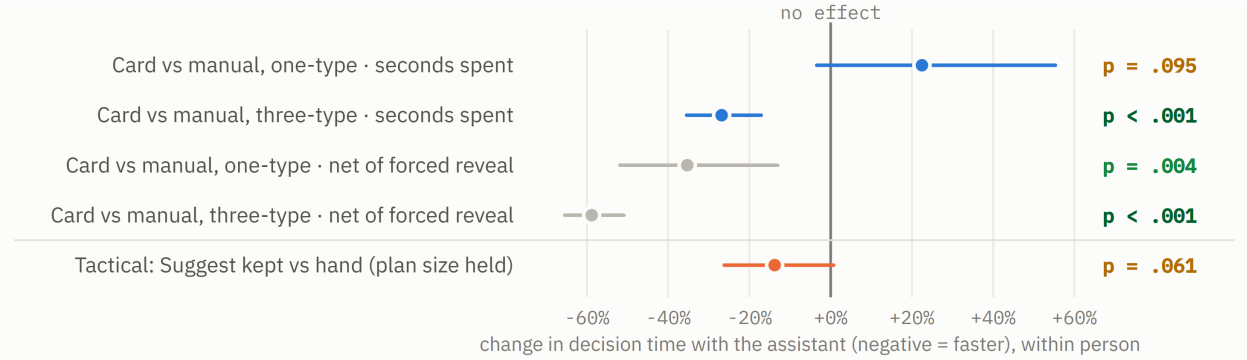


Figure 38: The speed difference as estimates

Method. As in the section above. Each row is a within-person % change in decision time with its 95% interval.

C.2.4 The exception that proves the rule: the people who call the assistant slow

[Pass 5: extra: may make it in]

Six people complain that the assistant is slow (C06: P06, P15, P16, P26, P27, P29). On every measure the two groups' intervals overlap: none of the ten differences is reliable with six people. The directions are worth a sentence, not a claim. They take cards and request a tactical suggestion at the same rate as everyone else (58% vs 66%; 74% vs 76%), and they lean *towards* the size rule: they decide before the cards load more often (30% vs 13%, p = .055) and their three-type minus one-type gap is larger (+40 vs +19 points, p = .09). Read that way, the complaint and the rule are the same observation (the fixed wait is not worth it on small jobs), and P29 says as much. But this is a reading of six people, not a finding. (Figure 39)

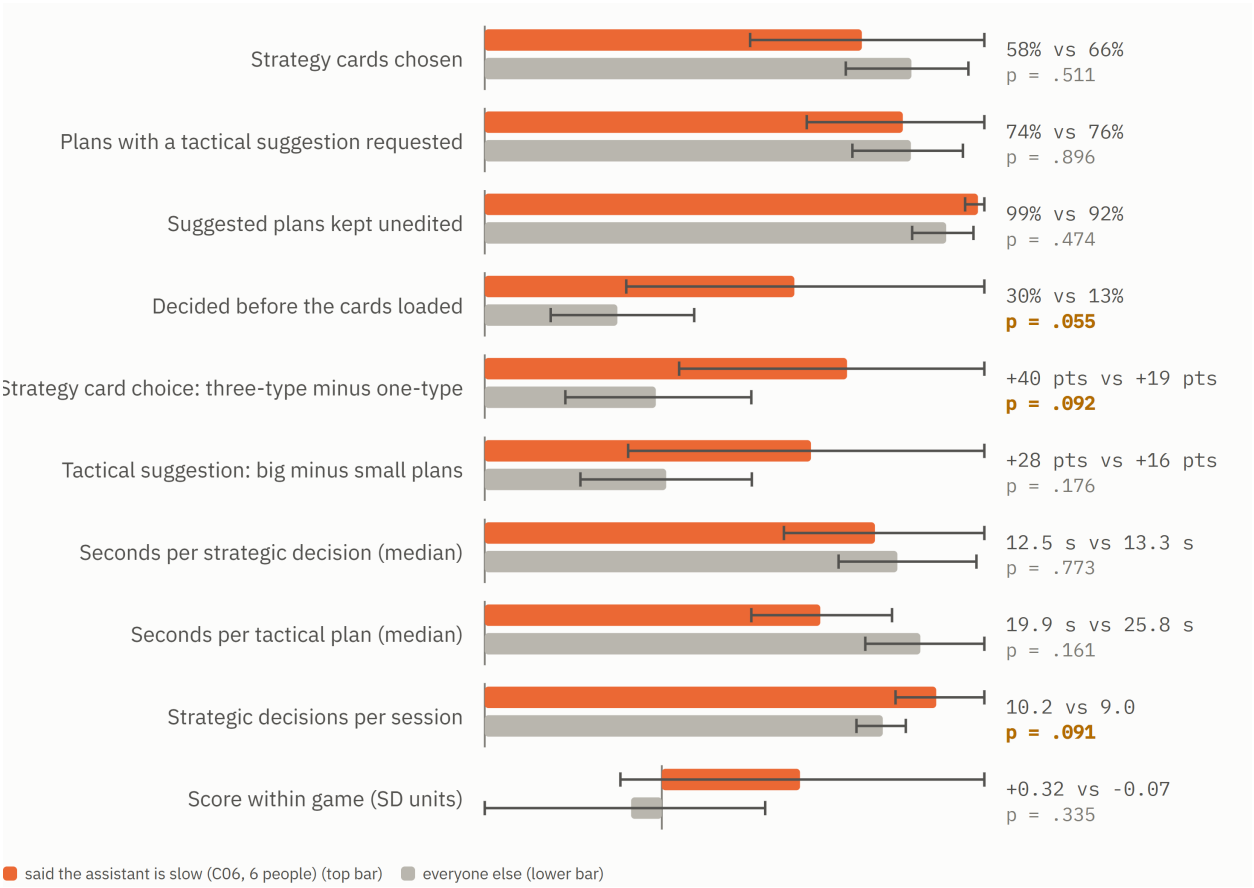


Figure 39: The exception that proves the rule: the people who call the assistant slow

Key numbers.

- Pre-emptive share 30% vs 13% (p = .055); three-type minus one-type card gap +40 vs +19 points (p = .09); decisions per session 10.2 vs 9.0 (p = .09)
- Tactical plan time 19.9 vs 25.8 s (p = .16); score +0.32 vs −0.07 SD (p = .34)
- Strategy card choice 58% vs 66% (p = .51); Suggest 74% vs 76% (p = .90) — reliance itself does not differ

Method. Per-person measures, averaged over the person's two sessions, for the six C06 people against the other 27. Bars are group means; whiskers are 95% intervals from resampling people within each group. Score = session score (P01/P02 rescaled for their half task weights) standardised within each of the four games. p = two-sided permutation test (group labels reshuffled 20,000 times); 10 comparisons, uncorrected, so expect about one p below .10 by chance.

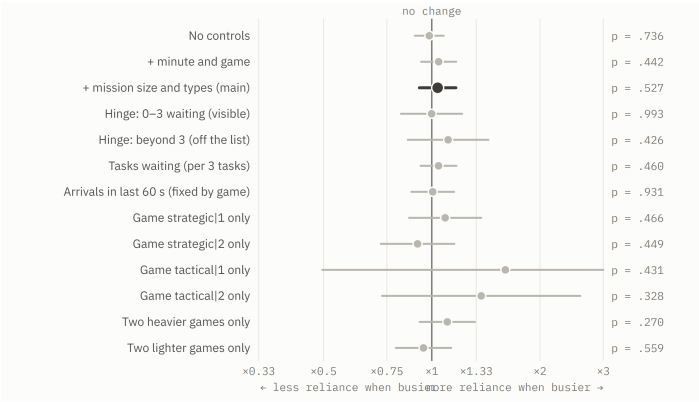
C.3 A busier moment does not push people onto the assistants — even though they say it does (supplementary analyses)

Supplement to Section 7.6.

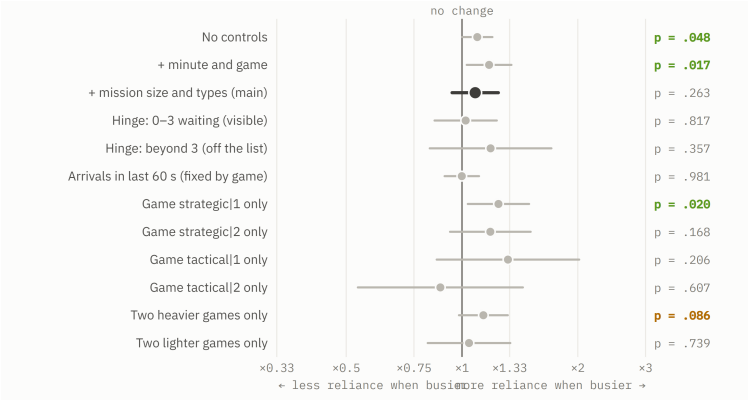
C.3.1 Every way of asking it gives the same answer

[Pass 5: numbers only]

The within-person change in the odds of relying per extra mission waiting, under every specification tried — including each game on its own, the heavier vs the lighter pair of games, a 'hinge' that lets the effect differ beyond the visible list, tasks waiting instead of missions, and missions that had just arrived (which the operator cannot cause). Strategic: all 13 are compatible with no change. Tactical: the main model is flat, but two of twelve variants reach $p < .05$ (without mission size; Strategic Heavy game 1 alone) — a small upward lean that disappears once mission size is held fixed and is absent for the arrivals measure. (Figure 40)



(a) Strategic tier.



(b) Tactical tier.

Figure 40: Every way of asking it gives the same answer

Key numbers.

- Strategic main: OR 1.04 [0.92, 1.17], p = .53 per extra mission waiting; range across 13 specs 0.91–1.60, none $p < .05$.
- Strategic hinge: 0–3 waiting OR 1.00 [0.82, 1.22]; beyond 3 OR 1.11 [0.86, 1.44] — no change at the visible limit.
- Tactical main: OR 1.08 [0.94, 1.24], p = .26; + minute and game only: 1.18 [1.03, 1.34], p = .017; arrivals in last 60 s: 1.00 [0.90, 1.11].

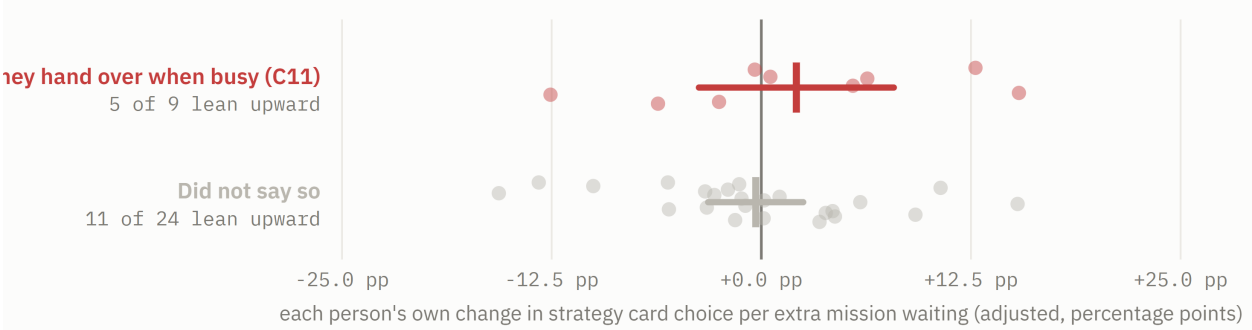
- Heavier games only (Strategic Heavy 1, Tactical Heavy 2): strategic 1.10 [0.93, 1.32], tactical 1.14 [0.98, 1.32].

Method. GEE logistic regression (decisions clustered within person, bias-reduced standard errors). 'Within person' means the queue is centred on that person's own average, so the estimate compares a person when busier with the same person when less busy; their average queue is kept as a separate term. 'Main' adds mission size, tasks, drone types, minute and game. An odds ratio of 1.04 means the odds of choosing a strategy card are 4% higher per extra mission waiting. To make the interval concrete: at the main model's upper limit (1.17) uptake would rise from 65% at 0 waiting to 80% at 5; at its lower limit (0.92) it would fall to 56%. So the data rule out a large effect but not a modest one either way.

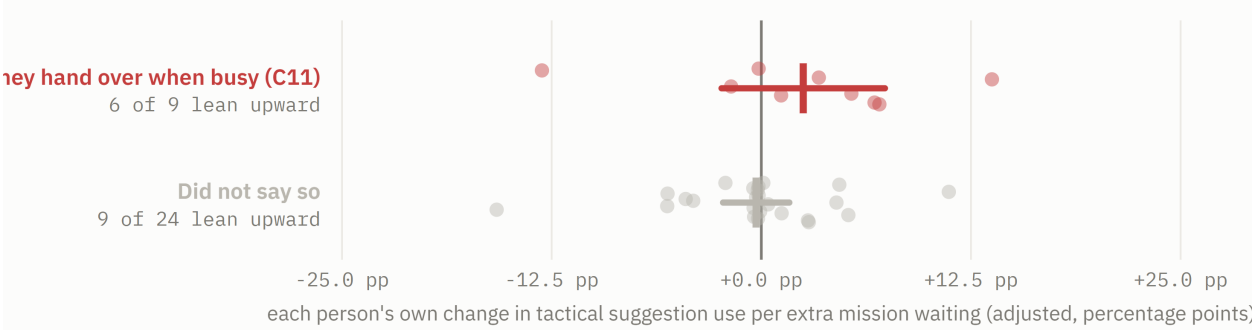
C.3.2 Said vs did: the people who said they hand over when busy

[Pass 5: numbers only]

Nine operators (P05, P06, P11, P18, P20, P25, P32, P33, P35) said, unprompted, that they lean on the assistant more when busy, overwhelmed or rushed. Each dot is one person's own slope: how much their (adjusted) reliance changed per extra mission waiting. If what they said described what they did, the red group would sit clearly right of zero. It does not: on average they move about +2 points per extra mission, not distinguishable from zero or from the other 24 people. Their claim usually compared sessions ('the first one was so fast I used it a lot') — so the second view checks that version: reliance in their heavier game minus their lighter game. Again no difference. (Figure 41)



(a) Strategic tier, moment to moment.



(b) Tactical tier, moment to moment.

Figure 41: Said vs did: the people who said they hand over when busy

Key numbers.

- Strategic slopes: said +2.1 pp per extra mission (5 of 9 upward) vs not said −0.3 pp (11 of 24 upward); p = .41.
- Tactical slopes: said +2.5 pp (6 of 9 upward) vs −0.3 pp (9 of 24); p = .09 — the closest thing to support, and not significant.
- Queue × said interaction: strategic OR 1.13 [0.83, 1.53] p = .44; tactical 1.18 [0.91, 1.53] p = .20. Within the 'said' group alone the queue OR is 1.03 (strategic) and 0.97 (tactical).
- Heavier minus lighter game, adjusted: said +3.8 pp strategic, +1.5 pp tactical; everyone −1.1 / +1.2 pp; all intervals span zero.

Method. **Moment to moment:** for each person with 6+ decisions and some spread in queue depth, a straight line through their (observed − expected) values against queue depth; the slope is in percentage points per extra mission waiting. Groups compared with a Mann–Whitney test; also tested as a queue × 'said' interaction in the GEE above. **Busier session:** the four games' load measured by total drones demanded per session (Tactical Heavy 2: 115; Strategic Heavy 1: 89; Strategic Heavy 2: 88; Tactical Heavy 1: 63); for each person, their mean (observed − expected) in the heavier of their two games minus the lighter. With 9 people

in the 'said' group, only a large difference could have been detected — this is absence of evidence for the claim, and the point estimates are small.

C.3.3 In play: overload talk comes at slightly busier moments, reliance does not follow

[Pass 5: extra: may make it in]

Using the narration recorded during play (30 people, 60 sessions), we found every moment someone said an overload-type phrase ('too many', 'can't keep up', 'oh no', 'struggling'…): 51 moments from 24 people. All nine of the 'said' group produced at least one. At those moments slightly more missions were waiting than at a random utterance (2.4 vs 2.1; p = .09) — so what people feel as overload is weakly tied to the queue. But in the minute either side, their strategy card choice and tactical suggestion use were exactly what their missions and habits predict. The felt pressure is real enough; it just does not turn into handing work over. (Figure 42)

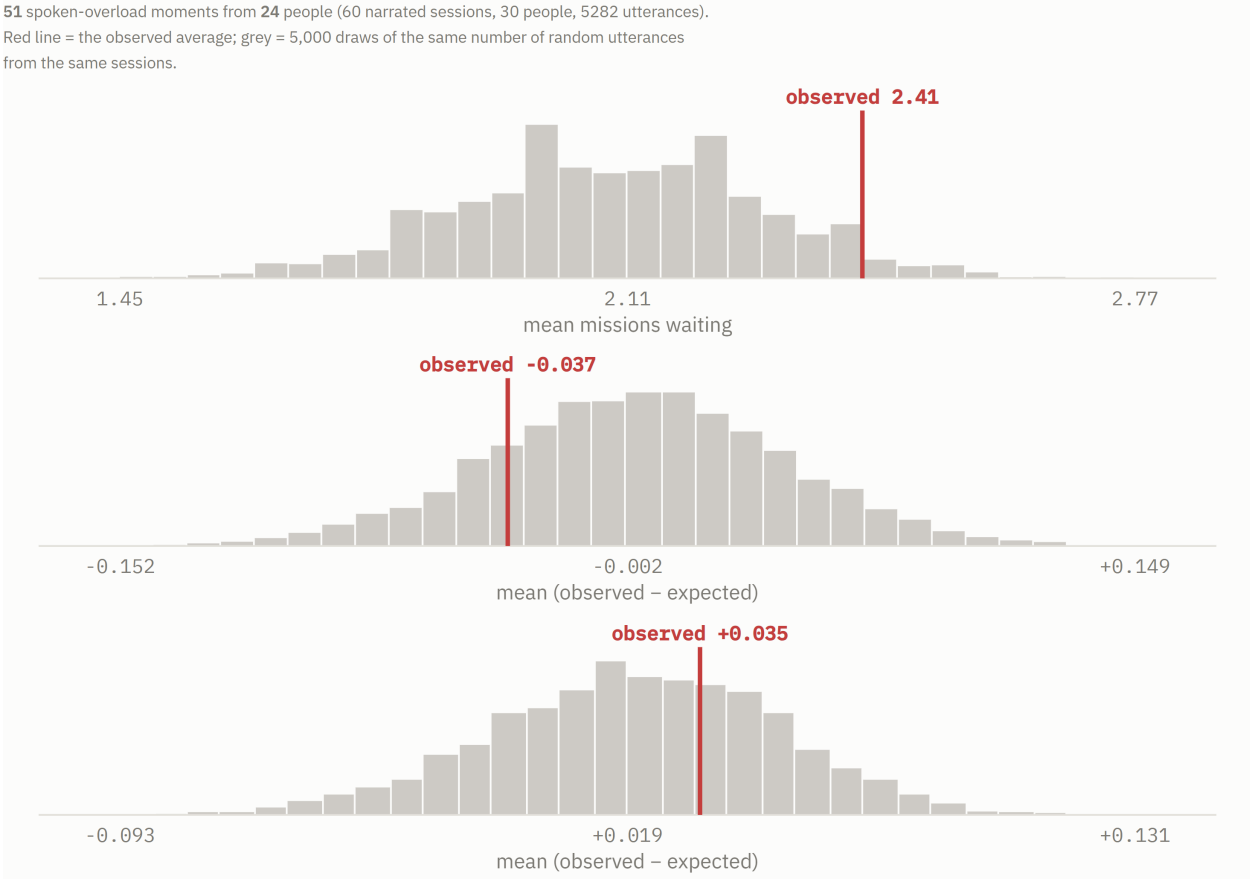


Figure 42: In play: overload talk comes at slightly busier moments, reliance does not follow Top: missions waiting at the overload moments; middle: strategy cards taken within ±60 s (observed − expected); bottom: Suggest pressed within ±60 s (observed − expected). Red line = observed mean; grey = 5,000 draws of the same number of random utterances from the same sessions.

Key numbers.

- Missions waiting at overload moments 2.41 vs 2.09 at random utterances (95% of random draws 1.71–2.45), p = .09.
- Card-taking within ±60 s: −3.7 pp vs expected (random utterances +0.1), p = .38. Suggest: +3.5 pp (random +2.1), p = .65.
- 16 of the 51 moments came from the 9 people coded C11; 35 from others.

Method. Phrase list fixed in round 2 (before any behaviour was looked at), applied to the transcript of each session window

(from segments.json; utterances 20 s apart merged into one moment). For each moment: missions waiting at that second (rebuilt from the log), and the mean (observed − expected) of strategic decisions and tactical plans started within ±60 s. Null: the same number of utterances per session drawn at random from ALL that session's speech, 5,000 times — so the comparison is 'overload talk' vs 'any talk', in the same people and sessions. Caveats: no speaker separation (researcher speech is in the transcripts); in round 2 about half of matched phrases were genuine stress on reading; alignment uses file times for 24 of 30 people (±2 s).

C.4 The dip in strategy-card use belongs to the people who later said they noticed the illogical card, early in the session; it shows up whether or not they had met that card yet (supplementary analyses)

Supplement to Section 7.8.

C.4.1 Strategic flaws are visible; tactical flaws are hidden

[Pass 5: numbers only]

Neither assistant had errors injected for anyone analysed (the ε manipulation ran only for the excluded P09). Each tier still has a natural flaw. On the **strategic** side, it is the illogical extra drone type. It sits on the card face next to the mission's own requirements, everyone saw it, and 12 people mentioned it in debrief or interview (15 counting in-play remarks). On the **tactical** side, it is the wrong refusal: Suggest answers 'No workable plan' when a fix existed using a drone busy on another task. To the operator it looks exactly like a correct refusal. 14 people met one (23 wrong refusals out of 39). Four of them reacted aloud and accepted it, most saying they would have to abandon. Only P28 named the cause. Only 3 people ever described a tactical-plan flaw (C13), and just one of them had been wrongly refused. 'The strategic assistant is distrusted more' may partly be 'its mistakes are the ones you can see'. (Figure 43)

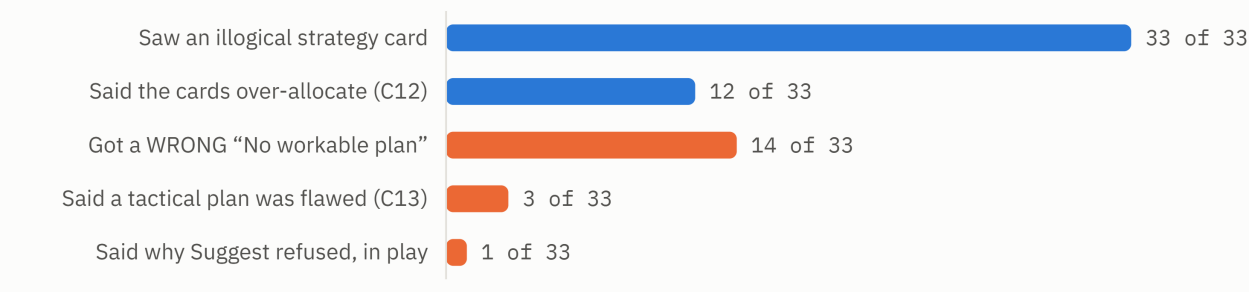


Figure 43: Strategic flaws are visible; tactical flaws are hidden

Key numbers.

- Strategic: 33 of 33 saw the illogical card (132 of 606 decisions); 12 coded C12; 15 incl. strict in-play remarks
- Tactical: 14 of 33 got $\geq$1 wrong refusal (23 of 39 refusals); 3 coded C13, 1 of them among the 14; 4 reacted aloud to a wrong refusal, 1 named why

Method. Strategic exposure: people with $\geq$1 illogical-card decision. Tactical exposure: people with $\geq$1 Suggest press in a recovery that returned 'No workable plan' or a partial plan while a fix using a busy drone existed (the 'blind' flag in h3_recovery). Noticing: qualitative codes C12 and C13 (debrief or interview), plus the hand-coded in-play remarks (extras.SPOKEN_STRAT / SPOKEN_TAC), each matched to the nearest preceding Suggest press.

C.4.2 On the missions that carry the flaw, noticers stop using the cards

[Pass 5: extra: may make it in]

This is the cleanest behavioural trace, and it is where you would expect one. On their 1st illogical-card mission, noticers chose a strategy card 47% of the time and never-said people 56%. With each further illogical-card mission, noticers chose a strategy card about 14 points less often. By their 4th–6th they took one **0 times in 13** (7 people), going manual instead, mostly *before the cards had loaded* (62%). Non-noticers moved the other way: by their 4th–6th they chose a strategy card 74% of the time. Most of that was the illogical Conservative card itself (52%), which noticers took 0 times in 13. The handover analyst's check fits this: when people build a one-type mission by hand rather than take the Conservative card, nearly all of the 'trim' is dropping the unneeded-type spares (−1.2 drones overall, +0.05 on the types the mission needs). So hand-builders strip out exactly what is illogical; see Section 7.4. The manual-card-error card's version (people who mention a card error go manual more on one-type missions) holds in direction. But the same people also go manual more on three-type missions, where nothing is wrong (37% vs 22%). The one-type-*specific* part does not hold between people (interaction OR 1.12, p = .78). What is specific to the flawed missions is the *trend*: noticers learn to skip the cards there. (Figure 44)

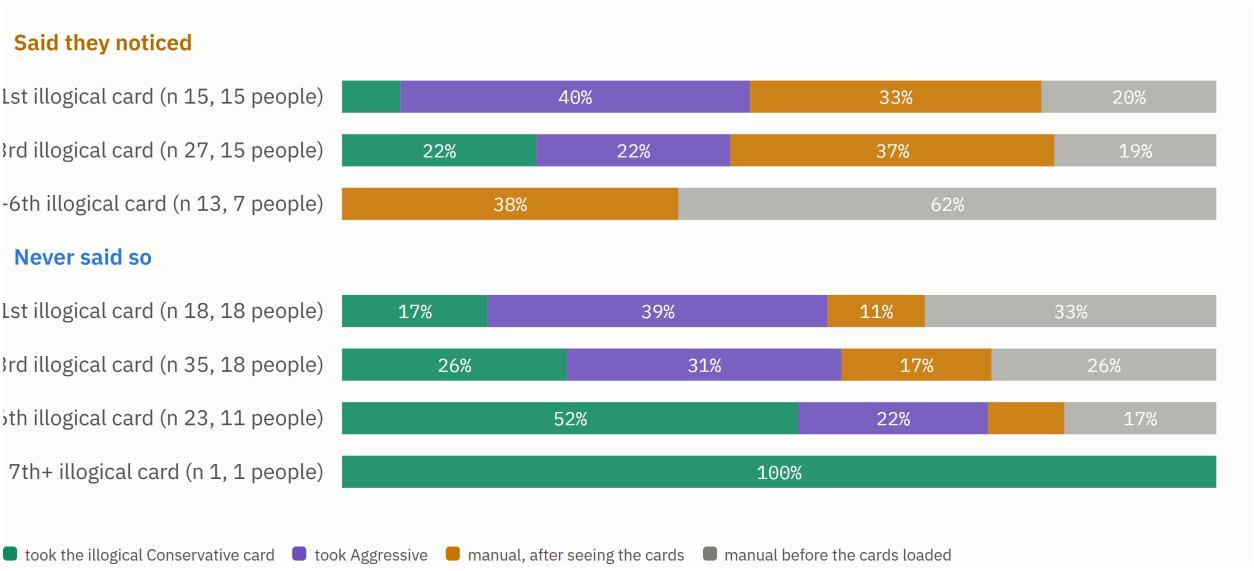


Figure 44: On the missions that carry the flaw, noticers stop using the cards

Key numbers.

- Took any card, per extra illogical card: noticers −13.9 points [−25, −3]; never-said +5.3 [−2, +12]; difference p < .001
- Took the illogical Conservative card: noticers −3.5 points per card; never-said +9.3 [+1, +18]; difference p = .001
- Manual before the cards loaded: noticers +15.4 points per card [+2, +29]; never-said −5.8; difference p = .012
- 4th–6th illogical card: noticers chose a strategy card 0 of 13 (7 people); never-said 74% of 23 (11 people), 52% the illogical card
- Manual on one-type missions: noticers 63% vs 37%, OR 2.41 [0.92, 6.32], p = .07; on three-type 37% vs 22%, OR 2.29, p = .06; one-type × noticed OR 1.12, p = .78

Method. Rows: the 132 illogical-card decisions, numbered per person (1st, 2nd, . . .). Each is coded by what happened: took the Conservative card (the illogical one), took Aggressive, manual after the cards loaded, or manual before they loaded (a pre-emptive choice that never saw the cards). Trend = a linear model of each outcome on the card's number (capped at 7) × group, with game

held fixed and people clustered. The interaction p is whether the two groups' slopes differ. Between-person check (manual-card-error): manual ~ said + game on one-type missions, then on three-type, then the said × one-type interaction (logistic GEE, people clustered).

C.4.3 What 'expected' and 'adjusted' mean

[Pass 5: numbers only]

Expected is the chance a model gives that this person chooses a strategy card on this decision. It knows only what the decision was (drones, tasks and drone types the mission needed), when it was (minute of the session, whether it was a re-queued mission, how many others were waiting), which of the four games was being played, and how often *this person* took cards on all their *other* decisions. **Adjusted** = the group's overall rate + the average of (observed − expected) at that position. In plain terms: 'how far above or below their usual self, given the mission in front of them, were people here?' The person term does most of the work (odds ×2.84 per unit; the model ranks takes above non-takes 81% of the time). So 'adjusted' mostly means 'compared with that person's normal habit'. Of the mission features, only the number of drone types matters (×1.51 per extra type). Because adjusted is a shifted average residual and not a probability, it can go above 100% at a small dot that sits far above its expectation. The old not-said +2 dot read 122% for this reason. (Figure 45)

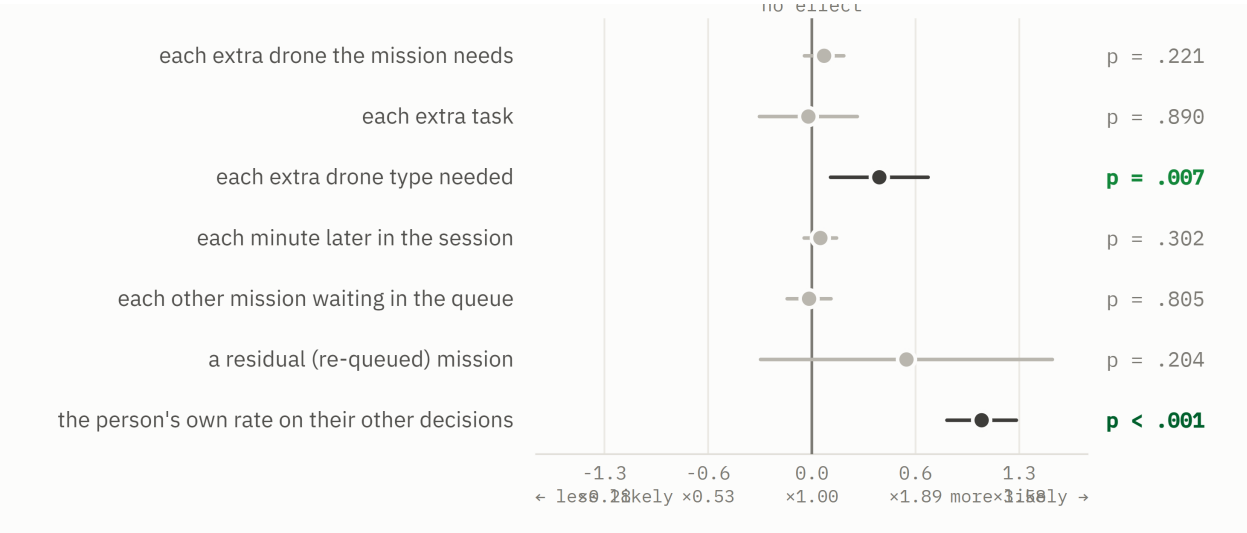


Figure 45: What 'expected' and 'adjusted' mean

Key numbers.

- Person's own rate elsewhere: odds ×2.84 [2.29, 3.51] per logit unit
- Each extra drone type needed: ×1.51 [1.12, 2.04]; drones ×1.08, tasks ×0.98, minute ×1.05, queue ×0.98 (all n.s.)
- Model ranks takes above non-takes 81% of the time (AUC 0.81)

Method. Logistic regression over all 606 decisions: took ~ drones + tasks + types + minute + game + queue length + residual flag + logit(person's leave-one-out rate, shrunk with +1/+2). The same specification as extras2.add_expected; the script checks the fitted values are identical. The chart shows each term's odds ratio with a 95% interval.

C.4.4 Abandonment by what Suggest said

[Pass 5: numbers only]

Every drone failure that needed the operator is one recovery episode. Grouped by what Suggest answered: a **wrong refusal** ended in abandonment 53% of the time (9 of 17). When Suggest gave a plan it was 6% (3 of 51), and when operators fixed things by hand without Suggest it was 13% (9 of 69). A **correct refusal**, where no fix existed with the drones on the mission, ended in abandonment every time (11 of 11), as it should. The striking row is the first one. These were saveable missions, and the same 14 people abandoned only 6 of their 61 other attempted fixable failures. Abandonment was not their habit; the refusal is what is different. When nothing at all was tried, every failure was abandoned or left open (35 + 11). (Figure 46)

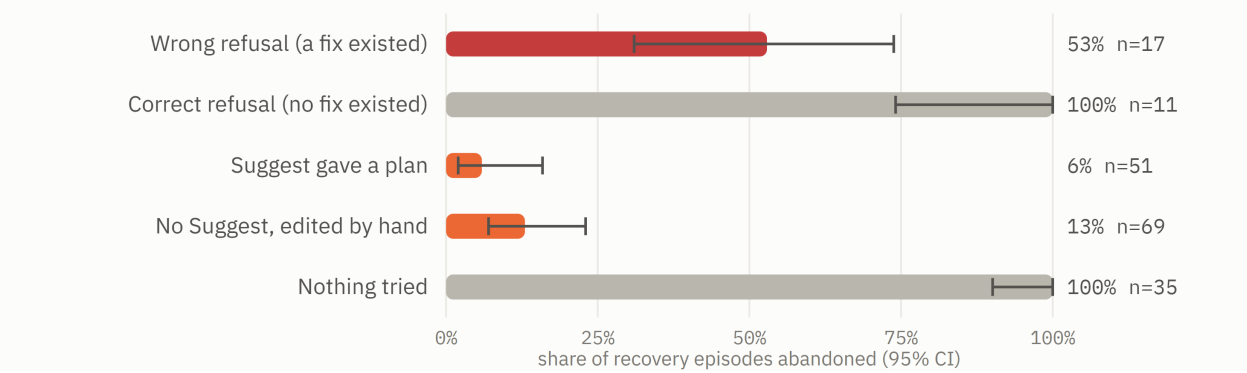


Figure 46: Abandonment by what Suggest said

Key numbers.

- Wrong refusal: 9 of 17 abandoned (53% [31%, 74%]), 14 people; all 17 fixable with drones on the mission
- Correct refusal: 11 of 11 abandoned (100% [74%, 100%]), 6 people
- Suggest gave a plan: 3 of 51 (6%); hand edits only: 9 of 69 (13%); nothing tried: 35 of 35 (+11 left open)
- Wrong refusal vs other attempted fixable failures: 9/17 vs 10/118; odds $\times$12.2, Fisher $p < .0001$; GEE $\times$12.3 [3.2, 47.6], $p = .0003$
- The same 14 people on their other attempted fixable failures: 6 of 61 abandoned

Method. Recovery episodes are rebuilt by h3_recovery.load_all (194 episodes, 33 people). It is a time-local rebuild of which drones were on the mission, idle or executing, at each Suggest press. A press is a **refusal** when Suggest could not cover every task the operator's live plan left short. A refusal is **wrong** when those tasks could have been covered by drones still on the mission, counting ones busy on another task (chaining them on). Suggest only offers drones that are not executing, which is its blind spot. An episode counts as a wrong refusal if it had at least one wrong-refusal press; as a correct refusal if it had refusals and none was wrong. Rates are abandoned $\div$ (abandoned + recovered), with Wilson 95% CIs. The comparison counts wrong-refusal episodes against every other fixable episode where something was attempted (Suggest gave a plan, or hand edits): Fisher exact test, plus a GEE logit clustered by person (odds ratio 12 means the odds of abandoning were twelve times higher).

C.4.5 Do they check after a refusal? What the log shows between the refusal and the close

[Pass 5: extra: may make it in]

The timeline puts every refusal episode on one row, with 0 as the moment Suggest first said no. After a **wrong** refusal that ended in abandonment, not one operator dragged a drone in the planner. Three requested a tactical suggestion again, got the same answer, and abandoned. None did anything on another mission in between, so they were not called away. 8 of the 9 abandoned within 10 s of the first refusal, at a median 3.8 s after the last one. That is time for a glance, not a re-plan. The 8 who **did not** take the refusal at its word look completely different: 7 of 8 started dragging drones, took a median 21 s, and saved the mission. After a correct refusal people behaved the same way as after a wrong one (abandoned in a median 4.4 s, no drags). That is right in those cases, and it shows operators get no cue to tell the two apart: the message is the same either way.

The verified case: P19, session 1, M003. Camera-2 failed. The Supply Drop needed a Camera. Two Lifters sat idle, and the only remaining Camera was busy on Search & Service, due to finish in about 45 s. Suggest said "No workable plan", and the operator abandoned 3 s later. Chaining that Camera on would have saved it (checked frame by frame on the recording). P19 later blamed themselves for failed missions (C19). No wrong-refusal abandoner is among the three who blamed the assistant (C20). (Figure 47)

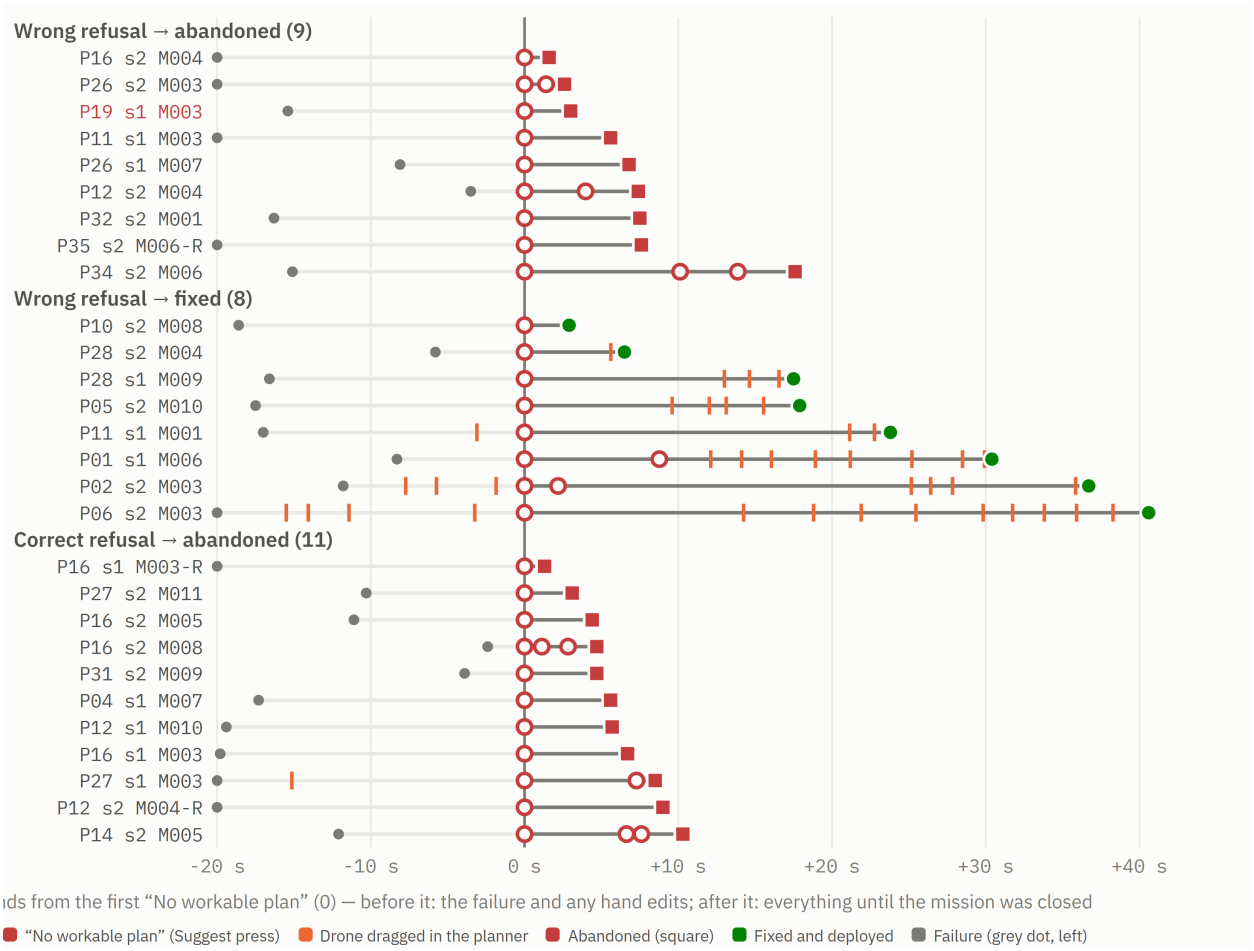


Figure 47: Do they check after a refusal? What the log shows between the refusal and the close

Key numbers.

- Wrong refusal → abandoned (9 episodes, 8 people): drags after the refusal 0 of 9; re-requested a tactical suggestion 3 of 9; actions elsewhere 0 of 9; dragged before the refusal 0 of 9

- 8 of 9 abandoned within 10 s of the first refusal; median 6.8 s from the first refusal, 3.8 s from the last
- Wrong refusal → fixed (8 episodes, 7 people): 7 of 8 dragged drones after; median 20.9 s to deploy
- Correct refusal → abandoned (11, 6 people): 0 drags after, 3 retries, median 4.4 s from the last refusal
- Verified: P19 s1 M003, abandoned ~3 s after the refusal

Method. For each refusal episode the analysis takes the time of the first refusal and lists every logged operator action from then until the mission was abandoned or redeployed. That includes drags in the recovery planner (tactical_assignment_changed), further Suggest presses, and any action on another mission (strategic decisions, tactical confirms, recalls, other recoveries), from the raw event log. "Checked" here can only mean a *logged* action. Looking at the strategic map or the planner leaves no trace, so a fast abandonment could still hide a quick look, and the narration recordings are the place to check that. Seconds are session time; the session pauses if the primary window is hidden, so these are real play seconds.

C.5 The Tactical Assistant is trusted more, and most people say so too (supplementary analyses)

Supplement to Section 7.9.

C.5.1 Is the tactical plan given less attention in Strategic Heavy sessions?

[Pass 5: extra: may make it in]

A check of the idea that tactical trust is higher in Strategic Heavy sessions because the tactical plan gets less attention there. Raw, it does: in Strategic Heavy sessions people request a tactical suggestion less, edit less and spend less time in the planner. But Strategic Heavy plans are much smaller (5.3 vs 8.1 drones). Compared at the same plan size and session order, the planner is open about 12% less long and the checking is about 9% shorter, but neither is distinguishable from no difference, and editing is the same. The one clear difference is the other way round: on same-size plans people request a tactical suggestion *more* in Strategic Heavy (odds $\times 1.5$). People whose planner time drops more do not clearly trust the Tactical Assistant more ($\rho = -.24$, p = .19). And tactical trust itself barely differs between the scenarios (+0.15, p = .25); the gap comes from strategic trust. So the attention account is not supported, though a small effect cannot be ruled out. (Figure 48)

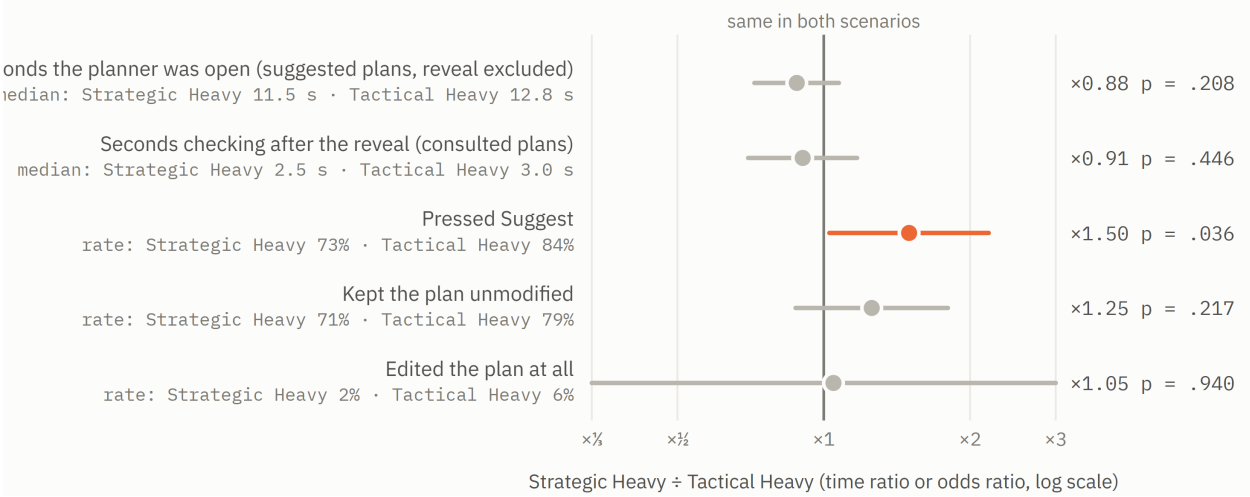


Figure 48: Is the tactical plan given less attention in Strategic Heavy sessions?

Key numbers.

- Planner time (suggested plans): median 11.5 s Strategic Heavy vs 12.8 s Tactical Heavy; adjusted $\times 0.88$ [0.72, 1.07], p = .21
- Checking time after the reveal: median 2.5 s vs 3.0 s; adjusted $\times 0.91$ [0.70, 1.17], p = .45
- suggestion requested: 73% vs 84% raw (OR 0.56, p = .005); same-size plans OR 1.50 [1.03, 2.18], p = .036
- Kept unmodified: 71% vs 79% raw (OR 0.67, p = .046); adjusted OR 1.26 [0.88, 1.80], p = .22
- Edited at all: 2% vs 6% raw (OR 0.44, p = .033); adjusted OR 1.05 [0.32, 3.47], p = .94
- Attention drop vs tactical-trust rise, person by person: $\rho = -.24$ [−.56, .12], p = .19, n = 31

Method. 630 tactical plans, 33 people (P09 excluded): 381 in Strategic Heavy sessions, 249 in Tactical Heavy. Five measures per plan. **Planner time**: from the planner opening to Deploy, minus the forced one-second-per-pair reveal, on plans where a tactical suggestion was requested (486), so it is time spent with the assistant's plan. **Checking time**: last Suggest press to Deploy, minus the reveal (the "checking" measure of the section above). **suggestion requested**, **kept unmodified** and **edited at all** on every plan.

Each is compared Strategic Heavy vs Tactical Heavy with a GEE model clustered by person: times on the log scale (so the result is a ratio of typical seconds), yes/no measures as odds ratios. **Raw** has scenario only; **adjusted** adds plan size (drones $\times$ tasks) and session order, so it compares like-for-like plans. The trust link: each person's average size-and-session-adjusted log planner time in each scenario, the Strategic Heavy minus Tactical Heavy difference, against their difference in tactical trust; Spearman ρ with a bootstrap 95% CI (31 people with suggested plans in both scenarios).

C.5.2 A bad first session with the cards does not explain who loses strategic trust

[Pass 5: numbers only]

If strategic trust fell because people had bad experiences with the cards, then those who met more illogical cards in session 1 should lose more trust. They do not. The raw correlation even points the other way ($\rho = +.40$), but that comes entirely from scenario order: people who played Strategic Heavy first see the odd cards early and lose the least trust (-0.29 against -1.26 for those who played it second). Once order is held constant, the correlation is +.14 (p = .42). Round 1's pre-registered version (missions that failed after a card) was also null: $\rho = -.05$ [$-.39$, .30]. Read this as "no evidence that how much went wrong drives it", not as proof that experience doesn't matter.

Key numbers.

- Illogical cards seen in session 1 vs change in strategic trust: $\rho = +.40$ [+.03, +.70], p = .02 — order artefact
- Same, holding scenario order fixed: $\rho = +.14$, p = .42
- Mean change in strategic trust -0.76; Strategic Heavy first -0.29, Tactical Heavy first -1.26
- H5 primary (card failures in session 1): $\rho = -.05$ [$-.39$, .30]

Method. One row per person (33). The predictor is the count of decisions in session 1 where an illogical strategy card was on screen (from load_decisions' odd flag). The outcome is session-2 minus session-1 strategic trust. Spearman ρ uses a 4000-draw seeded bootstrap CI. The partial version ranks both variables, removes the part explained by which scenario was played first, and correlates what is left. The $\rho = -.05$ is H5's primary test (docs/analysis/round1/h5_trust_drop.md).

C.5.3 What they said agrees with what they rated

[Pass 5: numbers only]

In the debriefs and interviews, 17 operators named the Tactical Assistant as the one they preferred or trusted, 7 the Strategic one, and 9 gave a conditional, "both" or no answer. Those who said tactical had a questionnaire gap of +1.36 on average; those who said strategic were at −0.04. For 18 of the 24 who named one side, the questionnaire points the same way. The coding is from one coder (an AI model) and has not been second-coded, so treat the counts as suggestive. (Figure 49)

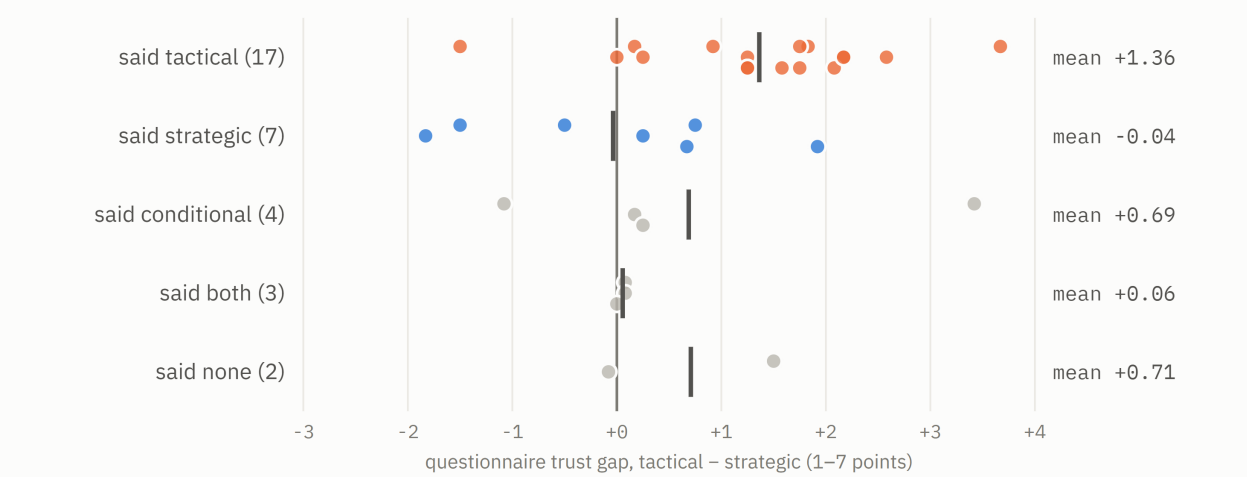


Figure 49: What they said agrees with what they rated

Key numbers.

- Said tactical 17, strategic 7, conditional 4, both 3, none 2
- Sign test 17 of 24: p = .064
- Questionnaire gap: tactical-sayers +1.36 [+0.76, +1.97] vs strategic-sayers −0.04 [−1.27, +1.19]; Mann–Whitney p = .033
- Questionnaire agrees with the stated side for 18 of 24

Method. Stated preference is the round-2 coding of each person's debrief and interview (docs/analysis/round2/wE_codes.csv, field pref). Its sign test asks whether 17 of 24 is more lopsided than a coin toss (exact binomial, two-sided, p = .064). The other test compares the questionnaire gap of tactical-sayers with strategic-sayers (Mann–Whitney, p = .033). Round 1's independent coding agreed poorly with it (κ = .27), and round 2's κ = .96 came from the same AI model. A human second coder is still needed (meth-qual-coder).

C.6 A practitioner's reading: P36, a participant with operational military experience (supplementary analyses)

Supplement to Section 9.1.

C.6.1 Passages not picked this pass

The passages from the same participant that were not picked for the main section, kept so nothing is lost. Each carries the links back it was first given; they were not re-read against a trimmed version.

D Literature Suggestions

[Suggested additions keyed to the results, for the Related Work and Discussion. Not added to `refs.bib`: verify each reference before citing.]

- **Stable individual differences in reliance** (personal style, Section 7.2): Merritt & Ilgen (2008), "Not all trust is created equal: dispositional and history-based trust in human–automation interactions", *Human Factors* 50(2); Hoff & Bashir (2015), "Trust in automation: integrating empirical evidence on factors that influence trust", *Human Factors* 57(3).
- **Attitude scales not predicting behaviour**: Grassini (2023), "Development and validation of the AI Attitude Scale (AIAS-4)", *Frontiers in Psychology* 14 (the scale used); Schaefer et al. (2016), "A meta-analysis of factors influencing the development of trust in automation", *Human Factors* 58(3).
- **Task complexity and delegation** (mission size, Section 7.3): Lee & Moray (1994), "Trust, self-confidence, and operators' adaptation to automation", *International Journal of Human-Computer Studies* 40(1); Dzindolet et al. (2003), "The role of trust in automation reliance", *IJHCS* 58(6).
- **Anchoring on advice** (manual builds track the cards, Section 7.4): Tversky & Kahneman (1974), "Judgment under uncertainty: heuristics and biases", *Science* 185; Furnham & Boo (2011), "A literature review of the anchoring effect", *Journal of Socio-Economics* 40(1).
- **Workload and reliance** (the null, Section 7.6): Parasuraman & Manzey (2010) (already in the bib); Bailey & Scerbo (2007), "Automation-induced complacency for monitoring highly reliable systems", *Theoretical Issues in Ergonomics Science* 8(4).
- **Visible errors and reliance** (illogical card, Section 7.8): Dietvorst, Simmons & Massey (2015), "Algorithm aversion: people erroneously avoid algorithms after seeing them err", *Journal of Experimental Psychology: General* 144(1); Madhavan & Wiegmann (2007), "Similarities and differences between human–human and human–automation trust", *Theoretical Issues in Ergonomics Science* 8(4).
- **Automation bias and the cost of checking** (wrong refusals; tactical plans kept unchecked, Section 7.9): Bahner, Hüper & Manzey (2008), "Misuse of automated decision aids: complacency, automation bias and the impact of training experience", *IJHCS* 66(9); Goddard, Roudsari & Wyatt (2012), "Automation bias: a systematic review of frequency, effect mediators, and mitigators", *JAMIA* 19(1); Mosier & Skitka (already in the bib).
- **Measuring trust; stated vs revealed**: Kohn et al. (2021), "Measurement of trust in automation: a narrative review and reference guide", *Frontiers in Psychology* 12.
- **Methods**: Liang & Zeger (1986), GEE, *Biometrika* 73(1); Mancl & DeRouen (2001), bias-reduced sandwich SEs, *Biometrics* 57(1); Holm (1979), *Scandinavian Journal of Statistics* 6(2); McFadden (1979) on pseudo-R^2 (in Hensher & Stopher, eds., *Behavioural Travel Modelling*); Rousseeuw (1987), silhouettes, *Journal of Computational and Applied Mathematics* 20; Hubert & Arabie (1985), adjusted Rand index, *Journal of Classification* 2; Ericsson & Simon (1993), *Protocol Analysis* (think-aloud).